Optimal Public Transportation Networks: Evidence from the World's Largest Bus Rapid Transit System in Jakarta *

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Abstract

Designing public transport networks involves trade-offs between coverage, service frequency, and direct service. We use the expansion of the bus system in Jakarta, Indonesia, to study these trade-offs. We analyze how new direct connections, changes in bus travel time, and wait time reductions affect bus ridership and aggregate flows, and estimate a transit network demand model by matching the route launch events. Commuters in Jakarta are 2-3 times more sensitive to wait time than bus time, and inattentive to long routes. We develop a flexible framework to characterize optimal networks. A less concentrated network would increase ridership and commuter welfare.

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1 Introduction

Governments are investing heavily in public transport in the megacities of low- and middle-income countries (LMIC), giving city governments a growing role in planning and operations. For example, the number of LMIC cities with a Bus Rapid Transit (BRT) system went from 14 cities in 2000 to 100 in 2020 (BRT+ Centre of Excellence and EMBARQ, 2023). Bus networks are particularly important because they account for a large share of ridership. For example, Delhi's public bus system carried 3.3 million passengers per day in 2017, much more than its more famous metro system (2.7 million, Hindustan Times 2018). Ensuring that these transport systems are well-designed is key to sustainable and efficient mobility.

Designing bus networks involves trade-offs. Because the fixed cost of a bus route is relatively small (compared to, say, building a subway or light-rail line), there is great latitude in designing a system. For example, for a given number of buses, one can choose between having a more *direct* network (i.e., more direct routes, rather than a hub-and-spoke system that requires more changes between routes); a more *intensive* route network (i.e., more frequent service to key locations); and a more *expansive* route network (i.e., many routes serving more destinations). These trade-offs may overlap in complex ways.

Fundamental to these trade-offs is an understanding of public transport preferences and demand, which in turn affects bus ridership: how commuters value bus travel time, wait times in stations, and transfers between routes. Different preference parameter configurations may result in very different-looking optimal transportation networks, but centralized public transport planners face an information barrier in learning about these complex demand patterns. Estimating preferences over these non-price attributes is challenging because it requires plausibly exogenous variation at an aggregate (route) level.

In this paper, we study the expansion of the TransJakarta bus system in Jakarta, Indonesia. TransJakarta operates a public Bus Rapid Transit (BRT) network, which is the largest such network in the world (Institute for Transportation and Development Policy, 2019). From 2016-2020, TransJakarta launched a total of 93 routes across the city in a staggered fashion for both BRT routes and non-BRT (feeder) routes. These new routes changed the attractiveness of bus travel in different ways for different potential journeys, which we use to estimate preferences and study the implication for network design.

Our paper proceeds in three steps. First, we estimate the reduced form impact on bus ridership and overall commuting flows from improvements to the TransJakarta network. To do so, we show how the 93 new bus routes create different types of improvements in the availability of direct connections, travel times, and wait times for different origin-destination pairs, and analyze these 'events' using a differences-in-differences framework. Second, we estimate a model of commuter travel behavior, which allows us to back out the underlying preference parameters that match the reduced-form estimates from step 1. Third, we simulate what optimal transportation networks would look like given the preferences from step 2, and contrast their characteristics with that of the existing network.

For the first step, we define three types of 'events' that correspond to different types of service improvement for origin-destination (o, d) pairs. The first two events focus on new direct routes for (o, d) pairs that were previously served only via transfer connection. The first event type (hereafter, Event 1) considers cases where the total time on the bus is unchanged with the new direct route. This introduces variation in the degree to which the network is *direct*. Event 2 considers new direct routes where the total time on the bus also falls, creating additional variation in service *speed* in addition to becoming more direct. Event 3 considers (o, d) pairs that receive a new direct route that overlaps with the old direct route(s) from o to d. This leads to more frequent service, varying the degree to which service is *intensive*, without substantially changing time spent on the bus or the directness of the route.

We use several new datasets to analyze these events. To measure (o, d) ridership flows on the bus network, we use TransJakarta's detailed administrative data for both BRT routes and non-BRT feeder routes. TransJakarta collects fares using a tap-to-pay smart card system and our data includes every tap — over 500 million taps in the study period — along with an anonymized card identifier.[1] Second, we measure bus travel times and the distribution of wait times that passengers face using GPS data that tracks the position of every TransJakarta bus every 5-10 seconds. Third, we measure overall commuting flows on each (o, d) pair (regardless of transportation mode) using anonymized smartphone location data. We aggregate the outcome data at the weekly level and at the level of 1km-wide hexagonal grids, and show robustness to other grid shapes and sizes.

Using a unified difference-in-differences design, we find that bus ridership is responsive to each of these three types of service quality improvements.[2] For BRT routes, a new direct route between an existing (o, d) pair increases ridership by 0.16 log points. Event 2, which substantially reduced travel times in addition to a new direct route, increased ridership by 0.27 log points. Event 3, which reduced wait times by 0.31 log points (through adding buses via an incidental new route), increased ridership by about 0.09 log points, implying that ridership also responds substantially to wait times. Event-study graphs show clear jumps in

[1] Since the system enforces tap-in everywhere in the network, but only uses tap-out at a subset of BRT stations, we use an algorithm to infer each trip's likely destination. We validate this approach using the subset of trips for which we directly observe both the origin and destination (Appendix A.3.4).

[2] We include a large set of "never-treated" origin-destination pairs, which largely alleviates the class of problems with two-way fixed effects specifications (see, e.g., De Chaisemartin and D'Haultfoeuille 2022).

ridership immediately after the launch of new routes, and no pre-trends. The results for non-BRT routes are typically larger than those for BRT routes in proportional terms, but similar in levels. Our results are stable when varying grid cell size from 500 meters to 2 kilometers, implying that the events lead to new bus ridership rather than displacing passengers from neighboring TransJakarta stations.

We next look at the impact of the same set of events on the aggregate volume of trips between a pair of locations (regardless of whether people take the bus or an alternative), measured using the smartphone location data. We do not find observable increases in all trips for any of the events in our benchmark specification.

The results imply substantial increases in ridership when service frequency improves on all three dimensions — directness, speed, and wait times — and with this being driven by substitution towards bus trips, rather than by completely new journeys. This increase in ridership may have spillover effects of other important social outcomes. For example, we estimate that every 1 percentage point increase in the mode share of TransJakarta lowers air pollution by 2.2% and lowers road travel times by 0.3%.[3]

With our reduced form estimates in hand, in the second step of our paper, we develop a model of commuter travel behavior to estimate underlying preference parameters. The model has a new formulation of how commuters choose routes within a geographically realistic transit network, in the presence of stochastic wait times. A commuter traveling from o to d chooses between direct and transfer bus options that differ in terms of type (BRT or non-BRT), total travel time on the bus, and the necessity of a transfer. Buses on each route arrive according to a Poisson process. A commuter who decides to take the bus gets a draw of wait times for all routes in her choice set and selects the best option overall. Because the idiosyncratic term in the model comes from random wait times given by Poisson processes, the model is invariant to aggregating identical routes, a property that allows us to bypass spurious gains from variety when we return to the optimal network design problem.[4]

Importantly, commuters sometimes face large choice sets with dozens of options within TransJakarta's network, many of which are too slow to consider. Therefore, we allow for (and estimate the degree of) partial inattention to bus route options that have high travel time on the bus, relative to the quickest bus option in the choice set (Larcom et al., 2017).

In a higher nest, commuters choose between using the bus network, taking expectations

[3]Converting road traffic lanes to dedicated bus lanes can increase road traffic congestion (Gaduh et al., 2022), yet the TransJakarta network expansion we study here did not reduce road traffic lanes.

[4]In typical discrete choice models such as logit, splitting a bus route into two identical routes with half the original bus allocation is the source of an expected utility gain because the new routes have independent idiosyncratic errors; this is the famous 'red-bus, blue-bus' problem discussed by McFadden (1974). We show that the general version of our model, which appears in Daganzo (1979) under the name "negative exponential distribution model," does not have this issue.

over the wait times they may face, or a private transportation outside option (e.g. motorbike, car), based on a logit specification. We assume that origin-destination flows are fixed.[5]

We estimate the model for the discretized Jakarta metropolitan area using 1km hexagonal grid cells. We use smartphone data to estimate commuting flows across all possible combinations of these grid cells. We then use the model to compute predicted ridership over each (o, d) pair and examine how it changes over time as the bus network changes, by running the same reduced form analysis as in the actual data. This allows us to estimate the preference and attention parameters that match the actual event-study estimates.

Using this approach, we find a high disutility for waiting time relative to time traveling on the bus. Specifically, we estimate that travelers find wait time 2.1× more costly than travel times for BRT, and 3.4× most costly for non-BRT. Waiting may be more stressful and less comfortable compared to riding the bus (for example, bus stations are not air conditioned and have very limited seating), and these estimates may also capture costly wait time uncertainty. In our model, commuters dislike bus options that involve transfers due to the additional wait time and typically longer time on the bus. However, our estimates show there is no additional transfer penalty, and for BRT we find that commuters dislike transfers slightly less than implied by the costs from the additional wait time and time on the bus. Commuters pay significantly less attention to bus options with bus travel time more than 36%-43% longer than the quickest option in their choice set. We also find that commuters from low poverty areas are more elastic to bus network service improvements.

What do these preferences mean for the optimal design of the network? In the third part of the paper, we introduce a theoretical and computational framework that allows us to characterize optimal networks, compare them with the current network, and perform comparative statics with respect to structural model parameters. To do this, we transform the planner's optimal network problem into a tractable problem of sampling from a probability distribution over all networks. Our main contribution is establishing this link between describing optimal policies and sampling. We can then use existing, established algorithms to sample in high-dimensional settings.

We ask how the network of bus routes should be configured across the city and how the existing set of buses should be allocated to these routes to maximize a utilitarian welfare objective informed by our estimated demand model.[6] This involves balancing trade-offs across reach (having enough bus stops to cover origins and destinations for which people

[5]While it is possible to further embed the model into a destination-choice nest or in an urban general equilibrium model (Tsivanidis, 2024), the null reduced form effect on aggregate trips suggests that over the period we study, the preferences we estimate over wait time, transfers, and travel time are not biased by changes in aggregate trip patterns.

[6]We later vary the planner's objective in sensitivity analysis.

would like to travel), wait times (putting more buses on a given route), and the topology of the network (which stations should be connected directly versus by transfer). We hold fixed TransJakarta's current BRT infrastructure, which allows for faster travel times along the 13 BRT corridors, but we allow routes to enter and exit the BRT corridors, to connect them in new ways, and to travel on existing streets. We study this problem using realistic geography of Jakarta at the level of 2km-wide square grids.

Our problem displays divergent substitution and complementarity forces at the same time. This precludes us from using existing methods to characterize optimal allocations when policies are either always complements or always substitutes (Jia, 2008; Arkolakis and Eckert, 2021). This is also a very high-dimensional problem, and the number of possible configurations is extremely large, making it difficult to identify the (generically unique) welfare-maximizing network.[7]

Therefore, we set up a slightly modified problem: the social planner chooses a network N that maximizes utilitarian welfare $W(N, \theta)$ from our estimated demand model — average expected utility for all commuters — plus an idiosyncratic shock ϵ_N for each possible network. The shocks capture network-level factors that the social planner cares about, such as cost shocks or benefits from certain networks, outside of our demand model. As such, in our framework, the social planner has a discrete choice problem over the space of all possible networks. A network with high model welfare carries a large probability of being chosen, because small idiosyncratic shocks may tip the planner into preferring it rather than the global optimal network. We parameterize the problem as a multinomial logit with parameter β. We bound the welfare loss under this probability distribution relative to the global optimum.

This approach allows us to transform a global optimization problem into a problem of sampling from a large-scale multinomial logit distribution. Several algorithms based on the Metropolis-Hastings algorithm have theoretical guarantees and good practical performance in sampling from such distributions.[8] We derive formulas for properties of optimal networks and comparative statics. We can estimate these objects using their sample analogs. Our framework is general and may be used in other settings where a social planner's problem is high-dimensional.[9] In particular, this framework can be used to describe optimal bus net-

[7]As we outline in Appendix section A.7.7, the number of feasible networks is on the order of $e^{26,000}$.

[8]We use a version of the simulated annealing (SA) algorithm, which we stop at a finite temperature (β^{-1}) and run multiple times to obtain a sample of networks drawn (asymptotically) from the planner's distribution. Our approach offers a new interpretation of the output from this algorithm.

[9]While the details of the implementation of the sampling algorithm will be application-specific, the theoretical framework can be applied to problems such as optimal spatial policy (Fajgelbaum and Gaubert, 2020), optimal transportation infrastructure (Allen and Arkolakis 2022; Fajgelbaum and Schaal 2020), and firms' global sourcing decisions (Antràs et al., 2017). The setting is appropriate when the welfare difference

works when considering changes to the BRT infrastructure, or in the presence of additional margins of adjustment, by embedding our demand model in a general equilibrium quantitative spatial urban model where agents choose vehicle ownership, home, work and shopping locations (Tsivanidis, 2024; Miyauchi et al., 2021). This may generate additional insights. For example, transit hubs may have an additional benefit due to agglomeration forces.

Using this approach, we find that optimal networks, based on the preference parameters that we estimated, lead to 62% higher ridership compared to the current network, and commuter welfare gains equivalent to reducing bus travel times by 11 minutes for all current riders.

Optimal networks have substantially lower wait times than the current network. They are only slightly more expansive, an effect that is concentrated in the areas just outside downtown Jakarta, and they have significantly fewer bus lines. Bus frequency is more evenly spread across routes in the optimal networks, i.e., slightly lower in the city center and higher in the other areas. Because bus headway on a route is a convex function of the number of buses allocated to the route, lowering the dispersion of the bus allocation also tends to decrease average wait times.

Our exercise posits a demand model and a planner's objective function and compares the current network to optimal networks based on that model and objective. We report results from a model with heterogeneous preferences and discuss several alternate model assumptions, such as crowding and agglomeration, and how they may impact our results. We also show the impact on our findings of alternate planner objectives such as exclusive focus on the welfare of residents of the city of Jakarta as opposed to outlying areas, and using bus ridership instead of commuter welfare as an objective.

In our last exercise, we explore how certain characteristics of the estimated optimal networks would change if preferences differed. For example, doubling the wait time cost leads to more concentrated networks connecting only 9.8% percent of origin-destination pairs, 46% less than before. Increasing the transfer penalty by the equivalent of 15 minutes of wait time increases the share of connected location pairs that have a direct connection from 15% to 18%. This type of comparative static exercise connects micro parameters in the model to macro properties of optimal bus networks, which helps uncover which parameters matter for network design and to assess the sensitivity of our results to the parameters to estimate.

Our project connects to several literatures. First, we build on classic questions in the transportation economics literature: travel demand estimation and mode choice in particular (McFadden, 1974; Ben-Akiva et al., 1985), and increasing returns in public transport

(or ratios) can be computed for pairs of counterfactuals, for example, if exact hat methods are applicable.

(Mohring, 1972).[10] We build on the literature investigating how commuters value trip attributes, see e.g. Wardman (2004); Currie (2005); Abrantes and Wardman (2011) and Small et al. (2007, page 53). Studies in this literature tend to focus on high-income countries and often infer stated preferences from respondents' choices between hypothetical alternatives. In contrast, we estimate preferences using natural experiments. In that respect, our study is closest to Kreindler (2024), which uses a field experiment to estimate the underlying preference parameters required to calculate optimal congestion pricing in Bangalore, India. Here, by contrast, we use a variety of real-world changes in the attractiveness of the network given by different types of network expansion for identification. Our study is also related to Buchholz et al. (2020) and Goldszmidt et al. (2020), both of which estimate the value of wait time for taxis and ride-hail from choices made on ride-hail platforms in Europe and the United States.[11]

Second, by embedding travel demand preferences into a model of optimal bus route network design, our project also contributes to the growing trade-inspired literature that tackles questions of how to design a transport network (Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022; Balboni, 2021; Santamaria, 2022; Graff, 2024; Alder, 2023; Krantz, 2024). This literature has so far focused on road infrastructure (typically inter-city), rather than urban transit.[12] Our detailed microdata allows us to estimate fine parameters that are usually hard to incorporate in these studies, and shows how they are related to features of optimal networks.

Our paper is also related to the literature on the impact of transit systems in LMIC. Gaduh et al. (2022) studies the impact of the initial launch of the TransJakarta system in 2002 to show that at that point in time, the system had no impacts on public transport use but the conversion of road lanes into dedicated lanes for the BRT network worsened road traffic congestion. Majid et al. (2018) study the impact of a new BRT transit line in Lahore, Pakistan. Tsivanidis (2024), Zárate (2023), and Balboni et al. (2021) measure the impact of the BRT systems in Bogotá, Mexico City, and Dar Es Salaam, respectively, and quantify general equilibrium effects through the lens of quantitative urban models. Tsivanidis, for example, focuses on estimating the welfare and inequality effects of the system and how it affects the organization of the city. Hernandez-Cortes et al. (2023) compares the impact of BRT and subways in Mexico City. All these papers take the public transport system

[10] See also Coulombel and Monchambert (2019) for evidence on dis-economies of scale.

[11] Hörcher et al. (2017) estimate subway crowding costs in Hong Kong with a revealed preferences approach.

[12] Fajgelbaum and Schaal (2020) study a neoclassical trade model where the planner's problem is globally concave. Allen and Arkolakis (2022) study a tractable general equilibrium model with routing and congestion. Balboni (2021) and Santamaria (2022) compare different road investment plans in dynamic settings. Alder (2023) introduces a heuristic algorithm for finding the globally optimal highway network in the context of a quantitative trade model for India.

design as given. By contrast, we tackle the complementary problem, focusing on estimating underlying preference parameters, and solving for the optimal system design.

Meta-heuristic algorithms, including the simulated annealing algorithm that we use here, have been used extensively in the transportation and operations research literatures to *approximate* optimal designs in urban transit network design problems (Wei and Machemehl, 2006; Iliopoulou et al., 2019). Our contribution is to link in an asymptotic sense the output of such algorithms to a social planner's discrete choice problem over all possible networks. Instead of approximating an optimal network, we sample multiple networks, which allows us to estimate optimal network characteristics and comparative statics.[13]

The rest of this paper is organized as follows. Section 2 describes the TransJakarta bus network, its expansions, and our data. Section 3 presents the reduced-form event-study results that show how ridership responds to new routes and improved bus frequency. Section 4 introduces the model, describes how we estimate the model using the reduced form moments, and presents estimation results. Section 5 explores the implications for optimal network design.

2 Setting and Data

2.1 The TransJakarta Bus Network

Setting Description. TransJakarta operates an integrated bus system serving the central urban districts in Greater Jakarta — a metropolitan area of about 18 million people. Its routes are concentrated primarily in the province of DKI Jakarta (the Jakarta Special Capital Region), though some extend to surrounding districts. Established in 2004, TransJakarta serves hundreds of thousands of passengers daily, with daily ridership peaking around 1 million in early 2020. TransJakarta is a business entity controlled by the Provincial Government of DKI Jakarta whose mission is to provide transportation services to further the welfare of Jakarta residents (TransJakarta, 2019). In 2019, the subsidy from the Jakarta government accounted for 78.3% of total revenue.

As of August 2019, TransJakarta has over 139 bus routes, which are a mix of Bus Rapid Transit (BRT) and non-BRT routes. BRT routes operate on dedicated bus lanes along designated stations. Passengers pay at turnstiles to enter the BRT station and board the bus. It has a network length of more than 120 miles of BRT corridors, the longest such system in the world (Institute for Transportation and Development Policy, 2019). Non-BRT feeder routes operate partially on normal city streets, and partially along the BRT corridors,

[13] Daganzo (2010), Fielbaum et al. (2016) characterize optimal transit networks in idealized environments.

where they also stop in BRT stations. These non-BRT routes connect locations that are further away from the BRT system.

TransJakarta costs a flat fare of IDR 3,500 (USD 0.25) per trip, regardless of distance.[14] Payment is collected by tap-in smart cards, either at BRT stations or through fare machines on non-BRT buses; tap-out is enforced at a small number of BRT stations as well.[15] Free transfers are allowed at BRT stations.

TransJakarta is the primary public rapid transit system in the city. A commuter rail system (KRL CommuterLine) with about 1 million daily ridership serves outlying areas for longer trips that rarely overlap with TransJakarta trips. In 2019, a single 16 km subway line was opened to serve about 80,000 riders per day. In 2018, privately provided shared transport, using minibuses, accounted for 2.9% of all trips in the larger Jabodetabek region (JUTPI2, 2019). The primary alternative to TransJakarta is private transport, consisting of a mix of motorcycles, private cars, motorcycle taxis (*ojek*), and automobile taxis.

TransJakarta Network Expansions. We study the large expansion of the TransJakarta network, which added more than 93 BRT and non-BRT routes between January 2016 and February 2020. Only one entirely new BRT corridor was launched during this period.[16] The newly launched BRT routes run along existing BRT corridors, sometimes connecting two corridors or running an express route on certain sections. The new non-BRT feeder routes stop both in some BRT stations as well as in roadside non-BRT bus stops, connecting one or multiple BRT corridors to other areas of the city.

The number of operating buses more than doubled during this period, from about 700 to more than 1,600. Figure 1 shows the expansion of the system. These expansions — for both BRT and non-BRT routes — took place at different times throughout the city. New routes used newly available buses rather than reallocating buses from other routes.

New lines were launched based on a mix of external constraints and discretion with the goal of increasing access and ridership.[17] All non-BRT route launches were chosen from a list of routes that were pre-approved in 2016 by the Jakarta transportation department. For

[14] We focus in this paper on the optimal design of *non-price* system attributes: travel time, wait times, transfers. For optimal pricing, Mohring (1972) argues that an increasing to scale provider should set fares at short-run marginal cost and get subsidized for the deficit. (See also Almagro et al. 2024.) As a reference, the fare is equivalent to 4-7 minutes of travel time on the bus if bus riders value time on the bus at 50-100% of an hourly wage of 3.5 USD. We lack the cost data that are needed to study optimal pricing, but note that the relatively low TransJakarta fares and high subsidy rates are consistent with this idea.

[15] Cash payments are allowed on non-BRT buses, accounting for around half of all non-BRT transactions in 2019.

[16] Corridor 13 is an elevated busway, thereby not affecting the number of lanes available to road traffic.

[17] TransJakarta has three performance indicators under the "Route Expansion" strategic target category: (1) service range (in 2019, the goal was to cover 75% of Jakarta's population within 500 meters from a bus stop), (2) highest number of customers per day, and (3) number of passengers per year (TransJakarta, 2019).

BRT routes, TransJakarta created new routes by connecting existing BRT stations along the existing BRT dedicated lanes. For both BRT and non-BRT, TransJakarta chose routes and launch periods based on new bus fleet delivery dates, bus availability based on bus-operator contracts, inputs from field reports, and other factors.

2.2 Data

We use three different datasets: **TransJakarta ridership** data from administrative data on TransJakarta's smart card transactions; origin-destination **aggregate-trip flows** data from anonymized smartphone location data; and **bus locations** data, which we obtain by processing detailed GPS data on every TransJakarta bus throughout our period.

We focus on the central urban districts of Greater Jakarta.[18] We use consistently defined geographical environments throughout our analysis. We divide space into identical grid cells and aggregate stations at the grid cells level, and trips at the grid cell pair level.[19] Throughout the reduced form and demand estimation analysis. We use a regular hexagonal tiling with 1,766 grid cells where adjacent grid-cell centroids are 1,000 meters apart, and then show robustness to using 500-meter and 2000-meter square grid cells. For the optimization section, we use 2,000-meter square grid cells to reduce the problem's dimensionality. We aggregate outcomes (e.g. ridership, trips, bus supply) at the weekday or work-week level, focusing on the 5AM to 10PM interval.

Ridership Data. We use the administrative ridership data that is captured electronically by smart cards to construct a highly granular TransJakarta origin-destination (o, d) ridership matrix at each point in time since 2016 (since 2017 for non-BRT). In BRT stations, passengers tap smart cards at turnstiles to enter the boarding area. In 36 percent of BRT stations, passengers also tap smart cards upon exit. In non-BRT bus stops, passengers board the bus directly from the street and tap to pay for the bus ticket upon boarding.[20]

We observe the time of each tap, as well as an anonymized identifier for the smart card used, which allows us to link transactions from the same smart card over time.[21]

[18] This consists of the province of DKI Jakarta and all the surrounding urban districts: Tangerang, South Tangerang, Depok, and Bekasi (see Figure A.1).

[19] We collapse the TransJakarta network to the level of grid cells. For transfer options, we consider that origin o and destination d are connected by a transfer option by routes r_1 and r_2 if these two routes intersect in an actual station.

[20] Passengers have the option to pay by cash on non-BRT routes, which accounted for 57% of transactions in 2019. We apply weights to the ridership patterns computed using the smart card data to match total ridership for each origin station.

[21] Individuals can purchase new smart cards, but do so infrequently. The median smart card in our data is active for over 4 months, and the median tap belongs to a smart card that is active for over 20 months.

While we observe each passenger's trip origins, we only observe destinations (d_{it}) for the trips that end at stations where tap-out is enforced. In all other cases, we construct a proxy destination station based on an algorithm that uses the origin station of the next trip for the same smart card, or another frequent origin for that card. We validate this algorithm using the existing tap-out data.[22] We describe the algorithm and ridership data cleaning process and coverage in Appendix A.3.4.

Aggregate Trip Flows. We augment our administrative data with anonymized historical smartphone location data to measure overall trip behavior, regardless of whether a given commuter uses the TransJakarta system or not. We have daily smartphone location data from March 2018 through March 15, 2020 from Veraset, a private data provider. These data cover 35 million weekday trips that belong to 2.3 million unique devices in our study area. (For reference, our study area had 14 million individuals over 15 years old at the 2020 census.) We use these data to: (1) create a panel of trip flows separately for each week in the data as an outcome variable when studying the impact of the network's expansion, and (2) construct a cross-section of typical (o, d) trip flows throughout the period in order to compute choice probabilities for using TransJakarta. See Appendix A.3.5 for details.

The smartphone location data captures rich travel patterns within Jakarta. Figure A.32 shows that the Veraset data is broadly representative across areas with different levels of poverty and density.

Bus Location and Allocation, and Network Expansion. We have data on the planned and realized daily bus allocations for each route, as well as exact GPS locations for every bus every five to ten seconds for the near-universe of routes. These data cover approximately 1,800 buses and 16,000 bus trips per day since January 2017. We process this raw GPS data to compute median bus travel times on all routes, between every origin and destination stations, and to measure the empirical distribution of wait time for each route. See Appendix A.3.2 and A.3.3.

To reconstruct the TransJakarta network at different points in time, we combine data from TransJakarta and data from a mobility planning app available in Jakarta during the study period. We cross-check launch dates inferred from bus allocation data with route- and date-specific TransJakarta ridership data, as well as directly with TransJakarta staff.

[22] For the set of destination stations where we have actual exit transactions, bivariate regressions of imputed and actual daily ridership shares have $R^2 = 0.85$ (Appendix Figure A.30).

3 Reduced Form Results: the Impact of Service Quality

We begin by estimating the 'reduced form' impacts of improved service quality on TransJakarta ridership and on all trips (measured using smartphone location data). We use the fact that the launch of a new bus route changes different dimensions of how attractive it is to take TransJakarta for different (o, d) pairs, based on how it affects the choice set of direct and transfer bus routes between those locations.

We focus on three types of 'events' induced by new route launches. The first two capture the launches of the first *direct* route between two locations that are already connected by *transfer*. In Event 1, the new direct route is not faster than the fastest existing transfer connection, in terms of total time on the bus. In Event 2, the new direct route is faster. Event 3 captures the addition of buses between two already directly connected locations because the new route overlaps with the existing direct route(s) between o and d. This increases the bus arrival rate and lowers wait times for those traveling from o to d. We examine these event types separately for BRT and non-BRT connections.

3.1 Estimation Framework

For each of the three event types described above, for both BRT ($E \in \{1B, 2B, 3B\}$) and non-BRT network expansions, ($E \in \{1N, 2N, 3N\}$), we estimate the following exponential equation:

$$\mathbb{E}Y_{odt} = \exp\left(\alpha^E Post^E_{odt} + \alpha^E_{-10} L^E_{\leq -10,odt} + \alpha^E_{10} L^E_{\geq 10,odt} + \mu^E_{od} + \nu^E_{ot} + \xi^E_{dt} + \varepsilon^E_{odt}\right) \tag{1}$$

Our dataset is at the origin (o) $\times$ destination (d) $\times$ week (t) level, so we include all two-way fixed effects. Specifically, μ^E_{od} are origin$\times$destination fixed effects — i.e. fixed effects for every combination of start and end grid cell — and ν^E_{ot} and ξ^E_{dt} are origin$\times$time and destination$\times$time fixed effects. These fixed effects flexibly capture differences in ridership across origin-destination parts, as well as arbitrary time-related shocks for each origin and destination.

The key variable of interest is $Post^E_{odt}$, an indicator variable for the event having taken place on the od route in the previous 10 months.[23] We control for $L^E_{\leq -10,odt}$ and $L^E_{\geq 10,odt}$, which are indicators for whether an event between o and d takes place 10 or more months in the future, and in the past, respectively. The coefficient α^E in equation 1 thus captures the overall effect of an event of type E for the pair (o, d) in the first 10 months after it occurs, relative to the 9 months prior to the event.

[23]For Event 3, we focus on the first event if there are multiple events for a given o, d pair.

The key identifying assumption is that absent the event induced by the new route launch, and conditional on the included fixed effects, the outcome for treated origin-destination pairs would have evolved on parallel trends relative to non-treated pairs. In the event study version of (1), we will test visually for the presence of pre-trends. We also show that treated and untreated grid cells have substantial common support on distance to CBD, log population, and the poverty rate (Figure A.2).

We estimate (1) using robust Poisson Pseudo Maximum Likelihood (PPML). We cluster standard errors two-way by both origin and destination, which allows for arbitrary serial correlation over time for each origin and each destination, as well as arbitrary correlation among destinations for a given origin, and vice-versa (Cameron et al., 2011).

The sample includes only *odt* observations that are connected at time t within the TransJakarta network, implying that both o and d contain TransJakarta stations before each type of event. The sample of pairs od for event type E is restricted to all origins treated at least once ($Post^{E}_{od't'} = 1$ for some d', t') and to all destinations treated at least once. Most of the origin-destination pairs included in the sample are never treated. (For example, for BRT event type 1 this number is 78.4 percent).[24]

We use two outcome variables Y_{odt}. First, we examine the total TransJakarta ridership between grid cells o and d during week t, summing up over all station pairs in o and d. Second, we also measure impact on all trips between o and d in week t, computed based on smartphone location data.

We use the full time period of data that we have for each type of event, but since the BRT, non-BRT, and smartphone trip data all begin on different dates, the data start at different times.[25] All data end in mid-March 2020, prior to the COVID-pandemic shutdown.

For each of the events described below, we present the overall effects from estimating the corresponding version of equation 1. We also present event study graphs, where we add month-by-month leads and lags of the key explanatory variables.

In Events 1 and 2, a route launch creates the first direct connection between o and d, a location pair that already has a transfer connection. The two event types differ only in terms of whether the new direct route has faster travel time on the bus compared to the fastest pre-existing (transfer) option.

In Event 3, grid-cell pairs o and d that are already directly connected get more buses

[24] The fact that most of our *od* pairs are never treated largely alleviates the class of problems with two-way fixed effects specifications that have been highlighted recently (see, e.g., De Chaisemartin and D'Haultfoeuille 2022), which are primarily a concern when using previously-treated observations as a comparison group.

[25] For BRT events, the sample comprises January 2016 to mid-March 2020. For non-BRT events, we further restrict the sample to the period after mid-January 2017, when our non-BRT data begins. When we use smartphone trips as an outcome variable, the time periods span from March 2018 to March 2020. We always exclude May-July 2018 due to missing BRT ridership data.

from a new route because it overlaps with the existing route for the portion between o and d. Specifically, $Post^3_{odt}$ is an indicator for the first event of an additional direct route launched between o and d taking place, in the ten months before week t.

3.2 "First Stage" Impacts on Travel and Wait Times

We first study how the three different events affect the attractiveness of riding the TransJakarta network from o to d (Table 1).

We first estimate the impact on the log minimum travel time on the bus between o and d. We use the bus GPS data to measure bus travel time for each route and pair of stations (on average over the study period), and at each time t between 2016 and 2020 we take the minimum over all available bus options (direct or single-transfer). This measure captures the pure travel time on the bus — i.e. it does not include any time spent waiting for a bus to arrive or any time waiting for a connecting bus.

Adding a new direct line between o and d reduces travel time, but only when the new line is faster than the existing transfer connection (Event 2), yielding an average effect of about 0.29 log points with a standard error of 0.027 (column 2). While the fact that travel time reductions are limited to Event 2 is, of course, mechanical to the way we define the events, we show that the differences are quantitatively large and statistically quite meaningful.[26] Event 3 has a very small effect on log travel time.

Next, we examine the log number of buses arriving per hour at the origin, over all the direct and transfer options that connect the origin and destination. This is a proxy of waiting times at the origin station. Event 3 increases the bus arrival rate by 0.31 log points (Table 1, column 6).[27] Events 1 and 2 have small effects on the bus arrival rates at the origin. Note that these effects do not capture the reduction in wait time for transfer connections.

Results for non-BRT network expansion are broadly similar and typically larger in magnitude (Table 1, panel B). Importantly, baseline bus arrival rates for treated origin-destination pairs is 2–4 times lower for non-BRT routes compared to BRT (columns 4–6). This means that the absolute reduction in wait time is much larger for non-BRT events compared to BRT events. Results are qualitatively similar when using 500-meter or 2000-meter square grids as the base geography (Tables A.2 and A.3).

The key takeaway from Table 1 is that the three events affect the desirability of using

[26]The first type of event in fact has a small positive effect on log travel time. This is to some degree mechanical based on how we separate events 1 and 2. New direct routes are categorized under event 2 if the new direct route's travel time is strictly smaller than that of existing transfer routes, and event 1 otherwise.

[27]For each route, we use a bus allocation that is constant over time. Hence our variation is uniquely from new routes. Results are similar when we use actual (time-varying) bus allocation or actual number of trips that appear in the GPS data, although results with the latter are somewhat noisy (Figure A.3).

the bus system in different ways: adding a new and faster direct line reduces travel time *and* reduces waiting times, in addition to eliminating the need to transfer, whereas additional buses mainly reduces waiting times.[28] This variation will allow us to identify preference parameters in Section 4.

3.3 The Impact of New Route Events on Bus Ridership

Travel Times and Direct Connections: Impact of New Routes. The impacts of the three events on bus ridership, estimated using equation (1), are presented in columns 7–9 in Table 1 for BRT and non-BRT. Figure 2 shows event-study versions of each of these equations with monthly lags and leads from the date of the event.

Ridership is highly responsive to improvements in service quality. On average, adding an additional direct BRT route between o and d without a travel time improvement (Event 1) leads to an increase in ridership of 0.16 log points over the 10 months following the introduction of the new route (Table 1, column 7). A new direct route that also decreases travel time (Event 2) — in this case, by 0.29 log point (column 2) — leads to an increase in ridership of 0.27 log points.

The event-study version of equation (1), shown in Figure 2, Panel (a), left column, shows no pre-trends before the first and second events, and a discrete uptick in ridership at the time of the new route launch. The increase is notably higher for Event 2. The event-study graphs also show that riders take some period to adjust to the new route.

The non-BRT results paint a qualitatively similar picture of the impact of bus service improvements (right-side of Figure 2 and Table 1, Panel B). While for all events, the point estimates are larger than those for BRT, the implied *level* effects are much more similar. For Event 1, the BRT effect implies that a treated origin, destination pair has 19.1 ($= 111.3 \cdot (e^{0.158} - 1)$) more riders per week on average in the post period. This compares with an effect of 33.8 additional riders per week for non-BRT Event 1. For Event 2, the BRT and non-BRT level effects are 23.6 and 20.6 additional riders per week, respectively.

Wait Times and the "Mohring Effect." We next investigate the addition of direct buses between o to d caused by the launch of a new route that overlaps with existing direct routes (Event 3). Adding additional buses leads to a 0.09 log point increase in ridership (Table 1, column 9). Figure 2, Panel (b) shows that there is a large, discrete uptick in

[28] Both "first stage" variables are imperfect proxies for the attributes of interest. For example, the second outcome counts bus arrivals even for transfer options between o and d that have very long travel time, which in practice will be valued less by commuters. The minimum travel time between o and d shown in Table 1 might depend on a very infrequent bus line. The model in section 4 will lay out how commuters value these attributes jointly for all bus options in their choice set.

ridership exactly when the new route is introduced.

Given the first-stage results discussed above, we can interpret Event 3 as being almost entirely about the effect of wait times on ridership. Thus, combining the estimates from columns 6 and 9 of Table 1, we get an implied elasticity of ridership with respect to wait times of -0.29, i.e. a 10 percent decrease in wait times leads to a 2.9 percent increase in ridership.

These estimates speak to the so-called 'Mohring effect.' Mohring (1972) argued that if demand for public transit is responsive to wait times, then there is an externality from riding the bus — more bus ridership allows the bus operator to add more buses to the route, decreasing wait times for other riders. Our estimates show that, indeed, ridership is sensitive to bus frequency, suggesting that this effect operates in this case and that the optimal public transit subsidy is likely positive for this reason.

An extreme form of the Mohring effect is when the elasticity of ridership with respect to wait times is greater than 1 in absolute value over some range. In this case, the planner may have multiple local optimal levels of service frequency in ridership: a low-ridership, high-wait-time regime, and a high-ridership, low-wait-time regime. For BRT, we can reject an elasticity of -1 at conventional significance levels.

For non-BRT routes, however, we find a much larger implied elasticity of ridership with respect to wait times, of -1.08 ($= -0.450/0.417$, Table 1, panel B). This suggests that adding more non-BRT frequency on some non-BRT routes could increase ridership enough to maintain (or even increase) *average* ridership per bus.

Similar to the first two types of events, the *level* impacts of Event 3 are similar for BRT and non-BRT. Specifically, the BRT effect implies that a treated origin, destination pair has 20.3 ($= 210.4 \cdot (e^{0.092} - 1)$) additional riders per week on average in the post period, relative to an effect size of 14.78 for non-BRT.

Alternative Aggregation Levels. One possible concern is that our results are capturing displacement effects within the TransJakarta network. This would be the case if, for example, some commuters would switch from using a certain origin bus station located in a grid cell o' to one of the treated origin grid cells o after an event.

To examine this, we consider different aggregation levels, where we re-estimate all our results using both 500-meter squares (smaller) and 2000-meter squares (larger). If substitution was a central concern, we would find much larger effects when we use smaller units of aggregation than when we use larger units. Figure A.4 compares the magnitude of coefficients for all three events, separately for BRT and non-BRT, placing estimated $Post_{odt}$ coefficients and their 95% confidence intervals side by side.[29] There is no systematic pattern of decreasing coefficients and confidence intervals overlap significantly in all graphs. This implies that our treatment effects are not significantly contaminated by displacement effects. To address the possibility of spatial auto-correlation, we follow Bester et al. (2011) and cluster two-way at the level of larger origin and destination clusters, which yields very similar confidence intervals (Table A.4).

Jakarta opened its first metro (MRT) line in March 2019. As a robustness check, we re-run our main analysis dropping all observations after the line went into operation (Table A.5). Most point estimates are within the confidence interval of the main analysis, with the exception of the impacts of event 3 on non-BRT bus ridership, which loses its significance, yet still stays positive. Overall, these results imply that our results are not affected by the opening of the MRT (Hernandez-Cortes et al., 2023).

Heterogeneity Based on Poverty. One important question is whether preference parameters depend on local poverty levels. Table A.6 shows the impact of all six events on bus ridership when we include interactions for whether the origin grid cell has above-median poverty rate in the estimation sample; these regressions also control for interactions with log population. We use poverty and population data at the kelurahan (urban villages) from SMERU (2014) and the PODES 2010 village survey, respectively. Four out of six coefficients are negative, although only one statistically significantly. The negative sign is consistent with poorer areas being less responsive to service quality changes. The results do not show a consistent pattern based on poverty levels. We revisit this analysis when we estimate the demand model.

3.4 Impacts on Overall Trips

Having shown how the TransJakarta ridership responds to these events, we next turn to the Veraset smartphone-based trip data to examine whether *aggregate trip volume* — regardless of whether passengers take TransJakarta or not — is responsive to the TransJakarta network expansions.

We re-estimate equation (1) for the total number of trips from o to d at time t from Veraset (Table 2 and Figure 3). Our measure of aggregate trips is derived from individual-level smartphone traces, and it covers all types of trips, not only commuting trips between home and work. Since our Veraset data only begins in March 2018, for comparability we also re-estimate the effects on TransJakarta ridership for the same time period.

[29] The detailed 500 and 2000 meter results are shown in Tables A.2 and A.3 and Figures A.5 and A.6.

We do not see positive and significant impacts of the three types of events on aggregate travel volumes between pairs of 1km hexagon locations. For example, for BRT Event 1, we find a coefficient of -0.008 with a standard error of 0.051 (column 10). This allows us to reject at the 95% level a positive impact of approximately $+0.091$, compared to the precise 0.11 effect on bus ridership in the same sample (column 7). Thus, we can rule out moderately positive impacts on all smartphone location trips. Figure 3 shows no clear patterns before and after the events. We find qualitatively similar results for most event types, and they are robust to analyzing the data at the level of smaller, 500-meter square grids (Table A.7).[30]

These results suggest that our main results on bus ridership reflect an immediate and large mode substitution towards TransJakarta, without an increase in total trips between the affected origin-destination pairs. Of course, we cannot rule out that over a longer time period compared to the 10 months we focus on here, the pattern of trips will also change. However, based on these results, our model will focus on mode choice and hold (origin and) destination trip choices fixed.

New bus routes may affect traffic congestion, decreasing it as commuters switch to TransJakarta. This effect is likely small given TransJakarta's low initial mode share, the impact of the events we study, and an elasticity of travel time with respect to traffic volume of 0.3. See Appendix A.8.

Overall, the reduced form results suggest that riders are responsive to the three dimensions of service quality we consider: wait times, on-the-bus ride times, and direct connections. We do not find evidence that these changes affect the aggregate volume of trips.

The modal shift from other modes towards TransJakarta buses has implications for traffic congestion and for pollution. We estimate these effects in Appendix A.8, using additional data on modal shifts and congestion elasticities from Jakarta, and estimates from Gendron-Carrier et al. (2022), who study the impact of subway systems on air pollution in 42 large cities. We estimate that every 1 percentage point increase in the mode share of TransJakarta, equivalent to a 53% increase in TransJakarta ridership, lowers air pollution by 2.2% and lowers road travel times by 0.3%.

Since the events we study affect multiple dimensions of service quality, we next estimate a model of commuter demand, which allows us to jointly infer the underlying preference parameters that best match the responsiveness we observe in the data.

[30] As above, results are also robust to clustering standard errors at higher-level spatial units in the spirit of Bester et al. (2011), see Table A.8.

4 Model and Estimation

We now set up a model of demand for public transportation that describes how commuters choose bus routes within the TransJakarta network, and at a higher level, whether they use the TransJakarta network or an "outside option" that captures private transport modes.[31] We consider the problem of a commuter traveling between an origin grid cell and a destination grid cell who chooses between direct bus routes that link the origin and destination grid cells (we assume one bus station per grid cell), single-transfer bus connections, and an outside option. The exact choice set for bus options will depend on the state of the public transport network at a point in time.

How do commuters decide which bus routes to take? The core of our model is a new formulation for how commuters choose routes in a transit network. The model highlights the importance of wait times, which affect decisions in two key ways. First, overlapping routes between an origin and a destination effectively decrease wait times, because the traveler can take the first bus that arrives among these bus routes. (This is exactly the mechanism that we study empirically in Event 3.) Second, travelers sometimes forego short wait times for other route characteristics. For example, a traveler may decide to not take the first bus that arrives at the station if this option involves a transfer and instead wait longer for a direct (and shorter) route.

In the model, a traveler's utility from a given option is additive in wait times and several deterministic factors: travel time and the necessity to make a transfer. For each route and station, bus arrivals follow a Poisson arrival process. Commuters draw a vector of random wait times for the different routes in their choice set. This wait-time randomness induces idiosyncratic heterogeneity in route choices.

Our network routing model has four desirable properties. First, the model is invariant to aggregation of identical routes. Second, we confirm using our bus GPS data that the distribution of wait times is approximately exponential, as assumed in our model (Appendix Figure A.29). Third, our model has realistic predictions: overlapping routes lead to shorter wait times, and commuters sometimes choose to wait for a later bus arrival in order to use a better route. Fourth, we obtain tractable expressions for expected utility and choice probabilities.

We add to this baseline formulation the possibility that commuters are partly inattentive to certain options in their choice set. We assume that a route's arrival rate may be attenuated by an attention factor that depends on the travel time on the bus of options using that

[31] While is it possible to further embed these decisions into a model of destination choice, or a general equilibrium urban equilibrium model (Tsivanidis, 2024), given our null results on aggregate trips in this context, we hold these decisions fixed in the model.

route, relative to the minimum travel time on the bus in the choice set. The model with full attention is a special case.

In the upper nest, commuters make a logit choice between using the private mode or the TransJakarta network, where the benefit of the latter is given by the risk-neutral expected utility of choosing from among available TransJakarta options, where the expectation is over wait times and it includes the attenuation of true bus arrival rates due to partial inattention. The private option in the model includes private vehicles, such as motorcycles, taxis, motorcycle taxis, and ride-hail apps, as well as other types of shared transport, such as mini-buses outside the TransJakarta network. We assume that these modes are infinitely elastic to meet demand at a given cost, but their prices and attributes do not vary over time. We also assume that the TransJakarta network does not affect road traffic congestion.[32]

We estimate the model parameters leveraging the TransJakarta network expansion, focusing on commuters' values for travel time, wait time, and transfers, as well as attention parameters, which we allow to differ for BRT and non-BRT routes. We use a classical minimum distance approach whereby we search for the parameter vector that allows the model-predicted ridership to replicate the reduced form analysis documented above. In addition to the six moments (three events each for BRT and non-BRT), we also use a seventh moment, described below, to help pin down the attention parameters.

4.1 Bus Route Choice Model

A Random Utility Model with Waiting Times. The core part of our model is a flexible, static discrete choice over bus options, where idiosyncratic heterogeneity is given by exponentially distributed random wait times. Daganzo (1979) calls this the "negative exponential distribution model." We first describe the general structure and key properties of the model, before delving into the details that are specific to bus network routing.

Consider an agent making a static discrete choice among a finite set of options $k \in K$. Each option has a deterministic utility v_k and a random component $w_k \geq 0$, which measures wait time. The wait time for each option is governed by an independent Poisson process with arrival rate λ_k, which can differ by option. This means that the wait time w_k is drawn independently from an exponential distribution $\Pr(w_k > w) = \exp(-\lambda_k w)$. The agent observes the wait times for all options and picks the option with the highest combined utility $u_k = v_k - \alpha_{\text{wait}} w_k$, where α_{wait} governs the agent's preferences over wait times.

[32] BRT routes run on dedicated lanes separate from other road traffic, and new BRT routes during our study period were launched along existing BRT corridors, without changes in the supply of lanes for other road traffic. We assume that non-BRT routes did not affect road traffic congestion due to their low frequency and relatively low mode share.

We incorporate partial inattention by assuming that the agent notices each arrival from option k with independent probability p_k. In other words, with probability $1-p_k$ the agent fails to notice the first arrival for option k. This leads to an "effective" arrival rate for option k of $\widetilde{\lambda}_k = \phi(p_k)\lambda_k$ with $0 \leq \phi(p) \leq 1$ (see Appendix A.5).

This model has a convenient invariance property: it is unchanged when options that have the same indirect utility are combined into a single option with the sum of arrival rates.

Proposition 1. *Assume that the choice set contains two options $k = 1, 2$ with $v_1 = v_2$ and arrival rates λ_1 and λ_2. The model where options 1 and 2 are replaced by a single option 3 with $v_3 = v_1 = v_2$ and $\lambda_3 = \lambda_1 + \lambda_2$ is isomorphic to the original model in terms of choice probabilities and expected utility.*

Proof. This follows from the fact that the sum of two independent Poisson count processes with arrival rates λ_1 and λ_2 is a Poisson count process with arrival rate $\lambda_1 + \lambda_2$. □

This property implies that the demand model is unchanged if a route is split into two identical routes, with the total number of buses (which, as we will see, determine arrival rates) split between the two new routes. In other words, our model does not feature the 'red bus, blue bus' problem discussed by McFadden in the context of the multinomial logit and other random utility models. This will prove to be an attractive invariance property for the planner's optimization problem that we will later set up in section 5.

Choice probabilities and expected utility are as follows:

Proposition 2. *Assume that options are ranked $v_1 < v_2 < \cdots < v_N$. The probability π_k to choose option k is given by*

$$\lambda_k^{-1}\pi_k = \sum_{i=1}^{k} e^{-\alpha_{wait}^{-1} M_i} \frac{e^{v_i \alpha_{wait}^{-1} \Lambda_i} - e^{v_{i-1} \alpha_{wait}^{-1} \Lambda_i}}{\Lambda_i},$$

where $\Lambda_i = \sum_{\ell=i}^{N} \lambda_\ell$ and $M_i = \sum_{\ell=i}^{N} v_\ell \lambda_\ell$, and $v_0 = -\infty$ by convention. Expected utility is

$$\mathbb{E} \max_k u_k = v_N - \pi_N \frac{\alpha_{wait}}{\lambda_N},$$

where recall N is the index of the option with the highest deterministic utility.

The proof uses algebraic manipulations of the exponential distribution (Appendix A.6). The first part of this result ensures that computing choice probabilities is computationally tractable. The second part shows that expected utility has a particularly simple expression. Note that if the commuter only had the option N in their choice set, expected utility would be $v_N - \alpha_{\text{wait}}/\lambda_N$. In general, the influence of other options $k \neq N$ on expected utility is summarized by the probability π_N to choose the top option.

Bus Route Choice Model Setup. The travel demand model in the context of the TransJakarta network is as follows. Consider a commuter i traveling from a given origin grid cell o to a destination grid cell d. They have a choice set with a finite number of bus options $k \in M_{odt}$, where the choice set depends on the TransJakarta network at calendar date t, which in our application will range between 2016 and 2020. Each option has a utility level $u_k = v_k - \alpha_{\text{wait}} T_k^{\text{wait}}$, where v_k is the deterministic component that depends on characteristics such as travel time and transfer terms (in cases where the bus option k involves a transfer), and T_k^{wait} is the wait time at the origin determined by a Poisson arrival process that we describe in the next section.

The bus route choice set M_{odt} contains all direct and single-transfer bus connections between o and d available at t. The bus connections must start in the origin grid cell o, end in the destination grid cell d, and transfer in an intermediate TransJakarta station m. If a route r connects o and d directly, we do not allow connections that rely on r for the first or second leg.[33]

The utility for a *direct* public transit option k is:

$$u_k = \underbrace{-\alpha_{\text{time}} T_k^{\text{time}}}_{v_k} - \alpha_{\text{wait}} T_k^{\text{wait}} \tag{2}$$

where T_k^{time} and T_k^{wait} are travel time on the bus and wait time, respectively.

If passengers decide to take a *transfer* option $k \in M_{odt}$ that includes connecting in an intermediate station, we assume that they get the utility from the first leg k_1 of the route up to the intermediate station, and the expected utility from the second leg of the route, taking expectations over the best direct option k_2 connecting the intermediate station to the destination:

$$u_k = \underbrace{-\alpha_{\text{time}} T_{k_1}^{\text{time}} + \mathbb{E} \max_{k_2} \left[-\alpha_{\text{time}} T_{k_2}^{\text{time}} - \alpha_{\text{wait}} T_{k_2}^{\text{wait}} \right] + \mu_{\text{transfer}}}_{v_k} - \alpha_{\text{wait}} T_{k_1}^{\text{wait}}. \tag{3}$$

We assume that second-leg wait times are not known at the time when the commuter makes the initial decision, so the consumer needs to take expectations over the best route k_2 that they will take from the intermediate station to the destination. μ_{transfer} captures

[33] This excludes dominated connections such as going backward, disembarking, and then taking a route that goes back through the origin towards the destination.

the pure transfer penalty of taking a transfer *above and beyond* the travel time and the (expected) utility for the second leg. As 98.7% of observed bus trips are between origin-destination pairs connected directly or by a single transfer, we abstract from the possibility of agents taking routes involving two or more transfers.

Preference parameters are allowed to differ by BRT/non-BRT. For clarity, we do not track this distinction in notation now and discuss it when we set up the model's estimation.

We assume the following timing for the commuter's decision. Conditional on deciding to use the TransJakarta network, the commuter goes to the (unique) station in her origin grid o and observes wait times T_k^{wait} for all possible routes for options k in her choice set. For transfer routes, she observes the wait time for the first leg and forms expectations about the wait time for the second leg. She then chooses the bus option that maximizes utility.[34]

Poisson Bus Arrival Process and Exponential Wait Time Distributions. Buses on a route r arrive at the origin grid o according to a Poisson count process with arrival rate λ_r. The arrival rate is given by $\lambda_r = N_r/RTT_r$, the ratio between the number of buses allocated to that route, and the return travel time needed for one bus to make a full loop on route r.[35] We assume that the Poisson processes for different routes are independent.

Agents in the model are "non-planning" in that they cannot synchronize their trip departure time with a specific bus given an arrival schedule. This is realistic for TransJakarta, which does not publish a schedule and where very infrequent service is rare. Under this assumption, the wait time at o for the next bus from a given route r is exponentially distributed. Using bus GPS data, we show that exponentially distributed wait times are a good approximation for small and moderate wait times (Figure A.29, panels A and B). The exponential distribution in the model slightly over-estimates long wait times, suggesting that our estimate of α_{wait} captures steeper-than-linear costs for large wait times.

The mean and variance of wait time for a route follow a tight relationship, which is similar for BRT and non-BRT routes (Figure A.29, panel C). Without independent variation in these moments, we note that our estimate of α_{wait} will capture the combined preference for the wait time distribution.

Large Choice Sets and Partial Inattention. The model outlined so far assumes that commuters consider all possible bus route combinations. However, the TransJakarta bus network is sufficiently interconnected and complex that some choice sets are very large. While the median choice set has four bus options, a quarter of origin-destination pairs have

[34] During our study period, TransJakarta displayed actual arrival times at each BRT station (Figure A.28). Non-BRT stations did not have displays but users could look up wait times using an app.

[35] Bus allocation and bus speeds are mostly constant throughout the day (Appendix A.3.2).

choice sets with at least 14 options, and 10 percent of choice sets have at least 28 bus options.

Many choice set options — those with very long travel times — are ex-ante unattractive. For the choice set at the 75th percentile, the slowest option in the choice set (by total travel time on bus) is 2.8 times slower than the fastest option in the choice set. For 10 percent of choice sets, this ratio is at least 3.6.

We allow commuters in the model to be partially inattentive to options in their choice set that have a long bus travel time. This is a simple heuristic for reducing the size of the consideration set, bearing some similarity to models of rational inattention (Gabaix, 2019).[36]

To capture partial inattention, we consider for each route an effective arrival rate $\widetilde{\lambda}_{rodt} = \lambda_r \cdot \phi_{rodt}$, where $\phi_{rodt} \in [0, 1]$. This is equivalent to assuming that for any bus on route k, the commuter does not observe it with some probability, independently of other buses. Note that under this model, the commuter still draws a waiting time for each route, but this may be longer for some routes. In our parametrization, the ϕ_{rodt} terms will be close to zero or one, so essentially, the commuter will either fully ignore or pay full attention to the route.

We parameterize the attention factor as a function of travel time on the bus. Specifically, let $T_{rodt}^{\min}$ denote the shortest travel time on the bus among bus options that include route r, and $T_{odt}^{\min}$ the shortest time in the entire choice set between o and d at time t. We define

$$\phi_{rodt} = \phi\Big(\frac{T_{rodt}^{\min}}{T_{odt}^{\min}}\Big), \tag{4}$$

where $\phi(T_{\text{ratio}}) = \frac{\exp(\gamma(\eta - T_{\text{ratio}}))}{1+\exp(\gamma(\eta - T_{\text{ratio}}))}$ has a decreasing logistic shape in the travel time ratio T_{ratio}. We will estimate the parameter η, which is an attention cutoff. Commuters pay significantly less attention when the travel time ratio is above η, and this function approaches a step function around η as the shape parameter γ grows to infinity. We will calibrate γ and vary it in sensitivity analysis.

Partial inattention leads to important differences in how commuters respond to changes in the TransJakarta network. Consider the addition of a *faster* route on top of a set of existing bus options, as in Event 2. With full attention, the new route can significantly decrease the waiting time at the origin, which increases bus network ridership. This effect is particularly strong when time on the bus is not very costly (low α_{time}) because in that case, all routes are good substitutes for each other. With partial inattention, the commuter will pay less attention to pre-existing bus options after the new faster route is launched, because of the travel time difference. Partial inattention can rationalize low responsiveness to this type of change in the choice set.

[36] Larcom et al. (2017) use a short-term disruption due to a strike to show that some commuters in London are using suboptimal routes.

4.2 The Logit Choice Between Bus Network and Private Option

At the higher decision nest, a commuter traveling between o and d first decides between using the TransJakarta network or using an outside option, a catch-all for private modes (private motorcycle, for hire motorcycles, car, other private minibuses, etc.). The utilities for the two options for a trip i between o and d at time t are:

$$u_{it}^{\text{bus}} = \underbrace{\left(\mathbb{E} \max_{k \in M_{odt}} u_k\right)}_{v_{odt}^{\text{bus}}} + \epsilon_{it}^{\text{bus}} \tag{5}$$
$$u_{it}^{\text{private}} = \zeta_{od}^{\text{private}} + \epsilon_{it}^{\text{private}}$$

where the term v_{odt}^{bus} captures the expected utility of using the bus network over different realizations of wait time vectors, and includes the effect of inattention. The term $\zeta_{od}^{\text{private}}$ captures all time-invariant factors that make the private option more attractive for that specific origin-destination pair, which includes distance-varying monetary costs. The terms $\epsilon_{it}^{\text{bus}}$ and $\epsilon_{it}^{\text{private}}$ are Gumbel-distributed error terms with parameter σ_{od}, giving rise to logit probabilities. We scale $\sigma_{od} = \sigma \frac{\overline{D}}{D_{od}}$ where D_{od} is the straight line distance between grids o and d, and $\overline{D} = 8.5$km is the average distance, and we estimate σ. Using a constant $\sigma_{od} = \sigma$ leads to noisier logit choices for o, d pairs that are close because the travel time component of utility is generally increasing in straight line distance. Our normalization compensates for this mechanical effect. Estimation results are similar with a constant $\sigma_{od} = \sigma$ (Table A.9).

4.3 Model Estimation

Our estimation strategy is based on finding the vector of preference parameters that allows the model to match the reduced form results documented in section 3. We estimate

$$\theta = (\sigma, \alpha_{\text{wait}}^{\text{BRT}}, \mu_{\text{transfer}}^{\text{BRT}}, \eta^{\text{BRT}}, \alpha_{\text{wait}}^{\text{non-BRT}}, \mu_{\text{transfer}}^{\text{non-BRT}}, \eta^{\text{non-BRT}}).$$

We estimate separate BRT and non-BRT costs of waiting, transfer shifters, and attention cutoffs. Waiting in non-BRT stations and transfers between non-BRT and BRT bus lines may differ compared to the costs of using the BRT network. For example, the non-BRT bus stations are on the side of the road and differ significantly from the BRT stations, which are covered and where commuters pay at turnstiles to enter.

Because we lack price variation to pin down the level of utility, without loss of generality we normalize the cost of travel time on the bus $\alpha_{\text{time}} = 1$. Given that the physical buses used on non-BRT and BRT routes are similar, we use the same cost of time on the bus for both BRT and non-BRT. We set the attention shape parameter $\gamma = 30$, which means that attention drops quickly to close to zero above η^{BRT} and $\eta^{\text{non-BRT}}$. We estimate the origin-destination private option attractiveness terms $\zeta_{od}^{\text{private}}$ to match the average bus ridership between o and d over time.

Event Study Moments. We match the following seven moments. First, we match the six "Post" coefficients α^E on bus ridership for all three events, separately for BRT and non-BRT, from Table 1.

To help pin down the attention parameters, we add a seventh moment focused on trip duration. We estimate the impact of BRT Event 2 on the logarithm of the total trip duration from o to d.[37] We first measure BRT trip duration (inclusive of waiting time) in the data using the specific tap-in and tap-out times for the same smart card at the BRT entry and exit station. To do this, we restrict to stations that are tap-out compliant, defined as when tap-out transactions are at least 30% of all taps at that station. Thirty-six percent of all stations (92 stations) are tap-out compliant according to this definition.

Travel times between o and d go down sharply after the new direct and quicker route is launched between o and d (Figure 4 and Table 3). The magnitude is 0.06 log points, which is smaller than the impact on log minimum travel time (-0.29). This reflects a mix of two effects. Travel time on the bus indeed falls to the extent that more commuters substitute to the new direct route. However, realized wait times increase as commuters are more likely to wait for the new direct route. The model estimation will help disentangle the attention and value of travel time parameters that give this result.

Computing Event Study Moments in the Model. For a parameter vector θ, we compute model-predicted ridership for all origin-destination pairs and all time periods. Using the 1km hexagonal grids, we compute ridership for 33,880 o, d pairs (covering a maximum number of 350,493 bus route options), for all versions of the TransJakarta network between January 2016 and March 2020.

Given θ, for any o, d, t we compute choice probabilities

$$\pi_{odt}(\theta) = \frac{\exp(\sigma_{od} v_{odt}^{\text{bus}})}{\exp(\sigma_{od} v_{odt}^{\text{bus}}) + \exp(\sigma_{od} \zeta_{od}^{\text{private}})},$$

which depend on the bus route choice model through the expected utility term v_{odt}^{bus}. Combined with the (cross-sectional) smartphone commuting flow data V_{od}, this yields model-predicted ridership for each o, d, t. We also compute expected trip duration. The choice set

[37] We focus on Event 2 because it significantly changes travel time on the bus.

between o and d at t includes all TransJakarta routes already launched at t that stop in both o and d. Transfer options consist of a first-leg route r_1 that connects the o to an intermediate station m, such that there is at least one direct route between m and d.[38]

We use GPS data to measure travel times on the bus within the network. (Appendix A.3.2 shows that these are stable within days and across our entire study period.) In the model, we assume that each route has a bus allocation that is fixed over time, equal to the average bus allocation in the data. This determines the route's bus arrival rate.[39]

Given θ, we estimate $\zeta_{od}^{\text{private}}$ to match average bus ridership over time between o and d.[40]

We next run the reduced form analysis on model ridership, which we will then match to the empirical moments from Section 3. Given the high-dimensional fixed effects in equation (1), estimating PPML regressions at each iteration of the model is computationally prohibitive. To address this challenge, we proceed in three steps. First, we approximate the result of PPML estimation of the level of model-predicted ridership $R = R_{odt}^{\text{model}}(\theta)$ with a weighted least-squares regression on $\log(R)$.[41] Second, we obtain a significant computational speed improvement by pre-computing a vector x^E for each specification E, such that the coefficient of interest α^E from WLS is given by the dot product $\alpha^E = (x^E)' Y_{odt}$ for any outcome vector Y_{odt}.[42] Third, we use a slightly coarser set of fixed effects: origin-destination, origin-quarter-of-the-year, destination-quarter, and week fixed effects. We confirm that the reduced form results are very similar with this specification relative to (1).

Estimation and Inference. We use classical minimum distance estimation (CMD) to match the vector of seven numbers $\widehat{m} = (\widehat{\alpha}^{1B}, \widehat{\alpha}^{2B}, \widehat{\alpha}^{3B}, \widehat{\alpha}^{1N}, \widehat{\alpha}^{2N}, \widehat{\alpha}^{3N}, \widehat{\alpha}^{2B,\text{duration}})$, where $\{\widehat{\alpha}^{1B}, \widehat{\alpha}^{2B}, \widehat{\alpha}^{3B}, \widehat{\alpha}^{1N}, \widehat{\alpha}^{2N}, \widehat{\alpha}^{3N}\}$ are the empirical estimates for events 1, 2, and 3 for both BRT and non-BRT estimated in Section 3 above and $\widehat{\alpha}^{2B,\text{duration}}$ is the travel time moment

[38] Transfer options require a connection at the level of a real TransJakarta station m, not only a grid cell. We exclude transfer options that use a route r that directly connects o and d.

[39] We use a bus allocation that is constant over time to avoid reverse causality in model-predicted ridership.

[40] We compute the bus share as the ratio of bus ridership from TransJakarta data to total commuting estimated using smartphone location data. When this ratio is above 1, we use a Bayesian procedure to estimate the posterior over $(0, 1)$. We then find ζ_{od} to match the mean of the posterior ratio.

[41] In the presence of heterogeneous treatment effects, running PPML with outcome R and OLS on $\log(R)$ will weight treatment effects differently (Tyazhelnikov and Zhou, 2021). That paper shows that using R or a proxy of R as weights leads to a weighting of heterogeneous treatment effects that is similar to PPML. In our case, we use the measured TransJakarta ridership R_{odt}^{data} as weights, as a proxy for $R_{odt}^{\text{model}}(\theta)$. Unlike model ridership, which depends on θ, actual ridership is pre-determined before estimation, which is an advantage for the procedure discussed in the next paragraph.

[42] The vector x^E is a row of the WLS matrix $(X'WX)^{-1}X'W$ where X is the matrix of covariates, including all fixed effects, and W is the weighting matrix. We compute this matrix inversion only once before estimation, which relies on using pre-determined weights. To further reduce size, we apply the Frisch–Waugh–Lovell theorem to partial out the fixed effect with the largest number of categories.

discussed above.[43] We find the parameter vector θ that minimizes the objective function

$$\min_{\theta} \, (m(\theta) - \widehat{m})' \widehat{W} (m(\theta) - \widehat{m}), \tag{6}$$

where $m(\theta)$ is the vector of model moments, and $\widehat{W} = \widehat{\Omega}^{-1}$ is the optimal weighting matrix given by the inverse variance-covariance matrix of the moments $\widehat{m}$. The matrix $\widehat{\Omega}$ comes from estimating all reduced form event coefficients "stacked" in a seemingly unrelated regression framework. We cluster standard errors two-way by origin grid and by destination grid, which introduces dependence between the different regressions. To reduce the risk of finding a local minimizer θ of (6), we repeat the optimization routine starting from 20 randomly selected initial conditions and confirm that all converge to the same vector.

To obtain confidence intervals for $\widehat{\theta}$, we repeat estimation 500 times where we match the moment vector $\widehat{m}^k = \widehat{m} + \varepsilon^k, k = 1, \ldots, 500$, where $\varepsilon^k \sim \mathcal{N}(\mathbf{0}, \widehat{\Omega})$ are draws from a multivariate normal distribution centered at zero with covariance matrix $\widehat{\Omega}$ (i.i.d. over k). We use the resulting $\hat{\theta}^k$ estimates to construct confidence intervals.

Identification. Our classical minimum distance strategy has two main advantages. First, the reduced form moments that we established in Section 3 correspond to model comparative static exercises of exogenously varying wait time, travel time, and directedness. Second, these moments are also directly relevant for our counterfactuals, where we will be considering alternate networks that differ in terms of these attributes.

To shed light on how model parameters and the event study moments used in estimation are linked, we calculate the Jacobian matrix, which measures how, in the model, each moment's value changes when each of the parameters changes. Table A.10 reports the matrix of derivatives $dm_i(\theta)/d\theta_j$, each measuring how moment m_i responds to a marginal change in parameter θ_j, and Figure A.8 plots these relationships for larger parameter changes. For clarity, both exhibits focus on BRT moments and parameters. The value of travel time on the bus α_{time} significantly and positively affects Event 2, with muted impacts on the other events, which is expected given that travel time does not change significantly for Events 1 and 3. The transfer shifter μ_{transfer} affects Events 1 and 2 negatively, with an impact on Event 3 that is almost zero. Again, this is as expected, as treated (o, d) pairs in the third event are already directly connected.

All parameters are estimated jointly, yet given this approximately upper-triangular matrix format, we can describe intuitively how parameters are identified in the following recur-

[43] See Wooldridge (2010), section 14.6. CMD is closely related to indirect inference (Gourieroux et al., 1993; Smith, 2016).

sive manner. First, Event 3 depends almost only on the value of wait time, so its value pins down the value of wait time. Next, Event 1 depends on wait time and the transfer shifter, so its value pins then down the transfer shifter. Finally, with these two parameters in hand, Event 2 pins down the value of travel time on the bus.

Inattention is important for the model's ability to match the empirical moments. Figure A.9 shows that the model with full attention always overestimates the value of Event 2, even for very low values of travel time. While travel time on the bus affects both utility and inattention, the bottom row of Figure A.8 shows that the Event 2 total duration (wait time plus travel time) moment depends differently on these two forces. Intuitively, inattention restricts the choice set based on an attribute (travel time), which has different effects on choice compared to making that travel time more costly.

4.4 Estimation Results

Table 4 shows the estimated parameters and 95 percent confidence intervals. We normalize some parameters relative to travel or wait time to ease interpretation. We first estimate the model only for BRT parameters in column 1, using the three BRT events as moments, as well as Event 2 for log trip duration.

Commuters in Jakarta view time spent waiting for the bus as more costly compared to travel time on the bus — the estimated value of wait time $\alpha^{\text{BRT}}_{\text{wait}}$ is 2.1 times larger compared to the value of time on the bus $\alpha^{\text{BRT}}_{\text{time}}$, and we can reject at 95% significance level that the two parameters are equal.[44] The wait time parameter governs the degree to which people are sensitive to frequent service.

There are several reasons why commuters may dislike wait time more than time on the bus. In our model, α_{wait} implicitly captures the importance of the entire wait time distribution, including wait time uncertainty.[45] Waiting conditions — i.e., standing in line in non-air-conditioned BRT stations — may be less comfortable than riding the (air-conditioned) bus. Commuters may also mis-perceive wait time as being longer than it actually is.[46]

The pure transfer penalty $\mu^{\text{BRT}}_{\text{transfer}}$ is positive, and has a value of 4.8 minutes of wait time equivalent. The lower end of the 95% confidence interval is close to zero, and we cannot

[44]By comparison, existing studies of time valuation in public transport contexts focus on Western countries, and most are based on hypothetical choices. The few that use a revealed preferences approach typically use observational variation in attributes, unlike the quasi-random variation we use here. Papers in this literature typically find that the ratio of wait time to in-vehicle time is above one, but with significant dispersion (Wardman, 2004).

[45]At the route level, average wait times and wait time variability are strongly related, which makes it difficult to separate how commuters value these two moments (Figure A.29, panel (c)).

[46]For example, Fan et al. (2016) conducts a survey experiment in Minnesota and finds that people mis-recall wait times, reporting stated wait times that are larger than what they actually experienced.

reject relatively large positive values.

Recall that μ_{transfer} captures preferences for transfer routes *above and beyond* the cost of transfer routes in terms of additional wait time (e.g. when making the connection) and any additional time on the bus. Our results suggest that people are *less averse* to transfers than implied by these attributes. For example, this could happen if commuters have a lower distaste for wait time in the intermediate station, or if they systematically underestimate these second leg wait times.[47]

We find that commuters exhibit inattention to the longer bus options in their choice set. Specifically, a commuter with estimated preferences is less attentive to bus options that are more than 36% longer than the quickest option in the choice set. This cutoff parameter is precisely estimated between 25% and 47%.

In the second column, we jointly estimate BRT and non-BRT parameters, adding the moments for non-BRT events 1, 2, and 3. The general pattern of results is similar. The ratio of the cost of wait time to travel time is 3.4 for non-BRT, which is statistically significantly larger than the ratio for BRT. This means that waiting for buses in non-BRT stations is even costlier than in BRT stations. The higher wait cost is perhaps not surprising given that non-BRT stations are just stops by the side of the road, not enclosed stations as with the BRT. We cannot reject that the transfer shifter is zero. The attention parameter is $\eta^{\text{non-BRT}} = 1.43$, slightly higher but broadly similar compared to BRT.

We next explore the model fit. Both models in Table 4 are exactly identified and the model perfectly matches the data moments at the estimated parameters. However, we can learn more about model fit by comparing the time series patterns, which are not explicitly matched. Figure A.10 shows these results, replicating the event study graphs from Figure 2 using model-predicted ridership. In general, for all six moments, the model does a good job of replicating the lack of pre-trends and increase in ridership after the launch of a new route. The model also does a good job to fit aggregate ridership trends, including after an origin-destination pair first becomes connected by non-BRT (Figures A.11 and A.12), with the mention that ridership adjustments happen more gradually in the data.[48] Overall, these results help build confidence in the model predictions for ridership and commuter welfare of counterfactual routes, which we consider in the last section of the paper.

Preference parameters may differ by individual characteristics. Our demand model can

[47] As in the case of wait time, there is a lack of estimates of the pure transfer penalty based on quasi-random variation, as in this paper. Currie (2005) reviews existing studies, concentrated in Western countries and based mostly on hypothetical choice or, to a lower extent, observational variation, and reports a pure transfer penalty of 22 minutes of bus travel time on average over studies, ranging from 5 to 50 minutes.

[48] While these aggregate moments are not directly targeted, the terms $\zeta_{od}^{\text{private}}$ are estimated to match average bus ridership over the estimation sample time period.

accommodate this type of heterogeneity. To shed some light on heterogeneity, we propagate through the demand model the analysis from Table A.6, where we interacted our events with the poverty level of the origin grid of a trip.[49] In Table A.9, column 2, we re-run the entire reduced form and model estimation analysis only for origin grids with below-median poverty levels. In column 3 we do the same for origin grids with above-median poverty. The results show that $\sigma = 0.19$ $[0.07, 0.38]$ for low-poverty areas versus $\sigma = 0.03$ $[0.01, 0.07]$ for high-poverty areas. This means that commuters from low-poverty areas are more elastic to travel time on the bus, a finding consistent with higher income individuals having a higher value of travel time overall. Commuters from low-poverty areas have a lower ratio of wait time to travel time on the bus ($\alpha_{\text{wait}}/\alpha_{\text{time}}$), although overall are still more elastic to wait times ($\sigma\alpha_{\text{wait}}$), and they have a more positive transfer shifter.

Table A.9 reports two other robustness exercises. In column 4, we use a constant logit parameter $\sigma_{od} = \sigma$ instead of our baseline normalization $\sigma_{od} = \sigma \frac{\overline{D}}{D_{od}}$. In columns 5 and 6 we vary the attention parameter γ from equation (4). In all cases, the estimates are similar and we cannot reject the values in our benchmark specification.

5 Optimal Network Design

We now use the preference parameters estimated in the previous section to study the implications for the bus transit network design problem in Jakarta. We ask how the planner can optimally re-allocate bus routes and the bus allocation to routes, holding the total number of buses and the BRT infrastructure fixed, to improve commuter welfare given the estimated preference parameters. We focus on two key exercises. First, we compare the current network used by TransJakarta to the planner's solution. Second, we study how the shape of the network solution changes when structural preference parameters change.

This problem is discrete and high-dimensional. It does not appear to have an analytical solution and it resembles NP-hard problems such as the traveling salesperson problem. A key challenge is that it exhibits both substitution and complementarity forces, so we cannot use results that rely exclusively on one of these forces (Jia, 2008; Arkolakis and Eckert, 2021).

We introduce a general framework that allows us to characterize optimal allocations and perform comparative static exercises in our setting.[50] The key idea is to allow the planner to care both about model welfare, as well as idiosyncratic shocks that capture important factors

[49]Ideally, we would analyze the impact of our events on ridership of different groups. Unfortunately, our data on overall trips and bus ridership does not include any individual characteristics.

[50]The methods we develop may also be applied to quantitative models where optimal policies are high-dimensional, discrete, and analytically intractable, especially when "exact hat" methods make it possible to compare welfare between two counterfactual policies.

outside the model, which are defined for every possible network. We assume these shocks are unknown to the researcher. The key implication is that for two networks N, N' with similar model welfare $W(N) \approx W(N')$, small shocks may tip the planner into preferring one network over the other. This "smoothes" the planner's problem and leads to a *probability distribution* π over networks, where networks with higher model welfare have higher probabilities of being chosen by the social planner as the 'optimal' network. We assume that shocks are type-I generalized extreme value (Gumbel) distributed, giving a multinomial logit distribution.

We can then characterize certain properties of optimal networks by taking expectations over the planner's distribution. We derive analytic formulas for expected optimal properties and their local comparative static with respect to model parameters. These objects can be easily estimated with a sample of networks independently drawn from the planner's distribution.

To implement this in practice, we need to sample from the planner's multinomial logit distribution. We show that established algorithms, such as Metropolis-Hastings, parallel tempering, or a modified version of simulated annealing (SA), asymptotically sample from the planner's distribution. In our application, we run multiple independent SA simulations to obtain an i.i.d. sample of optimal networks.

How to choose the variance of the planner's idiosyncratic shocks, and how far is expected welfare under the planner's probability distribution from the global optimum? The appropriate variance of shocks is application-specific depending on the importance of the model versus non-model factors. We show how to compute robustness to higher variance of shocks at no additional computational cost. We also prove a sharp bound on expected welfare (computed excluding shocks) and design a statistical test for whether a sample of networks are drawn from the planner's distribution.

We find that, through the lens of the model, optimal networks achieve significantly higher ridership and commuter welfare and have lower average wait times. Increasing the cost of wait time leads to less expansive networks, while increasing the transfer penalty makes the network slightly more direct.

5.1 Bus Network Optimization Planner's Problem

The planner's measure of welfare is average commuter expected utility:

$$W(N;\theta) = \sum_{o,d} V_{od} \mathbb{E} u_{od}(N;\theta), \tag{7}$$

where V_{od} is the (fixed) number of commuters from o to d and $\mathbb{E}u_{od}(N, \theta)$ is expected utility based on the demand model, which depends on the network N and model parameters θ.

The planner decides the number of bus routes, where bus routes run, and how to allocate a fixed total number of buses to these routes. Formally, a network is a tuple $N = (K, (r_1, \ldots, r_K), (b_1, \ldots, b_K))$ where K is the number of routes, r_k is a route, and b_k is the number of buses on route r_k. We fix the total number of buses to 1,500. We now describe these objects in more detail.

We divide the greater Jakarta area into 418 2km $\times$ 2km square grid cells.[51] We construct a directed graph where grid cells are nodes, and each grid cell is connected to its eight adjacent grids, including diagonals. Bus stops are located at grid cell nodes. A bus route is a non-self-intersecting path on this graph with 418 nodes and 1,536 edges. Buses on a route stop at each node and travel in both directions.

Bus travel times, transfer opportunities, and wait times are determined as follows. We predict bus travel times for all possible edges using the existing bus travel time data, separately for BRT and non-BRT (see Appendix A.7.1). We then assign BRT travel times to edges that have BRT infrastructure at present, and non-BRT travel times to all other edges. We hold BRT infrastructure fixed in all counterfactuals. As is the case in the current network, routes may traverse both BRT and non-BRT edges.[52] Bus passengers can transfer at nodes that have BRT stations in the current network. Wait times depend on the bus allocation. The bus arrival rate on a route is the number of buses on that route divided by the return travel time (RTT). While it is possible to use the framework that we develop here to study optimal expansions of BRT infrastructure and additional buses, we leave those exercises for future work because we lack data on the relevant costs.

We use the demand model that we estimate in section 4. A commuter in the model going from o to d chooses among available transportation options given the proposed network N and estimated preferences $\widehat{\theta}$, including inattention parameters.[53] Commuting flows V_{od} are fixed, computed using the smartphone location data (see section 2.2).[54] The counterfactual networks we consider here comprise intensive and extensive margin changes that resemble the historical changes that our model is estimated to match, which increases our confidence

[51] We use the coarser resolution for computational reasons and focus on squares to improve the visual legibility of straight bus lines.

[52] We assume that bus fuel costs only depend on the how long buses are running each day and hence do not change in counterfactuals.

[53] We use estimated preferences (BRT vs non-BRT) based on whether the origin grid is BRT or non-BRT in the current network.

[54] We re-estimate $\zeta_{od} = \zeta(D_{od})$ as a non-parametric function of distance D_{od} such that the model evaluated at the current TransJakarta network matches the total ridership for each distance bin. This gives us an imputed private option attractiveness for all (o, d) pairs. It also accounts for the difference in grid size relative to demand estimation.

in the model's projections.

To account for potentially higher ridership levels in counterfactual networks, we introduce bus capacity constraints. We assume that when a bus has ridership above its capacity, passengers perceive travel time as flowing more slowly. This time-dilation effect grows steeply as ridership per bus exceeds bus capacity.[55] As ridership in excess of capacity is endogenous, we define a *ridership equilibrium* where commuter optimal choices given crowding are consistent with the assumed crowding levels. Appendix A.7.2 describes the model and our computational approach.

The space we are optimizing over is extremely large. Even the number of unique paths is exponential in the number of grid cells, and a network consists of any combination of routes, while the allocation of buses to lines adds even more combinatorial dimensions to the optimisation problem. As we derive in Appendix section A.7.7, the total number of feasible networks is on the order of $e^{26{,}000}$, making it infeasible to derive the global welfare-maximizing network through exhaustive search over all possible networks. Instead, we set up below a general framework for characterizing optimal policies in settings with high-dimensional policy spaces such as ours.

5.2 Characterizing Optimal Policies

The Social Planner's Problem with Idiosyncratic Factors. Consider a social planner who chooses from a finite set of policies $N \in \mathcal{N}$ (in our setting, networks with at most 1,500 buses). In a typical application, the set $\mathcal{N}$ is high-dimensional and extremely large. The planner chooses the network N that solves

$$\max_N W(N;\theta) + \epsilon_N,$$

where $W(N;\theta)$ is welfare according to a known, fully specified model, and θ is a vector of parameters.[56] ϵ_N is an idiosyncratic shock for network N, capturing factors not in the model that gives rise to $W(N;\theta)$, such as network construction cost shocks or preferences for specific networks. Assume that the ϵ_N are i.i.d. Gumbel with parameter β.

The probability that network N maximizes social welfare is

$$\pi(N;\theta) = \frac{\exp(\beta W(N;\theta))}{\sum_{N'\in\mathcal{N}} \exp(\beta W(N';\theta))}. \tag{8}$$

[55] We do not model dynamic imbalance in crowding and the decision to wait for the next bus.

[56] In our application, we will use the point estimates $\widehat{\theta}$ from demand estimation. In principle, the objective function can also be the expected value of welfare $\mathbb{E}_\theta W(N,\theta)$, taking into account estimated parameter uncertainty.

How far is model welfare under π from the global maximum welfare under the model? This depends both on β, which measures the inverse standard deviation of the shocks, and the number of possible networks in $\mathcal{N}$. We prove the following bound.

Proposition 3. *Denote $W^{\max} \equiv \max_{N\in\mathcal{N}} W(N;\theta)$ the global maximum welfare under the model. Then, expected welfare under π, computed excluding shocks, has the following bounds:*

$$W^{\max} > \mathbb{E}_\pi W(N,\theta) \equiv \sum_{N\in\mathcal{N}} \pi(N;\theta)W(N;\theta) > W^{\max} - \frac{\log(|\mathcal{N}|)}{\beta}. \tag{9}$$

As $\beta \to \infty$, expected welfare under π converges to the global maximum welfare. As the number of possible networks $|\mathcal{N}|$ grows, the bound becomes wider. The proof of a slightly stronger and sharp bound is in Appendix A.7.3.

A Statistical Test of Mixing. We use the above bound to construct a statistical test for whether a sample of networks are drawn independently from π.

Consider a sample of networks $N_1,\ldots,N_K$ purportedly drawn independently from π. Proposition 3 puts a bound on the distance between mean and maximum welfare in the sample, up to sampling uncertainty. If we see more dispersion in welfare levels, we can statistically reject that the networks are iid sampled from π.[57]

Denote $\widehat{W}^{\max}$ and $\widehat{W}^{\text{mean}}$ the sample maximum and average welfare. Consider an upper bound for the number of networks $\overline{n} \geq |\mathcal{N}|$. Define

$$\widehat{T} = \widehat{W}^{\max} - \widehat{W}^{\text{mean}} - \frac{\log(\overline{n})}{\beta}. \tag{10}$$

Proposition 3 implies $\mathbb{E}\widehat{T} < 0$. Asymptotically, $\widehat{T}$ is bounded from above by a normal distribution $N\left(0, \frac{\sigma_\pi}{\sqrt{K}}\right)$ where σ_π is the standard deviation of welfare under π. To test whether $N_1,\ldots,N_K$ are drawn independently from π, we use the one-sided t-test based on $\frac{\widehat{T}}{\widehat{\sigma}_\pi/\sqrt{K-1}} < 0$ where $\widehat{\sigma}_\pi$ is the estimated standard deviation.

Optimal Policies: Properties and Local Comparative Statics. We next show how to characterize optimal networks by taking expectations over π, and how to estimate these objects using a sample of networks drawn from π. We then introduce methods that allow us to asymptotically sample from the π distribution.

[57] This test may have low statistical power for two reasons. First, Proposition 3 is a bound that may be wide for a particular π. Second, welfare levels may be similar and we may fail to reject mixing if sampling is "stuck" in a specific region of the state space $\mathcal{N}$, such as in a particular basin of attraction. This means that rejection is a strong indication that β is too high or that sampling algorithm was not run long enough to approach the stationary distribution.

Consider a property defined by a function $f(N, \theta)$. For example, this could measure the number of stations for a bus network N, in which case f only depends on N and not on θ. Alternatively, it could measure model-predicted ridership, in which case it depends on both N and θ.

We define the *expected optimal property* as

$$f^*(\theta) = \sum_{N \in \mathcal{N}} \pi(N; \theta) f(N, \theta).$$

This is a weighted average of the property f over all possible networks N, with weights given by the probability that N is optimal. Given a sample of networks $N_1, \ldots, N_K$ drawn independently from π, we can estimate $f^*(\theta)$ using the sample counterpart $\widehat{f^*}(\theta) \equiv \sum_{k=1}^{K} f(N_k, \theta)$.

We can compute comparative statics in θ by estimating $\widehat{f^*}(\theta')$ for any other value $\theta' \neq \theta$. To compute *local* comparative statics of the expected optimal property, assume that $W(N; \theta)$ is differentiable in θ for all N. Appendix A.7.4 derives and interprets an analytic expression for $D_\theta f^*(\theta)$, the gradient of the expected optimal property with respect to θ. Once again, we can estimate this using a sample of networks from the π distribution.

5.3 The Simulated Annealing Algorithm

To operationalize our framework, we need a way to randomly sample a network with probability weights given by the π distribution. In principle, we can use any algorithm designed to asymptotically sample from distributions like the multinomial logit.

The algorithm we use builds on the classic Metropolis-Hastings (MH) algorithm (Metropolis et al., 1953; Hastings, 1970), which constructs a stationary, reversible Markov chain on the space of networks $\mathcal{N}$. MH begins at an arbitrary network N_1. Given network N_k at step k, it draws a candidate network N' from a probabilistic proposal distribution $\Psi(N' \mid N_k)$. The distributions $\Psi(\cdot \mid \cdot)$ are user-provided and application-specific. MH accepts the network N' with probability:

$$\Pr(N_{k+1} = N' \mid N_k) = \min\left(1, \frac{\exp\left(\beta W(N')\right) \Psi(N_k \mid N')}{\exp\left(\beta W(N_k)\right) \Psi(N' \mid N_k)}\right), \tag{11}$$

and keeps $N_{k+1} = N_k$ otherwise. That is, the candidate network is always accepted if it has higher welfare (with an adjustment for proposal probabilities), and otherwise it is accepted with a probability that is increasing in $W(N')$ and decreasing in β. The transition probability above is constructed to ensure that under mild conditions the stationary distribution of the chain is exactly π. This implies that as $k \to \infty$, the network N_k converges in distribution to

π. (Appendix A.7.5 discusses this section more formally.)

The key challenge is that MH may need many steps to get close to π, known as "slow mixing" (Levin and Peres, 2017). The main concern is that for some welfare functions W, N_k may stay concentrated for a long time in a subset of the state space $\mathcal{S} \subseteq \mathcal{N}$ that is separated from the rest of the state space $\mathcal{N} \setminus \mathcal{S}$ by a "well" of networks with low welfare, making transitions out of $\mathcal{S}$ and into $\mathcal{N} \setminus \mathcal{S}$ unlikely. This problem is worse for large β, because that makes accepting lower welfare networks less likely. Conversely, fast mixing is more likely for low $\beta > 0$. In the limit $\beta = 0$, the algorithm becomes a random walk on the space of networks, which in some cases mixes fast.[58]

The simulated annealing (SA) algorithm is designed based on the idea that mixing is "easier" for low values of β (Nikolaev and Jacobson, 2010; Rothlauf, 2011). The algorithm is similar to MH except that it runs for a fixed number of steps K and at step k the term β in (11) is replaced by β_k. The term β_k is called "inverse temperature" in the context of SA, and it increases exponentially from some small β_1 to $\beta_K = \beta$.

In the initial phase, the low inverse temperature implies that new candidate networks are accepted with high probability. This makes the chain approximately a random walk in the space of networks. This helps the algorithm explore widely the space of networks. In later phases, β_k is close to β. It is useful to think of SA as a sequence of M MH algorithms that run for K/M steps each, where the m'th algorithm has inverse temperature β_m that increases with m. As the number of steps K increases, the final state N_K converges in distribution to the distribution π, as for MH. Intermediate states converge in distribution to a version of π with correspondingly lower β, which allows us to do robustness for lower values of β at zero additional computational cost.[59] Overall, SA is designed to provide fast mixing to the target distribution π.[60]

The proposal function Ψ is important for the practical success of the SA algorithm. We use small changes such as adding or deleting one stop at the end of a route, local route re-routing, and minor changes in bus allocation between routes, and large changes such as deleting an entire bus route, adding an entirely new bus route, or re-allocating a large share

[58] For intuition, consider a problem where a planner decides where to make n lumpy investments. There are 2^n policies $a = (a_1, \ldots, a_n) \in \mathcal{A}_n = \{0, 1\}^n$. Consider a "lazy" random walk, which with 50% probability picks a location uniformly randomly and then flips its investment, and with 50% probability makes no change. Intuitively, this process mixes fast because it diffuses symmetrically in all directions and it can reach any policy within n steps. In fact, its mixing time is $\mathcal{O}(n \log(n))$, despite the exponential size of the state space (Levin and Peres, 2017, see section 5.3.1).

[59] Other algorithms, such as parallel tempering, also have the feature that they also sample from distributions π for lower values of β (Woodard et al., 2009).

[60] The design of MCMC algorithms with fast mixing is an active area of research in statistics and applied fields, see for example Syed et al. (2022). Proofs of fast mixing only exist for certain classes of problems (Woodard et al., 2009).

of buses from one route to other randomly chosen routes. See Appendix A.7.6 for details.

Parameters and Implementation. We use the commuter preference parameters estimated in section 4.3. We use a bus capacity of 84 people per bus based on TransJakarta data on its bus fleet in early 2020. The initial network is generated randomly with 10 routes, each with 5 stops and 10 buses.[61]

We obtain independent asymptotic draws from π using 200 independent SA runs. Each time, we run the SA algorithm for $K = 251,089$ steps.[62] Shorter runs with $K = 46,041$ steps produce overall similar networks with slightly lower ridership and welfare (Table A.12).

We denominate welfare such that the equivalent variation of one unit increase in W is one minute of travel time saved per trip for all bus users in the current TJ network.

To determine the value of β – the inverse standard deviation of the planner's shocks – we conduct the test of statistical mixing from equation (10). We let the algorithm run until $\beta = 4 \times 10^5$ and compute[63]

$$\widehat{T} = \underbrace{\widehat{W}^{\max} - \widehat{W}^{\text{mean}}}_{\text{1.2 minutes}} - \underbrace{\frac{\log(\overline{n})}{\beta}}_{\text{0.07 minutes}} .$$

In other words, among the 200 final networks, each obtained from an independent SA run with 251,089 steps that stops at $\beta = 4 \times 10^5$, the difference between the largest and the average welfare is 1.2 minutes (Figure A.14). This should be lower than the theoretical upper bound, which we bound from above with 0.066 (Appendix A.7.7). In fact, the empirical welfare "spread" is higher, and we can reject that $\widehat{T} < 0$ because the standard error on the first term is around 0.03. This means that the 200 networks are unlikely to be random samples from the planner's distribution with $\beta = 4 \times 10^5$. In other words, if we want samples for $\beta = 4 \times 10^5$, we need longer-running SA. We next compute $\widehat{T}$ at intermediate values of β and identify the largest β for which we do not reject $\widehat{T} < 0$ at 5% level. We obtain $\beta \approx 2.3 \times 10^4$, which is our preferred value of β.[64]

[61] Figure A.15 shows that initial conditions do not matter. Networks sampled in the first several thousand steps vary significantly in terms of number of lines, total mileage, average travel time, and the share of trips with a feasible connection. Only after around half of the run time does the algorithm begin to converge.

[62] The acceptance probability (11) depends on the ratio of proposal kernels Ψ. Calculating this ratio is computationally intensive, so we approximate it by a ratio of 1 when running the simulated annealing algorithm. For most types of network change proposals, this approximation is likely valid. A notable exception is when adding new routes, when this approximation will tend to favor less expansive networks. This effect works against our finding of slightly more expansive networks, described below. See Appendix A.7.6 for details.

[63] This value of β corresponds to $\beta = 10^{10}$ when welfare units are based directly on equation (7) and the estimated demand model parameters. With this scaling, we initialize the algorithm with $\beta = 1$.

[64] In practice, networks achieved through either of these two parameterizations or a full simulated annealing

Figure A.14 shows the evolution of the 200 SA runs in terms of welfare and ridership and Figure A.16 plots the entire distribution of other network statistics for the final network samples.

5.4 Results: Characteristics of Optimal Networks

The TransJakarta bus network at the end of our study period is relatively dense in the urban core of the city, with most of the 110 bus lines crossing through the downtown areas (Figure 5a). While the core of the city seems well connected by very frequent bus services, Jakarta's periphery has few buses passing through, and most locations of the city are not connected at all (Figure 5b).

Figure 5c shows one example of a network asymptotically drawn from the planner's distribution. (Figure A.17 shows several other examples.) This network covers slightly more locations compared to the actual TransJakarta network, extending more to the areas just beyond the central DKI Jakarta area. The network has noticeably fewer bus lines. The edge frequency map shows that the sampled optimal network has more uniform frequencies, slightly lower in the city center than the current network, but higher in the outlying areas (Figure 5d). While agents in our model value wait time highly, the central area frequency reductions are not very costly because the initial frequency was very high, often more than a bus every minute.

We next look at welfare, ridership and characteristics of optimal networks. Table 5 reports statistics for the 2016 version of the TJ network in column 1, the current TJ network in column 2 and the sample average over optimal networks in column 3.

Optimal networks lead to a significant increase in welfare compared to the current network (panel A). The average expected utility over all the trips in Jakarta is higher by the equivalent of 12.6 seconds of travel time on the bus, per trip. This may not sound enormous, but recall that this is over all trips, not just bus trips – so given that 1.88% of all trips use the bus with the current network, the equivalent variation improvement is a reduction of 11.2 minutes of bus travel time per initial bus trip. Through the lens of the model, the welfare improvement from optimal networks is almost two and a half times that between 2016 and 2020.[65]

Ridership increases from 1.88% of all trips with the current network to 3.04% on average with optimal networks. As a consequence, the number of passengers per bus roughly doubles. Bus trips are longer on average in optimal networks, and optimal networks have a slightly higher share of direct bus trips.

run are very similar, so we focus on results obtained with $\beta = 4 \times 10^5$ while showing robustness to other values in Table A.13.

[65]Note that the bus fleet increased from 2016 to 2020, while in our counterfactuals this is fixed.

While not directly captured in our demand model, a 1pp shift from other modes toward TransJakarta bus ridership is also likely to decrease road traffic congestion by 0.3% and lower air pollution, as measured by PM2.5, by 2.2% (Appendix A.8).

Optimal networks are slightly more expansive than the current TransJakarta network on average (panel B). Optimal networks connect 45% of all locations in the study area compared to 42% for the current network. This effect is concentrated in the area just outside DKI Jakarta, where network coverage jumps from 27% to 40%. Overall, the share of location pairs connected by a direct or transfer bus connection increases from 12% to 18%. The share of Jakarta commuter who have network access to their destination increases slightly.[66] Optimal networks have fewer bus lines, 21.3 on average instead of 110 in the current network, and the networks cover 466 km on average, down from 544 km now.

Optimal networks have significantly shorter wait times than the current network. We compute the mean headway – the time between consecutive buses – both for adjacent origin-destination pairs, as in Figures 5b and 5d, and for all origin-destination pairs. We compute headways pooling over all routes that travel direct between these locations. Headways in optimal networks go down by 65%-75% compared to the current network for these two measures.

Figure A.18 shows the distribution of headway for the current network and over all sampled optimal networks. Optimal networks have a much more concentrated distribution with lower mean, including when we weight by the number of trips. How can this be achieved while holding the number of buses constant? The key intuition is that expected headway on a route is inversely proportional to the number of buses on that route, a convex function. Optimal networks reallocate buses from the city center area, where frequencies were very high (often below one minute), towards routes that reach the outer areas where the marginal bus has a large impact on decreasing headways. (We develop this intuition in a simple model with two locations in Appendix A.7.8.) Agents in our model value wait time highly, and the central area frequency reductions are not very costly because the initial frequency was very high.

A possible reason for high bus frequency concentrated in the city center is to manage bus occupancy. Our model already includes penalties when bus occupancy exceeds the average bus capacity of 84 people per bus (Appendix A.7.2). These capacities rarely bind in our counterfactuals (Figure A.19d). In another counterfactual we lower all bus capacities by a factor of 1.85 to account for the fact that within the day, peak ridership is 85% higher than

[66]In our context, efficiency gains do not come with a distributional cost, as coverage also grows for poor areas. 44% of the locations with above-median poverty rate have access to the current TransJakarta network, which grows to 45% on average for optimal networks. Profitability also improves as fare revenues go up due to higher ridership, holding the vehicle fleet the same.

the daily average. Our key results continue to hold, e.g. headways are much lower under these assumptions (Figures A.19f and A.20).

Alternative policy objectives. These results are obtained through the lens of our model and hence depend on the assumptions we make. First, we rely on the demand model being an accurate depiction of commuter behavior and utility. Second, we use a planner's objective given by the averages expected utility for all commuters in the study area, as in equation (7).[67]

Our baseline model does not have systematic commuter preference heterogeneity. In Figure A.21, we use the demand model where preferences are allowed to differ based on below-/above-median poverty (Table A.9). Optimal networks are much more expansive than the current network, and continue to provide much lower headways than the current network. Ridership increases by a similar amount as in our baseline runs.

Our model abstracts from several forces that may affect the shape of optimal networks, such as longer term adjustments in residential, employment, and agglomeration spillovers. On the cost side, we do not model operational costs such as bus depot locations (Marra and Oswald, 2023). It is also possible that Jakarta planners may seek to maximize a different objective than we do here, or may place different weights on commuters based on residence or poverty level. We next consider several additional counterfactuals with alternative weights to investigate this. Across these counterfactuals, our findings that optimal networks achieve higher welfare and ridership and lower wait times are robust, while the expansiveness finding is sensitive to assumptions.

First, optimal networks that maximize TransJakarta ridership instead of commuter welfare are described in Figure A.22. They are more compact than the baseline networks, and even than the current TJ network. Figure A.23 next shows results where we assume that the planner puts three times as much weight on areas with below-median poverty relative to areas with above-median poverty. A third potential change to our assumptions reflects the fact that TransJakarta is controlled and subsidized by the government of Jakarta. To accommodate that, in Figure A.24 we impose zero welfare weights for residents outside DKI Jakarta (the light blue area in the maps in Figure 5). Optimal networks under this criterion continue to outperform the current network in terms of ridership and commuter welfare. As may be expected, they are less expansive than the current network. Optimal networks continue to have much lower bus wait times.

We also report results where we propagate the uncertainty from demand estimation

[67] An alternate strategy is to make inferences about a planner's objective function based on observed policy (Goldberg and Maggi, 1999; Adão et al., 2023; Fajgelbaum et al., 2023).

through the entire procedure described here (Appendix Table A.14 and Figure A.25). This corresponds to a setting where the planner knows the true parameters, but researchers have uncertainty. We achieve this by running 100 independent runs of our algorithm, each based on an independent draw from the joint bootstrapped distribution of structural preference parameters. Optimal networks again have lower headways (i.e. lower wait times), higher ridership, and a lower number of bus lines compared to the current networks. Other characteristics are more variable.

Comparative Statics. We next study how the optimal network shape changes when commuter preferences change (Table 6). This type of exercise helps us assess the sensitivity of our results to the parameters to estimate, and reveals how these micro preference parameters affect optimal network design. We focus on changes in the three key preference parameters that we estimated: the value of wait time α_{wait}, the value of travel time on the bus α_{time}, and the transfer shifter μ_{transfer}. We study the impact of large changes in parameters by sampling 50 optimal networks each for these different parameters using the simulated annealing algorithm.[68] We study the impact of these three parameter changes on three classes of bus network shape measures.

First, we find that network coverage shrinks as wait time costs increase (Table 6, panel A). For example, if commuters dislike wait time twice as much as we estimate, the share of location pairs connected by optimal networks goes from 18% with our baseline parameters to 9.8%.

Second, we consider the speed of bus connections (panel B). For each origin-destination pair with a direct bus connection, we consider the quickest bus connection in the network, and divide its duration by the duration of the quickest possible bus route between the two locations, over all possible paths in the geographic environment. In the current TransJakarta network, direct connections are on average 33% slower than the quickest possible bus connection, compared to 24% for the baseline optimal networks. This goes down further, to 16%, when using $2 \times \alpha_{\text{time}}$. This counterfactual also affects the coverage measures in panel A, that is, they lead to more compact networks, although this effect is smaller than that of doubling α_{wait}.

Third, we consider to what extent the network provides direct connections versus offering connections that require a transfer (panel C). In the current TransJakarta network, 21% of all location pairs that are connected by bus also have a direct connection. This number is lower for optimal networks (15%), and it goes up to 18% when we assume that the transfer

[68] In Table A.15 we report the *local* comparative statics from (19) around the baseline estimated parameters, which is computationally faster because we do not need to run the simulated annealing algorithm again. For most entries, we find the same direction for comparative statics.

penalty is higher by the equivalent of 15 minutes of wait time. This counterfactual also increases the (inverse) measure of network speed in panel B, likely because of additional location pairs that are connected by winding direct connections.

6 Conclusion

The trend toward centralized urban planning of public transport networks is likely to continue in the coming decades. For example, there are half a dozen BRT systems in African cities, with ten others are under planning or construction (The World Bank, 2022). As the number of large cities increases, this number will grow, impacting traffic congestion and pollution, commuting and work patterns, and more generally the shape of how cities grow.

In this paper, we studied the design of citywide public transport networks. Launching a public transport route affects the attractiveness of using the network throughout the city. Its impact depends on how the new route increases access, and whether it leads to faster, more frequent, or more direct connections, and how potential riders value those attributes. We used the large expansion of the TransJakarta public bus system between 2016-2020 to separate how specific types of network changes affected bus ridership, and then estimated separate commuter preference parameters for these service quality dimensions by matching the reduced form impacts on ridership.

We find that commuters in Jakarta value bus wait time 2-3 times more than time spent on the bus, transfers in BRT stations do not carry a utility penalty above and beyond the additional wait time and time on the bus, and commuters pay little attention to slow options in their bus network choice set. Characterizing the optimal networks in terms of coverage, speed, and directedness metrics, we found that the optimal bus networks in Jakarta would be less concentrated — even holding the resources of the bus company fixed — offering lower average wait times, and boosting overall usage. An optimally designed network that is based on these parameters thus has the potential to dramatically expand ridership and consumer welfare compared to the networks in place today.

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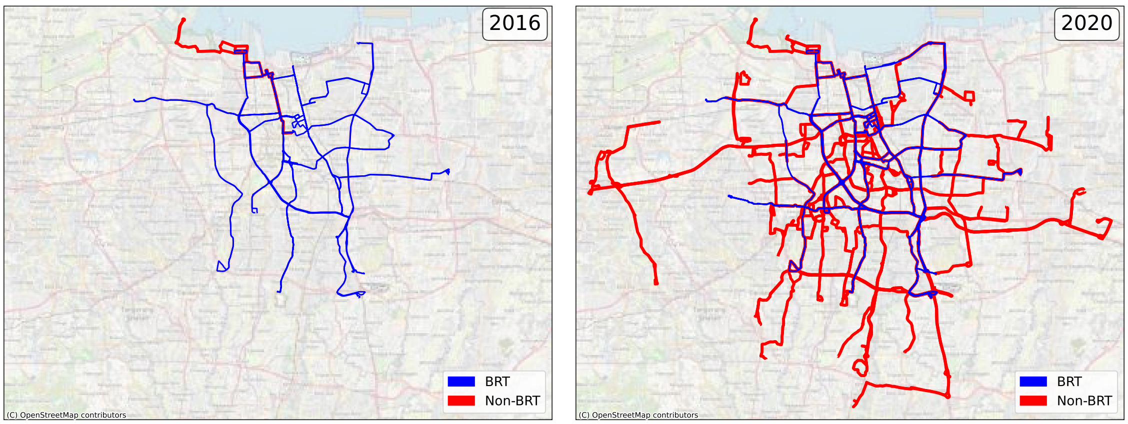
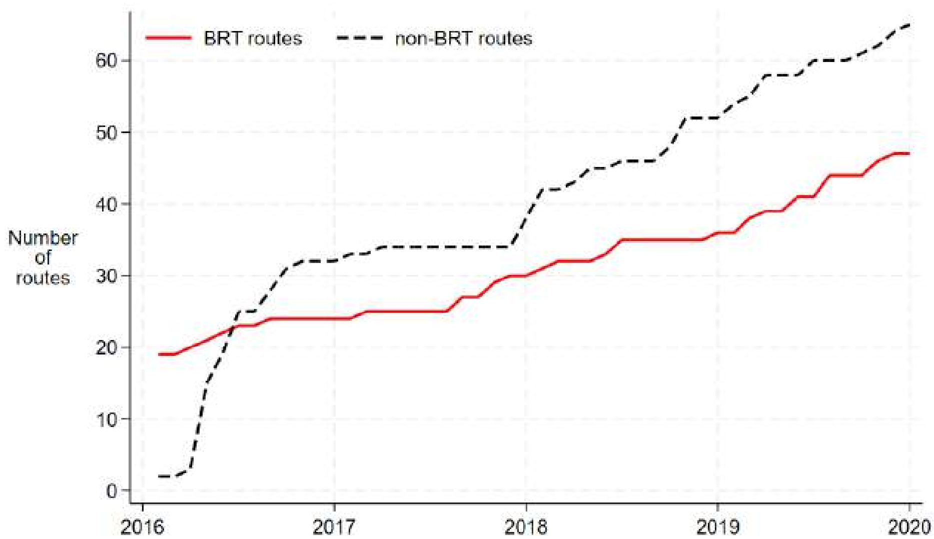
Figure 1: TransJakarta Network Expansion Since 2016

(a) Expansion of TransJakarta Route Network



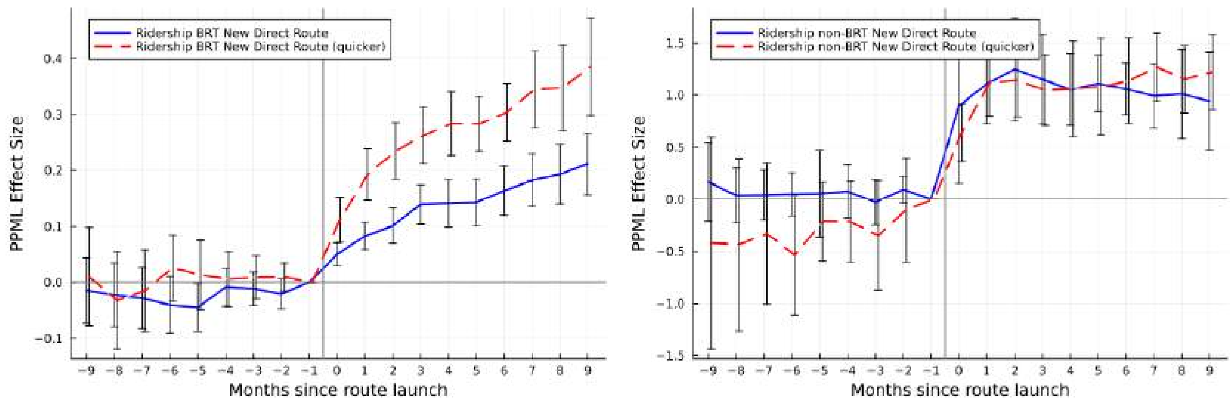
(b) Number of BRT and Non-BRT Routes Over Time



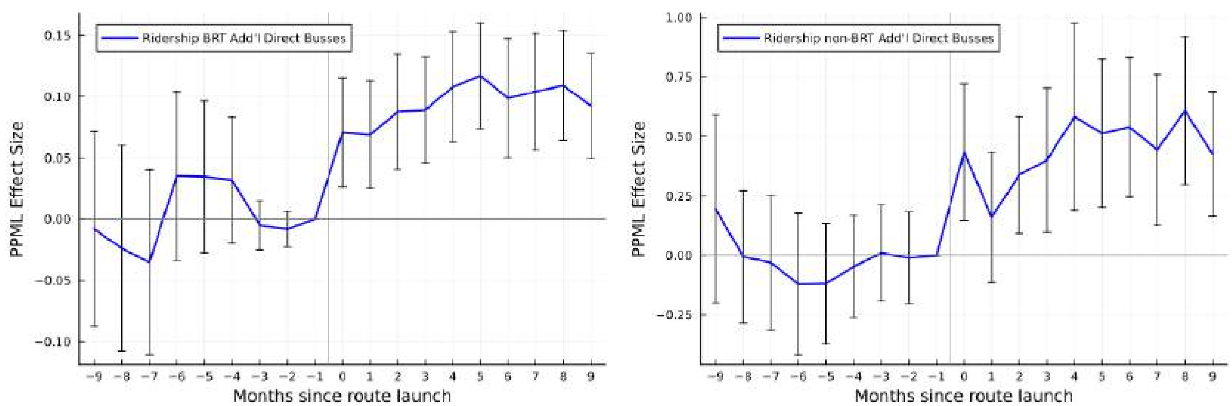
Notes: The top panel shows two snapshots of the TransJakarta network in 2016 and 2020. Blue lines indicate BRT corridors. Red lines indicate non-BRT routes. The bottom panel shows the number of active routes over time.

Figure 2: **Bus Ridership** Impacts of BRT and Non-BRT Network Expansions

(a) Event Types 1 and 2: New Direct Route



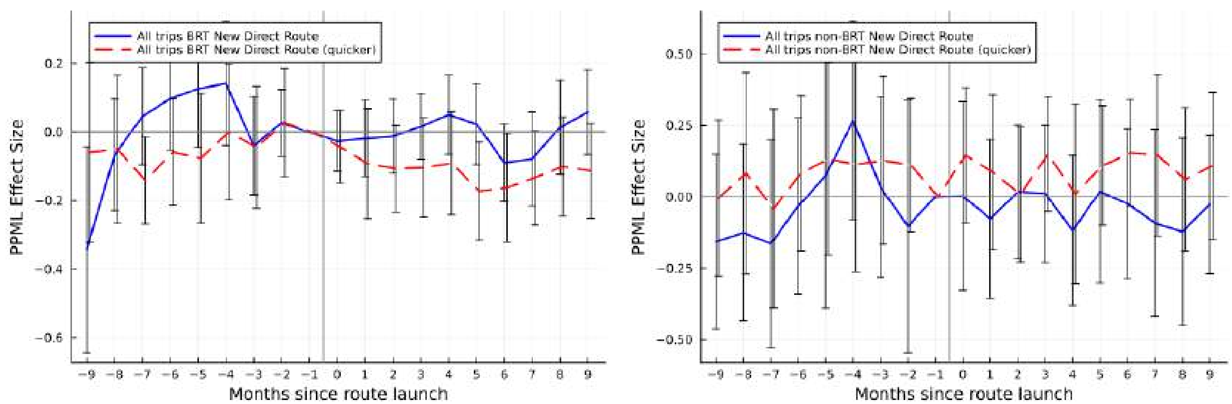
(b) Event Type 3: Additional Buses (Direct)



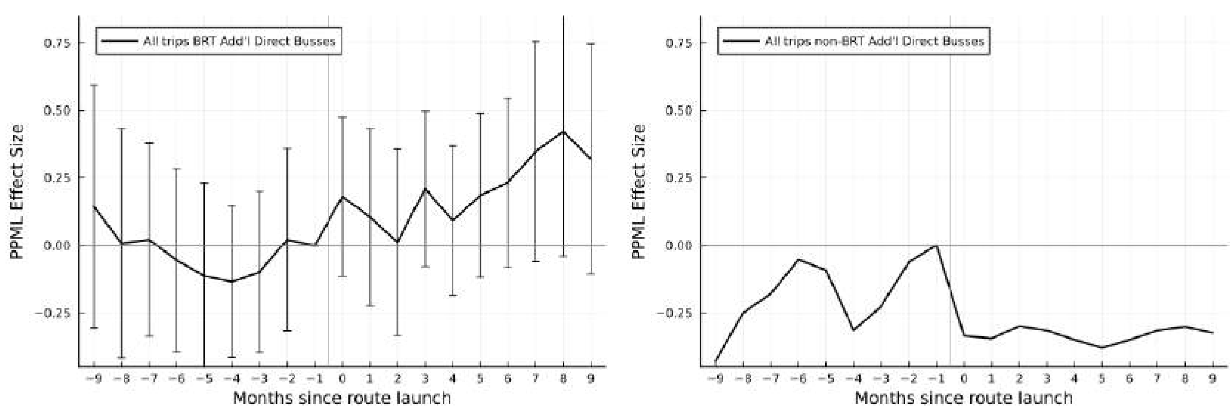
Notes: BRT graphs on the left, and non-BRT graph on the right. In the graphs on the first row, the blue, solid lines report monthly coefficients for the first event type (direct route with no change in travel time on the bus). The red, dashed lines report monthly coefficients for the second event type (quicker direct route). The graphs in the second row show the impact of the third event type. All specifications are the event study version of equation (1). 95% confidence intervals are based on standard errors clustered two-way at the origin and destination grid level.

Figure 3: **Aggregate Trip Volume** (Lack of) Impacts of BRT and non-BRT Network Expansions

(a) Event Types 1 and 2: New Direct Route

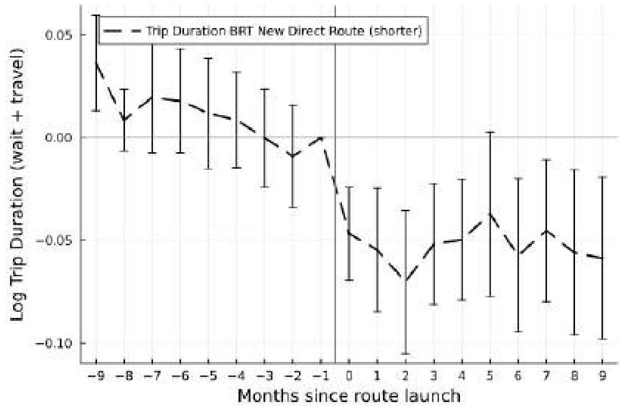


(b) Event Type 3: Additional Buses (direct)



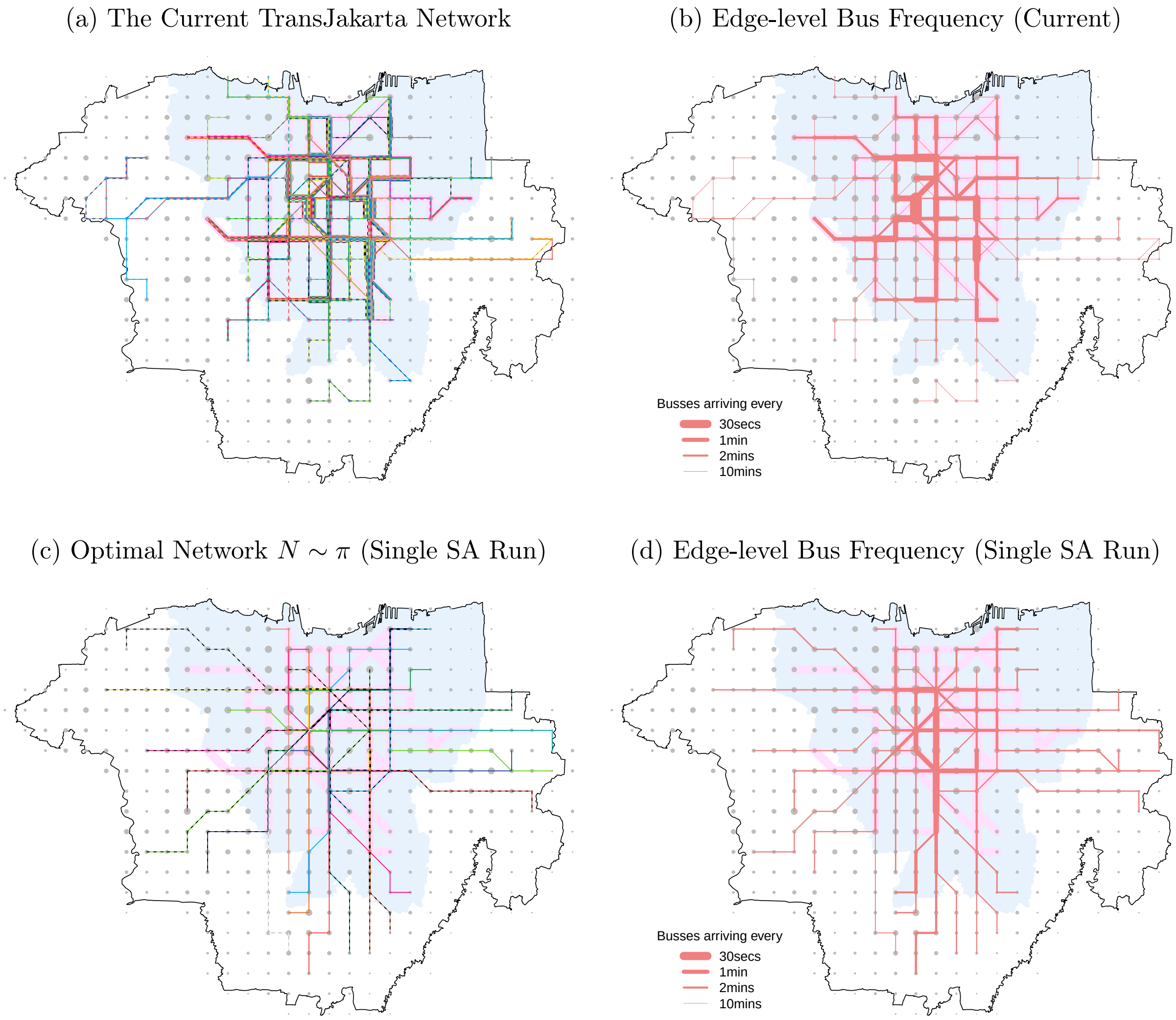
Notes: The outcome is aggregate trip flows from smartphone data. The time sample is restricted to after March 2018. The specifications are otherwise the same as in Figure 2. We do not report standard errors for non-BRT Event Type 3 due to a computational issue and refer the reader to the pooled effect in Table 2.

Figure 4: Trip Duration Impact of BRT Event 2 (New and Quicker Direct Route)



Notes: The outcome is the log of trip duration (wait time plus time on the bus), measured from linked tap-in and tap-out observations for the same smart card. The sample is restricted to BRT destination stations that are tap-out compliant, defined as stations where tap-out transactions account for at least 30% of all transactions. The specification is the same as for Event 2 in Figure 2. 95% confidence intervals are based on standard errors clustered two-way at the origin and destination grid level.

Figure 5: Current TransJakarta Network and Optimal Network Example



(a) The Current TransJakarta Network

(b) Edge-level Bus Frequency (Current)

(c) Optimal Network $N \sim \pi$ (Single SA Run)

(d) Edge-level Bus Frequency (Single SA Run)

Notes: This figure depicts the current TransJakarta network and one of the networks N obtained using the simulated annealing (SA) algorithm, asymptotically sampled from the planner's distribution π. The study area (black outline) includes the DKI Jakarta area (light blue) as well as the urban districts of Tangerang, South Tangerang, Depok, and Bekasi. Different bus lines are printed in different colors. Edges connected through BRT dedicated bus lanes are denoted by bright pink underlay. Panel (a) plots the current TransJakarta network (107 bus lines). Panel (b) shows bus frequency on each edge in the current TransJakarta network. To construct this graph, for each directed edge we compute the total bus arrival over all routes that share that edge. For each edge, we then take the average over the two directions. Panels (c) and (d) repeat the exercise for one of the 200 SA networks, chosen randomly.

Table 1: Impact of Network Expansion on Travel Time, Wait Times, and **Bus Ridership**

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.036***			-0.051*			0.158***		
	(0.007)			(0.025)			(0.021)		
E2: New Direct Line (quicker)		-0.294***			0.071			0.268***	
		(0.027)			(0.036)			(0.037)	
E3: Additional Busses			-0.027*			0.309***			0.092***
			(0.011)			(0.035)			(0.023)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	3,154,672	3,143,019	793,965	3,154,672	3,143,019	793,965	3,154,672	3,143,019	793,965
R^2	0.981	0.980	0.998	0.922	0.921	0.997			
Mean outcome	33.2	44.2	12.9	36.6	28.0	21.1	111.3	76.9	210.4
Median outcome	31.0	42.4	10.6	29.4	22.0	18.3	55.9	39.1	94.6
Unique origin x destination pairs	18,433	18,449	5,319	18,433	18,449	5,319	18,433	18,449	5,319
Unique origins	148	148	148	148	148	148	148	148	148

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.030**			0.294***			1.016***		
	(0.009)			(0.045)			(0.153)		
E2: New Direct Line (quicker)		-0.716***			0.208***			1.320***	
		(0.072)			(0.031)			(0.267)	
E3: Additional Busses			-0.089***			0.417***			0.450**
			(0.021)			(0.017)			(0.141)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	306,722	313,994	144,763	306,722	313,994	144,763	306,722	313,994	144,763
R^2	0.992	0.987	0.998	0.964	0.964	0.992			
Mean outcome	34.8	49.3	13.0	10.9	10.8	7.7	19.2	7.5	26.0
Median outcome	24.8	47.2	11.0	7.0	8.9	7.0	0.0	0.0	0.0
Unique origin x destination pairs	2,588	2,652	1,263	2,588	2,652	1,263	2,588	2,652	1,263
Unique origins	57	57	64	57	57	64	57	57	64

Notes: The two panels report results for BRT and non-BRT, respectively. Columns 1-6 capture the "first stage" impacts of the relevant events and are estimated using OLS. The bus ridership effects (columns 7-9) are estimated by PPML. "Min Travel Time" captures the time on the bus (excluding wait time) of the quickest route between an origin and a destination, given the routes available in the TransJakarta network at that time. "Buses per Hour" measures the total number of buses arriving at the origin grid over all the routes connecting the origin and destination. (The set of routes is restricted to direct routes for event 3.) These outcomes are computed using bus travel times defined at the route-origin-destination level and bus allocations defined at the route level, neither of which varies over time. This allows us to capture changes in choice sets. The coefficients capture the average effect in the first 10 months after the respective event. The origin and destination data is aggregated at the level of 1km hexagonal grids. Standard errors clustered two-way at the origin and destination grid level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: Impact of BRT and Non-BRT Network Expansions on **Aggregate Trip Volume**

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All Trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.057***			-0.157***			0.108***			-0.008		
	(0.008)			(0.031)			(0.016)			(0.051)		
E2: New Direct Line (quicker)		-0.214***			-0.107***			0.208***			-0.073	
		(0.026)			(0.030)			(0.032)			(0.049)	
E3: Additional Busses			-0.006			0.174***			0.046			0.166
			(0.036)			(0.042)			(0.041)			(0.117)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039
R^2	0.986	0.985	0.999	0.948	0.948	0.999						
Mean outcome	32.6	42.0	13.4	38.5	33.9	14.4	122.9	93.7	195.5	1895.5	1729.5	10117.3
Median outcome	30.6	42.2	10.9	33.5	29.3	12.5	61.9	54.8	153.7	0.0	0.0	6412.4
Unique origin x destination pairs	13,802	13,675	2,497	13,802	13,675	2,497	13,802	13,675	2,497	13,802	13,675	2,497
Unique origins	124	124	89	124	124	89	124	124	89	124	124	89

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.028*			0.260***			0.999***			-0.042		
	(0.011)			(0.050)			(0.199)			(0.060)		
E2: New Direct Line (quicker)		-0.737***			0.205***			1.337***			0.031	
		(0.093)			(0.036)			(0.322)			(0.060)	
E3: Additional Busses			-0.097**			0.375***			0.425*			-0.177*
			(0.031)			(0.023)			(0.176)			(0.073)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	156,656	160,409	61,377	156,656	160,409	61,377	156,656	160,409	61,377	156,656	160,409	61,377
R^2	0.995	0.989	0.999	0.973	0.973	0.996						
Mean outcome	35.9	44.3	12.4	11.2	11.6	7.9	13.0	7.6	40.7	3948.6	6060.7	6880.5
Median outcome	24.5	43.6	10.8	7.0	8.9	7.1	0.0	0.0	0.0	635.5	956.9	1017.5
Unique origin x destination pairs	1,906	1,950	860	1,906	1,950	860	1,906	1,950	860	1,906	1,950	860
Unique origins	49	49	53	49	49	53	49	49	53	49	49	53

Notes: This table reports the impact on the aggregate trip volume as measured from smartphone data (columns 10-12). The outcome is aggregate trip volume from smartphone data. The time sample is restricted to after March 2018. For comparison, columns 7-9 report the impact on bus ridership in the same sample. Standard errors clustered two-way at origin and destination grid level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Impact of BRT Event Type 2 on Total Trip Duration

	log Min Travel Time	log Bus/hr (origin)	Bus Ridership	log Trip Duration
	(1)	(2)	(3)	(4)
E1: New Direct Line	0.055***	-0.211***	0.110***	-0.003
	(0.012)	(0.038)	(0.025)	(0.005)
E2: New Direct Line (quicker)	-0.284***	0.167***	0.071*	-0.057***
	(0.038)	(0.046)	(0.036)	(0.012)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	Poisson	OLS
Controls	Yes	Yes	Yes	Yes
N	595,685	595,685	595,685	595,685
R^2	0.991	0.970		0.881
Within-R^2	0.063	0.012		0.001
Pseudo R^2	2.181	1.413		1.312

Notes: Column 4 in this table reports the impact on log trip duration (wait time plus time on the bus) of BRT Event 2. The outcome is trip duration measured from linked tap-in and tap-out observations for the same smart card. The sample is restricted to BRT destination stations that are tap-out compliant, defined as stations here tap-out transactions account for at least 30% of all transactions. We jointly estimate Events 1 and 2, so the "E2: New Direct Line (quicker)" coefficient captures the *additional* effect for Event 2. First-stage and bus ridership impacts are reported in columns 1-3. Standard errors clustered two-way at the origin and destination grid level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Estimated Demand Model Parameters

	(1)	(2)	
	BRT	BRT	non-BRT
Wait time $\alpha_{\text{wait}}/\alpha_{\text{time}}$	2.1	2.1	3.4
	[1.5, 3.6]	[1.5, 3.4]	[2.7, 5.0]
Travel time α_{time}	1.0	1.0	1.0
	[1.0, 1.0]	[1.0, 1.0]	[1.0, 1.0]
Transfer shifter μ/α_{wait} (minutes)	4.8	4.8	2.2
	[0.7, 7.5]	[0.8, 7.4]	[-1.3, 4.6]
Attention cutoff η	1.36	1.36	1.43
	[1.25, 1.47]	[1.25, 1.46]	[1.42, 1.44]
Logit parameter σ	0.13	0.13	
	[0.04, 0.32]	[0.04, 0.32]	
Moments:			
BRT events (1-3)	Yes	Yes	
BRT event 2 trip duration	Yes	Yes	
non-BRT events (1-3)		Yes	

Notes: We use a classical minimum distance with the optimal weighting matrix, and 20 random initial conditions. To construct the 95% confidence intervals, we re-estimate the model 500 times. Each time, we target a data moment vector $\widehat{m}^k = \widehat{m} + \varepsilon^k$, where ε^k is randomly drawn from the multivariate normal distribution $\mathcal{N}(\mathbf{0}, \widehat{\Omega})$. $\widehat{\Omega}$ is the estimated variance-covariance matrix of the reduced form analysis, jointly estimated in a seemingly unrelated regression framework. See Table A.9 for alternate model specifications.

Table 5: Properties of Current and Optimal Networks

Statistic	Network in 2016	Current Network	Optimal Networks	Difference Optimal – Current
Panel A: Ridership equilibrium properties				
Welfare (minutes travel time)	-3.9	0	11.21	11.21
				[0.45]
Ridership (%)	0.86	1.88	3.039	1.16
				[0.076]
Average bus crowding (pax/bus)	15.12	13.84	31.37	17.54
				[0.82]
Average bus travel time (hrs)	0.48	0.49	0.64	0.15
				[0.018]
Transfer trips (share of all bus trips)	0.81	0.88	0.85	-0.025
				[0.0077]
Panel B: Network and access properties				
Locations with a bus station (share)	0.17	0.42	0.45	0.022
				[0.019]
Locations with station just outside DKI (%)	0.0077	0.27	0.4	0.13
				[0.037]
Trips with access to bus (share)	0.43	0.73	0.75	0.019
				[0.018]
Location pairs connected by bus (share)	0.022	0.12	0.18	0.059
				[0.016]
Mean edge headway (minutes)	3.03	7.62	2.69	-4.93
				[0.19]
Mean location pair headway (minutes)	3.86	12.02	3.056	-8.96
				[0.25]
Total network mileage (km)	180.093	543.6	464.01	-79.59
				[21.98]
Number of bus lines	19	110	21	-89.005
				[2.046]

Notes: This table reports characteristics of the TransJakarta network in 2016 (column 1) and 2020 (column 2), as well as of optimal networks, on average over 200 independent simulated annealing (SA) runs (column 3). Panel A reports properties of the ridership equilibrium while Panel B reports network properties. Welfare in Panel A is denoted in equivalent variation terms of minutes of bus travel time for each bus trip in the current network. "Ridership" corresponds to the share of total trips across Jakarta using the bus network. "Average bus occupancy" is the occupancy level of the average bus, assuming buses are active for 17 hours each day (5AM-10PM) and using the model to predict total bus ridership for each route and pair of consecutive stations."Trips with access to bus" in Panel B denotes the share of all trips (measured using smartphone data and assumed fixed in counterfactuals) that can be completed using (have access to) the bus network. Grids within our study area at below-median distance to DKI make up the "just outside DKI" group. "Mean edge headway" reports the average across network edges of the headway between consecutive buses, pooling over all routes that travel on that edge. "Mean location pair headway" repeats this with location pairs rather than edges, pooling over all routes that travel direct between those locations. "Total network mileage" computes the total length of all edges traversed by the network. In column 4 we report the mean and standard deviation of the difference between optimal and current networks, computed over all optimal network draws.

Table 6: Optimal Networks Comparative Statics

Statistic	Current Network	Baseline Optimal	Large change $2 \times \alpha_W$	Large change $2 \times \alpha_T$	Large change $\mu_T - 15$mins
Panel A: Coverage measures					
Locations with a bus station (share)	0.42	0.45	0.33	0.36	0.48
		(0.0015)	(0.002)	(0.0023)	(0.0031)
Location pairs connected by bus (share)	0.12	0.18	0.098	0.11	0.21
		(0.0011)	(0.0013)	(0.0012)	(0.0028)
Total network mileage (km)	543.6	464.064	327.54	388.072	515.88
		(1.6)	(2.55)	(2.37)	(3.61)
Panel B: Speed measures					
Bus time relative to quickest possible bus route	1.33	1.24	1.2	1.16	1.24
		(0.0086)	(0.015)	(0.0083)	(0.02)
Panel C: Directness measures					
Connected directly (share of all connected pairs)	0.21	0.15	0.17	0.17	0.18
		(0.00063)	(0.0015)	(0.0011)	(0.0018)

Notes: This table reports how network shape measures (rows) change as a function of parameter changes (columns). Columns 3-5 report characteristics of optimal networks achieved with different sets of preference parameters, each obtained from 50 independent simulated annealing runs with altered parameter vectors: double value of wait time ($\alpha_{\text{wait}}^{\text{BRT}}$ and $\alpha_{\text{wait}}^{\text{non-BRT}}$) relative to our estimate, double value of time on bus (α_{time}) relative to our estimate, and an additional penalty of taking a transfer ($\mu_{\text{transfer}}^{\text{BRT}}$ and $\mu_{\text{transfer}}^{\text{non-BRT}}$) equivalent to 15 minutes of wait time. See Table 5 for variable definitions. To construct "bus time relative to quickest possible bus route," for each origin-destination pair that is directly connected, we consider the quickest direct bus connection in the network, and divide its duration by the duration of the quickest possible bus route between the two locations in the entire geographic environment. We then average this ratio over all directly connected origin-destination pairs. "Connected directly (share of all connected pairs)" computes the share of all location pairs that are connected by bus, for which a direct connection also exists. Bootstrapped standard errors are in parentheses. Table A.15 reports the *local* comparative statics version of this table.

Online Appendix

Optimal Public Transportation Networks: Evidence from the World's Largest Bus Rapid Transit System in Jakarta

Gabriel Kreindler, Arya Gaduh, Tilman Graff, Rema Hanna, Benjamin A. Olken

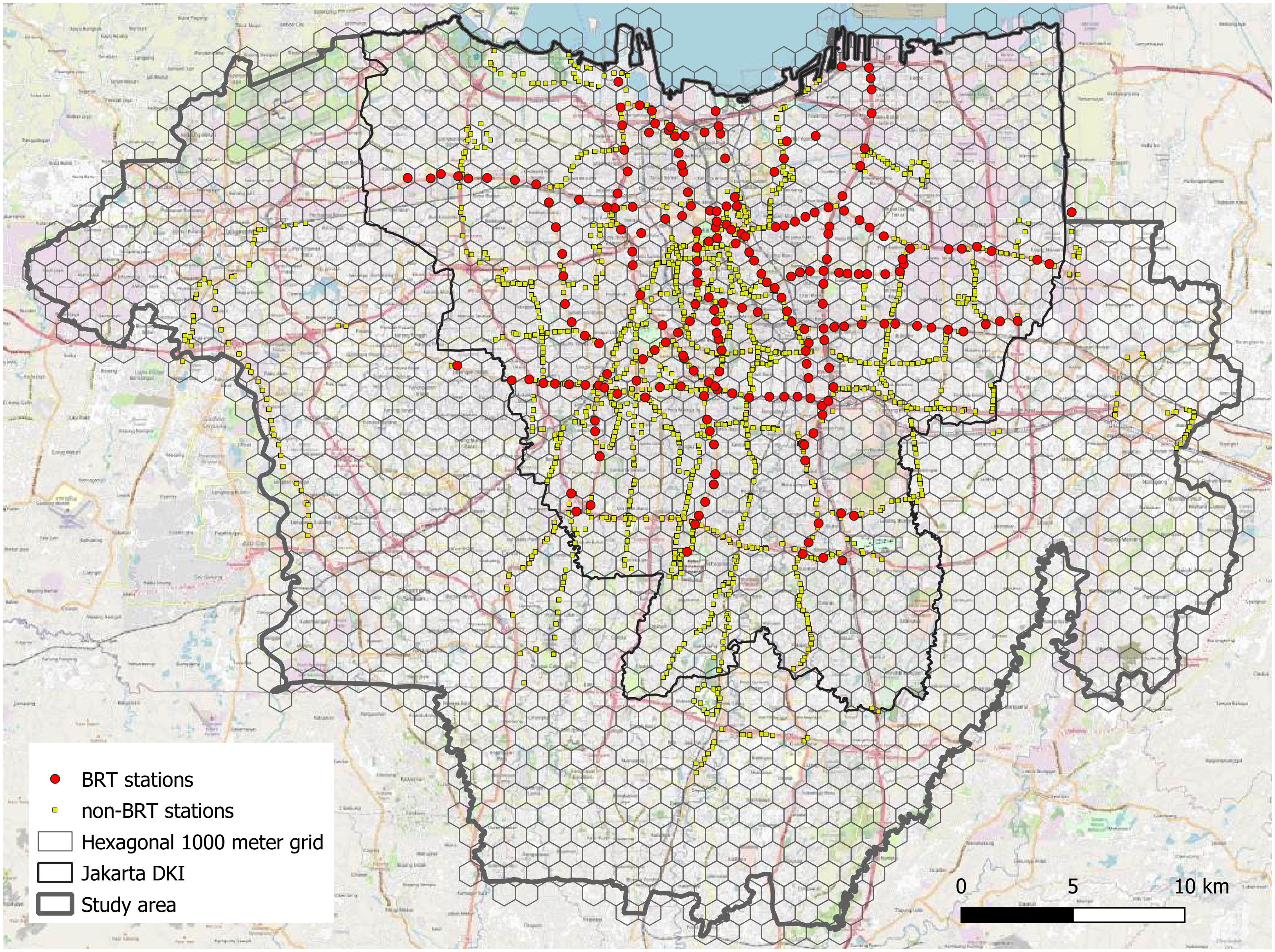
May 13, 2025

A Online Appendix

Contents

A.1 Appendix Figures

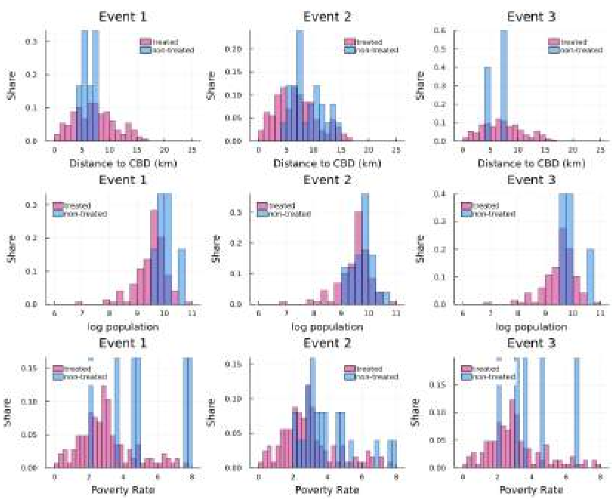
Figure A.1: Map of Study Area With 1km Hexagonal Grids and TransJakarta Bus Stations



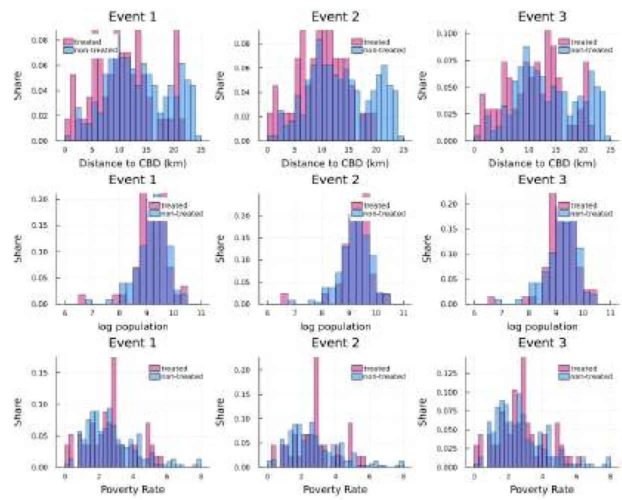
Notes: This figure shows the location of all TransJakarta stations (BRT in red circles, non-BRT in yellow squares) as of March 2020, together with the hexagonal grid with distance 1000 meters between centroids. The thin black link is the boundary of the Special Capital Region of Jakarta (DKI Jakarta), whereas the thicker gray boundary also includes the adjacent cities of (in counterclockwise order) Tangerang, South Tangerang, Depok, and Bekasi.

Figure A.2: Characteristics of Treated and Non-Treated Origin Grids

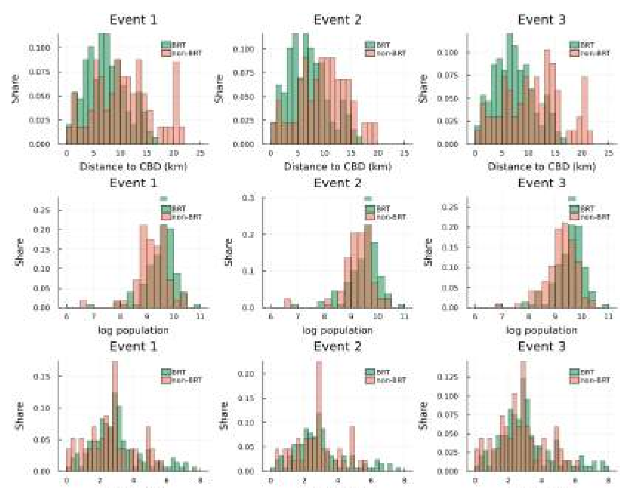
(a) BRT vs non-treated



(b) Non-BRT vs non-treated



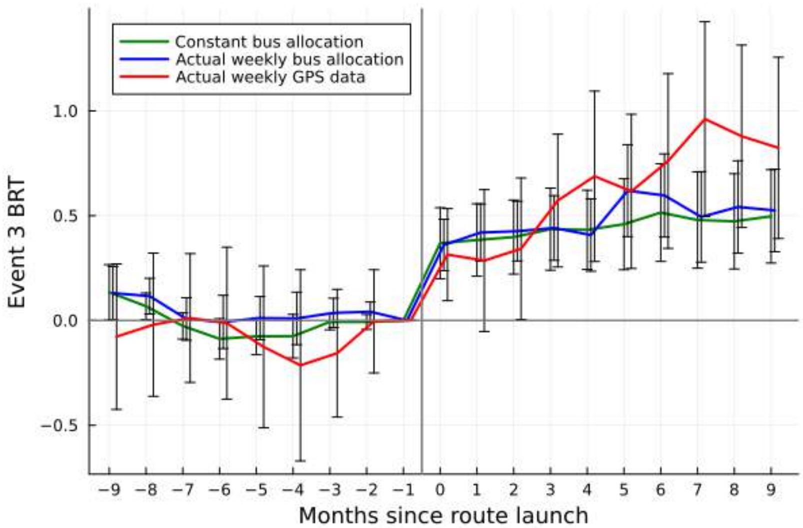
(c) BRT vs non-BRT



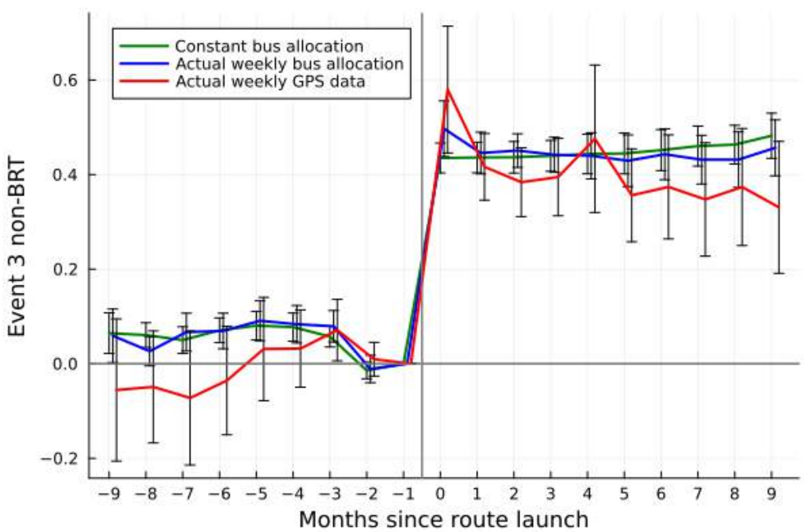
Notes: panels (a) and (b) display histograms of characteristics of treated and non-treated origin grids with bus access. For each event type and for BRT/non-BRT, we define an origin grid as treated if it ever has an event of that type (for some destination). The rest of the origin grids with bus access are the non-treated group. Panel (c) compares treated BRT grids and non-BRT treated grids.

Figure A.3: Event 3 "First-Stage" Robustness with Actual Bus Allocation and GPS Data

(a) Event 3 BRT

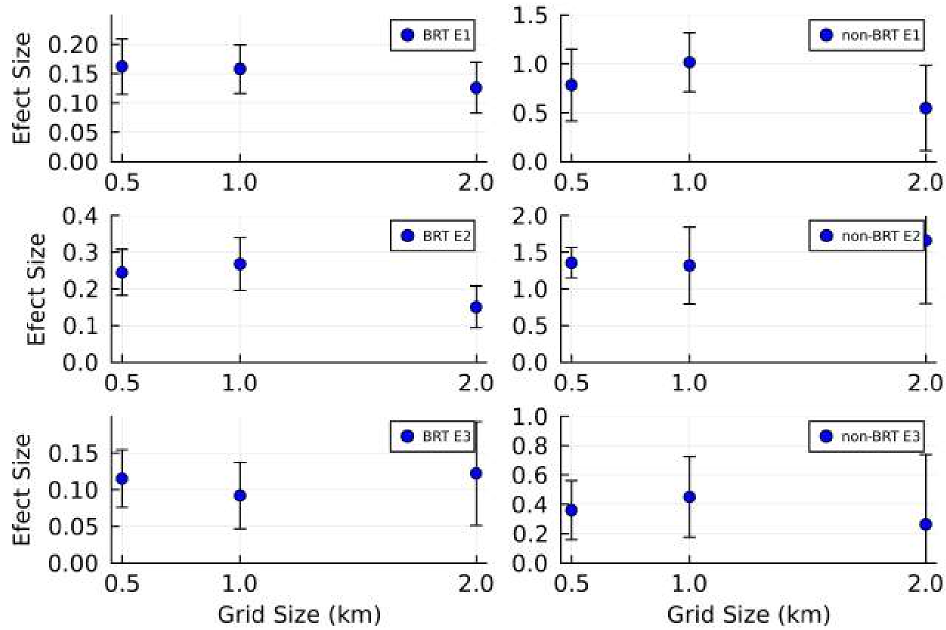


(b) Event 3 nonBRT



Notes: This graph plots the impact of Event 3 on the log of bus arrival rate. We compute the bus arrival rate in three ways. First, in green, our benchmark method uses for each route a bus allocation that is constant over time. (This avoids endogenous changes in bus allocation in anticipation of changes in demand.) Second, in blue, we use TransJakarta's actual bus allocation, which varies over time within route. Third, in red, we use the number of trips that appear in TransJakarta's bus GPS data. In all specifications, the outcome is the log average weekly bus arrival rate per hour between an origin and a destination, in a week, aggregated over all direct routes. We restrict the estimation sample to observations with coverage in the GPS data, which account for around 69% and 83% of all observations for BRT and non-BRT, respectively.

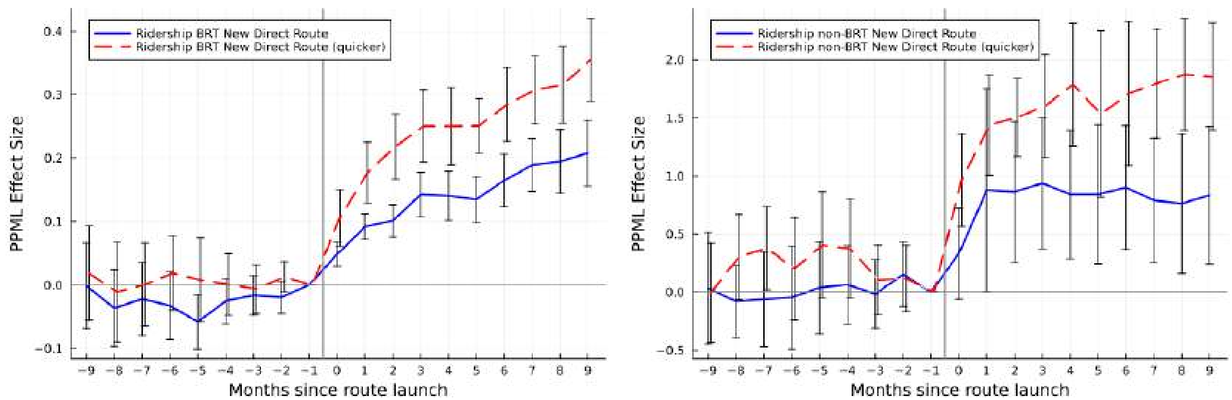
Figure A.4: Varying the Level of Aggregation: Impact of Network Expansion on Bus Ridership



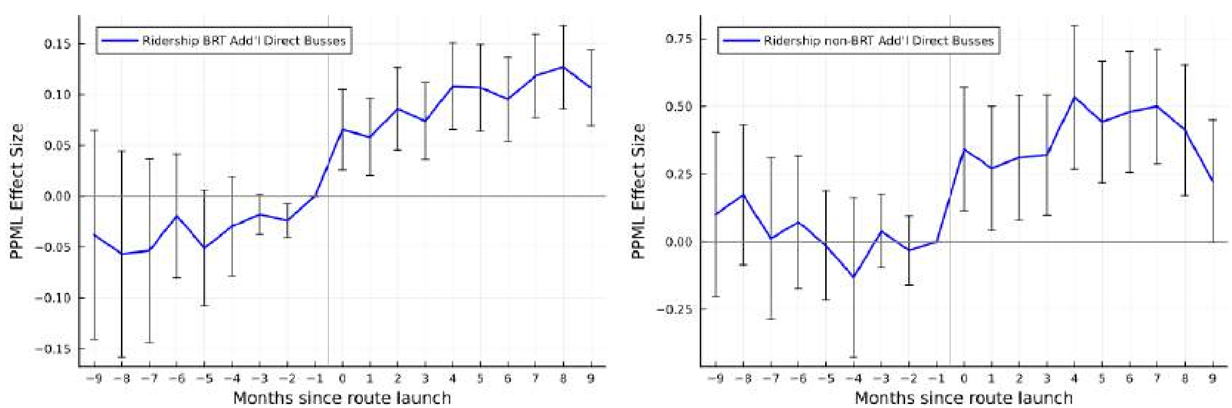
Notes: This graph plots the $Post_{odt}$ coefficients from our main specification with bus ridership as the outcome, for all six events, and for the three aggregation levels we consider: 500-meter square grids, 1000-meter hexagons, and 2000-meter square grids.

Figure A.5: Robustness With 500m Square Grids: **Bus Ridership** Impacts of BRT and Non-BRT Network Expansions (PPML)

(a) Event Types 1 and 2: New Direct Route



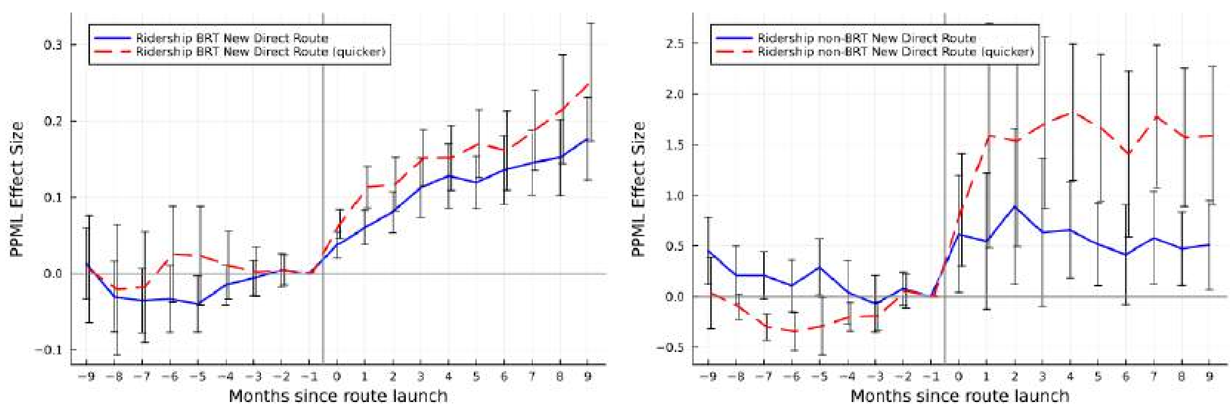
(b) Event Type 3: Additional Buses (direct)



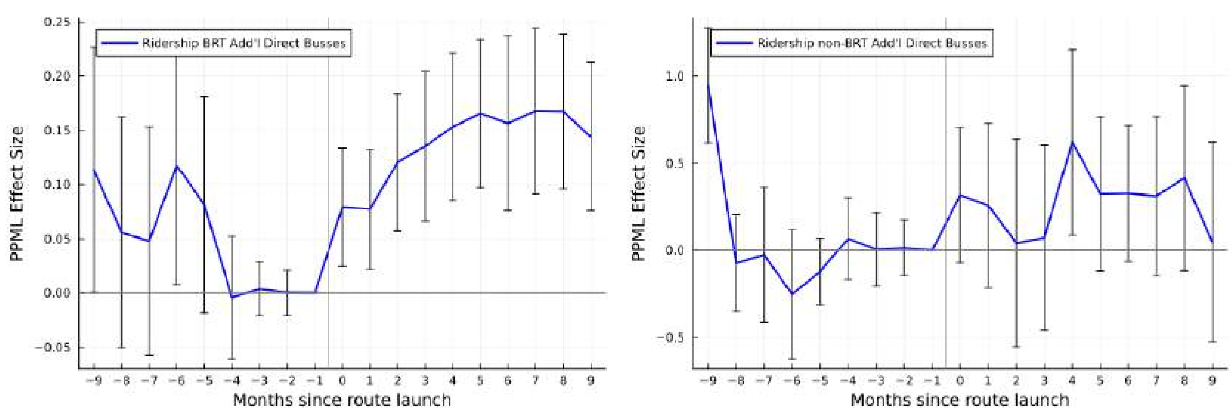
Notes: This graph replicates Figure 2 using 500-meter square grids instead of 1km hexagonal grids.

Figure A.6: Robustness With 2000m Square Grids: **Bus Ridership** Impacts of BRT and Non-BRT Network Expansions (PPML)

(a) Event Types 1 and 2: New Direct Route



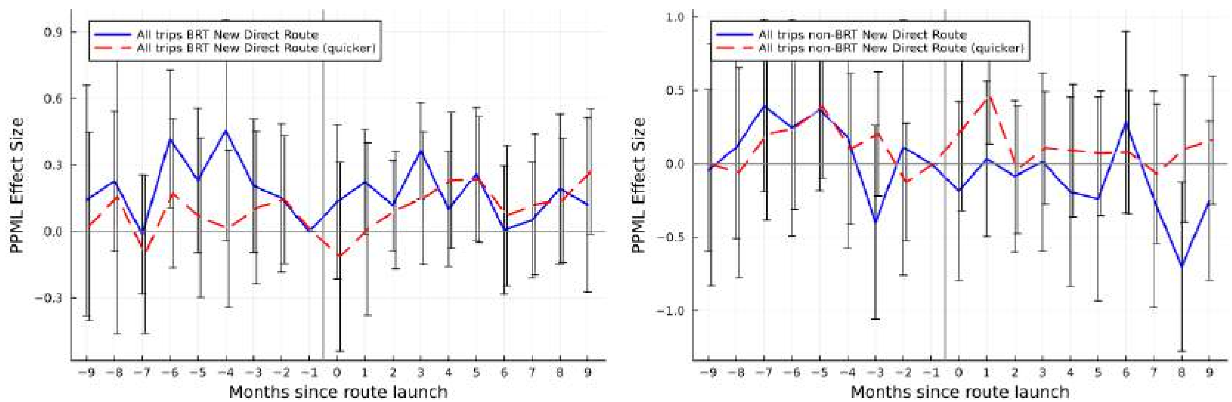
(b) Event Type 3: Additional Buses (direct)



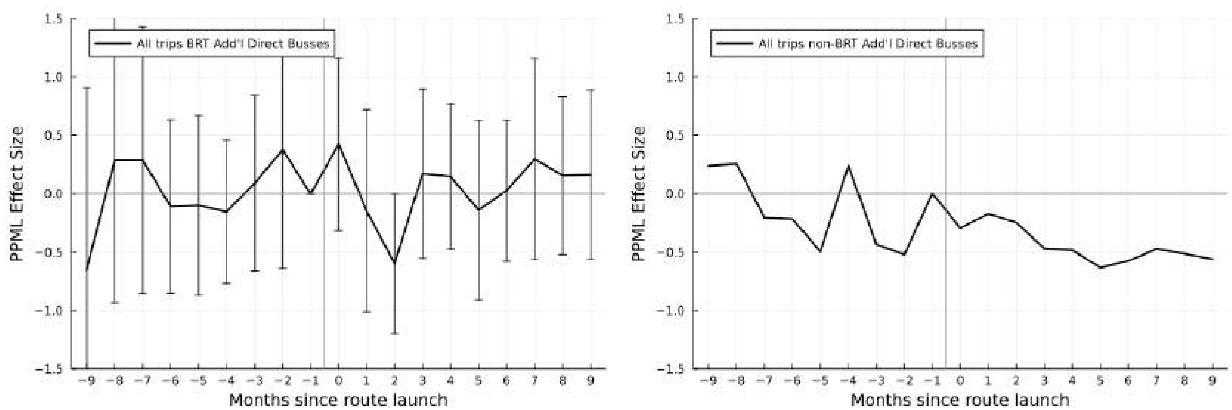
Notes: This graph replicates Figure 2 using 2000-meter square grids instead of 1km hexagonal grids.

Figure A.7: Robustness 500m Square Grids: **Aggregate Trip Volume** Impacts of BRT and Non-BRT Network Expansions

(a) Event Types 1 and 2: New Direct Route

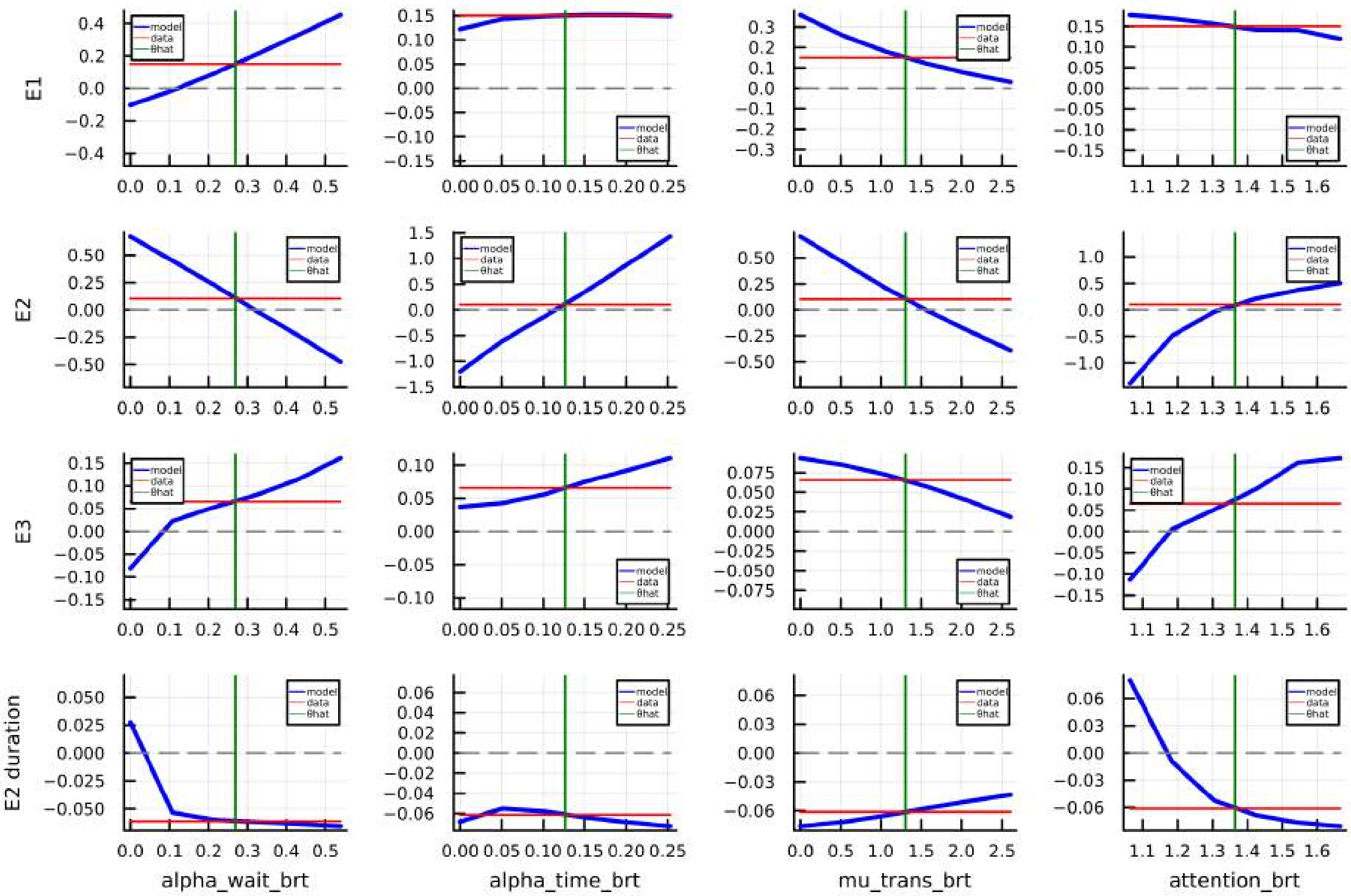


(b) Event Type 3: Additional Buses (Direct)



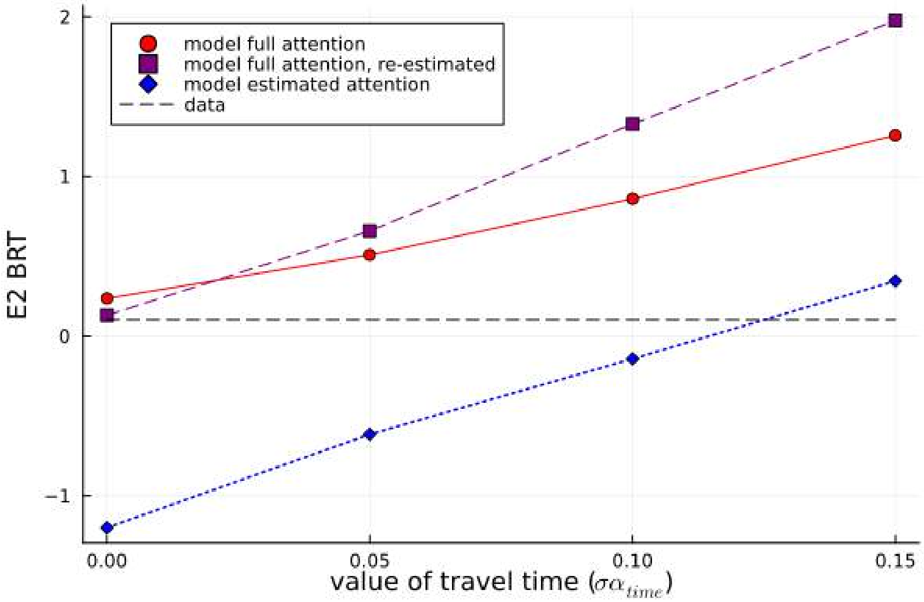
Notes: This graph replicated Figure 3 using 500m square grids. We do not report standard errors for non-BRT Event Type 3 due to a computational issue and refer the reader to the joint effect in Table A.7.

Figure A.8: Moment Dependence on Parameters (Jacobian)



Notes: This figure shows how, in the model, the moments we use in estimation (rows) vary as a function of parameters (columns). Each graph shows how the value of one of the moments $m_j \in \{\alpha^{1B}, \alpha^{2B}, \alpha^{3B}, \alpha^{2B,\text{duration}}\}$ depends on one of the parameters $\theta_i \in \{\sigma\alpha^{\text{BRT}}_{\text{wait}}, \sigma\alpha_{\text{time}}, \sigma\mu^{\text{BRT}}_{\text{transfer}}, \eta^{\text{BRT}}\}$. For each plot, we use 10 values of θ_i that bracket the estimated value $\widehat{\theta}_i$. The thick blue line plots the model's moment value $m_j(\theta_i)$. The green vertical lines indicate the estimated parameter $\widehat{\theta}_i$ (column 2 in Table 4). The red horizontal lines indicate m_j^{data}, the value of the moments in the data.

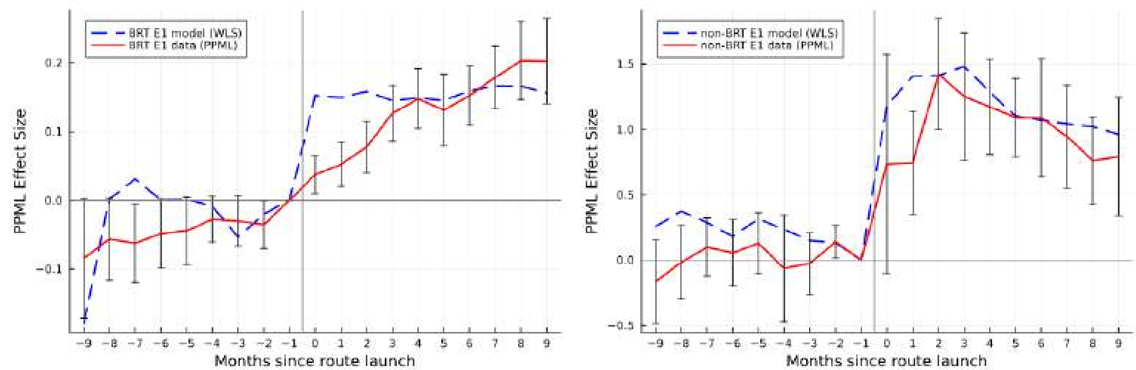
Figure A.9: Model Fit and Full Attention



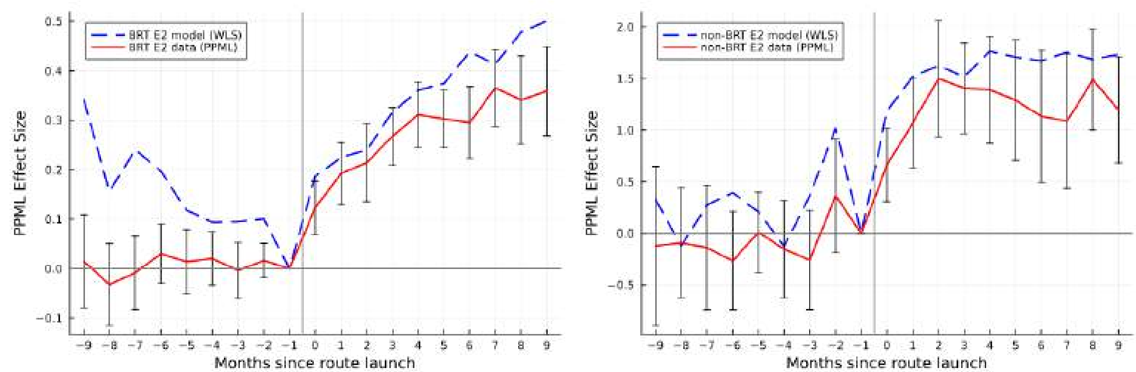
Notes: This figure shows that the model with full attention cannot fit the Event 2 BRT moment. The horizontal dashed line indicates the value of this moment in the data. The blue line (with diamonds) plots the model's predicted value of the event 2 BRT moment using the estimates from Table 4, column 2, where we vary the value of time α_{time} holding the other parameters fixed. (Recall that in the main estimation we normalize $\alpha_{\text{time}} = 1$.) The red line (with circles) repeats this exercise in the *full attention* model, i.e. setting $\eta^{\text{BRT}} = \infty$. The purple line (with squares) repeats the exercise for the full attention model where the other parameters $(\alpha^{\text{BRT}}_{\text{wait}}, \mu^{\text{BRT}}_{\text{transfer}})$ are re-estimated to match the 1st and 3rd events moments, given an assumed value of $\sigma\alpha_{\text{time}}$ and the previously estimated σ.

Figure A.10: Demand Estimation Model Fit (Targeted Moments)

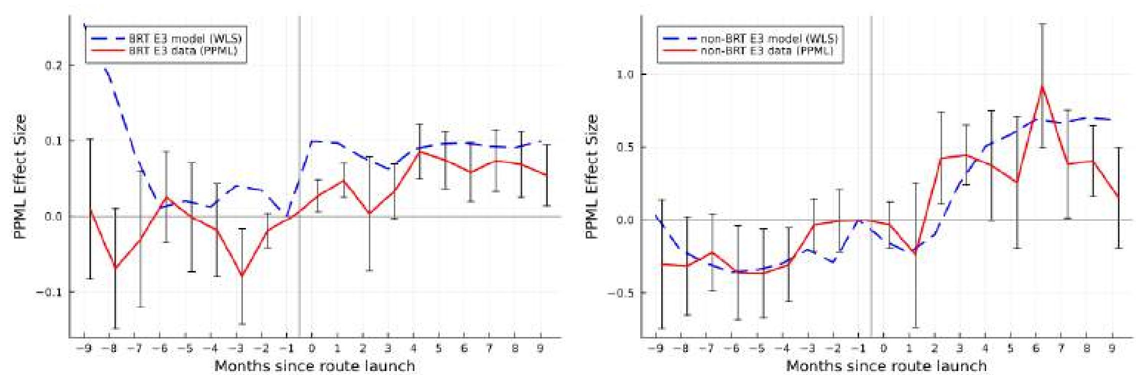
(a) Event Type 1: New Direct Route (Not Quicker)



(b) Event Type 2: New Direct Route (Quicker)



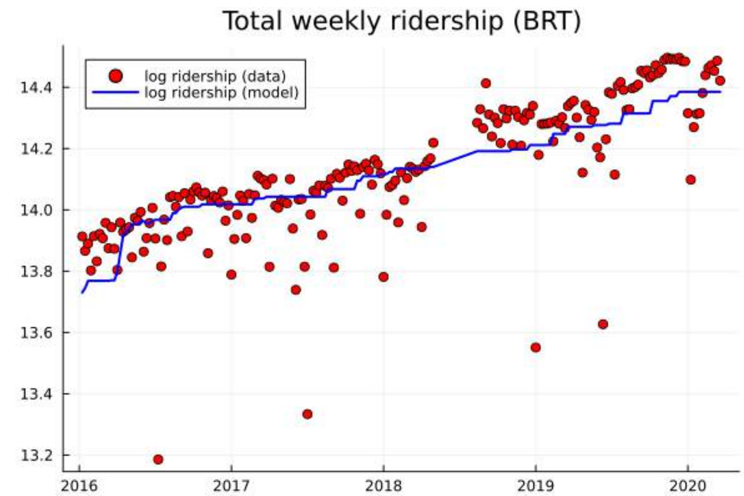
(c) Event Type 3: Additional Buses (Direct)



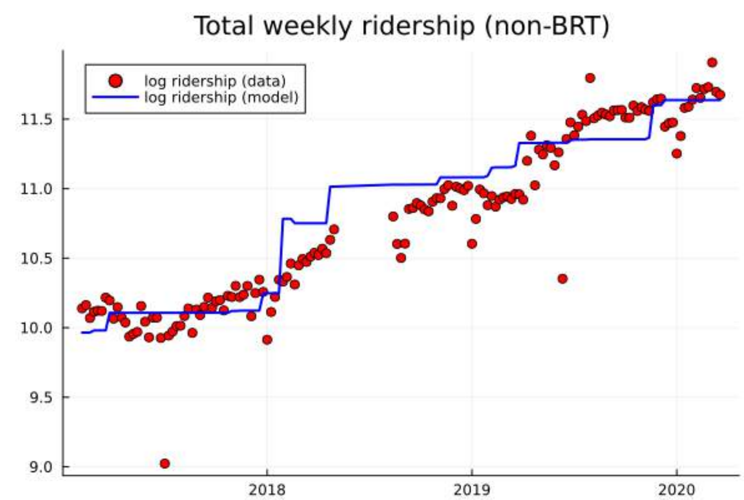
Notes: These figures plot the event impact on ridership data (red solid lines) and the same analysis on model-predicted ridership using the model estimated in Table 4, column 2 (blue dashed lines). For model-predicted ridership, we use weighted least squares on log model ridership.

Figure A.11: Demand Estimation Model Fit Over Time

(a) BRT Ridership



(b) Non-BRT Ridership

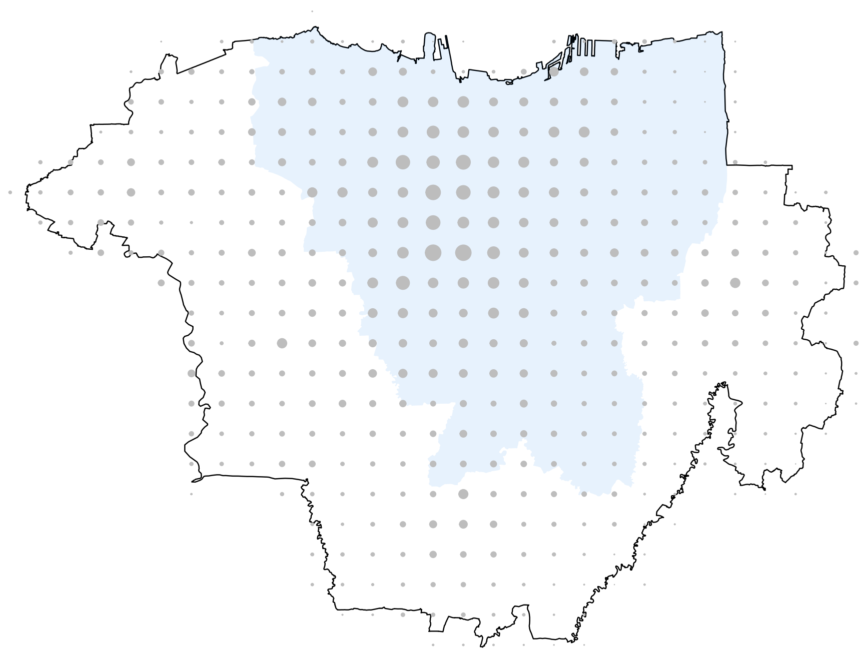


Notes: These figures report actual and model-predicted total ridership over the entire TransJakarta network over time at the weekly level, based on the model estimated in Table 4, column 2.

Figure A.12: Demand Estimation Model Fit for Non-BRT Route Launch

Notes: This figure reports actual and model-predicted ridership after the launch of the first non-BRT connection (direct or transfer). To construct this figure, we consider the 74 origin-destination pairs (using 1km hexagonal grids) that are part of the reduced form analysis and that first become connected through non-BRT after 2017, when our non-BRT ridership data begin. We then aggregate total bus ridership by semester after the route launch. (Note, the sample of origin destinations pairs changes by semester. However, 86% of the origin destination pairs appear in semester 3 after the route launch.). We then plot the inverse hyperbolic sine of total semester ridership in the data and in the model, based on the model estimated in Table 4 column 2.

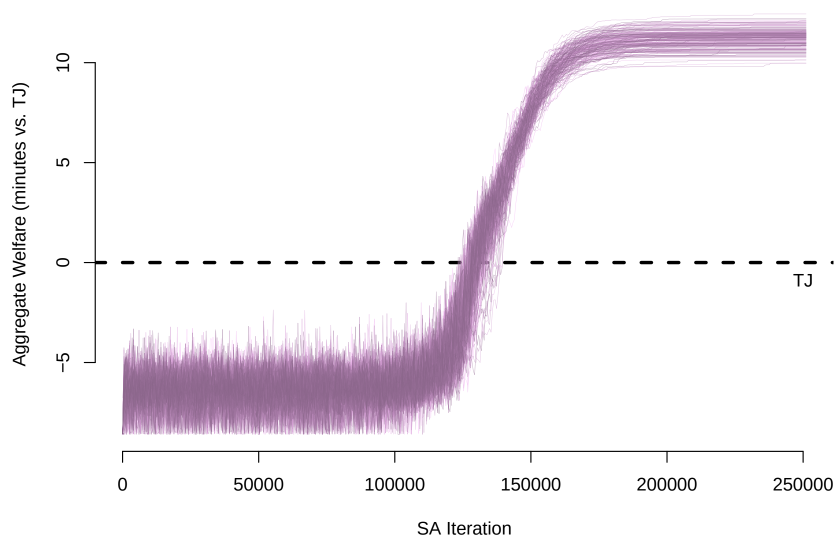
Figure A.13: Jakarta Geography Used for Optimal Networks



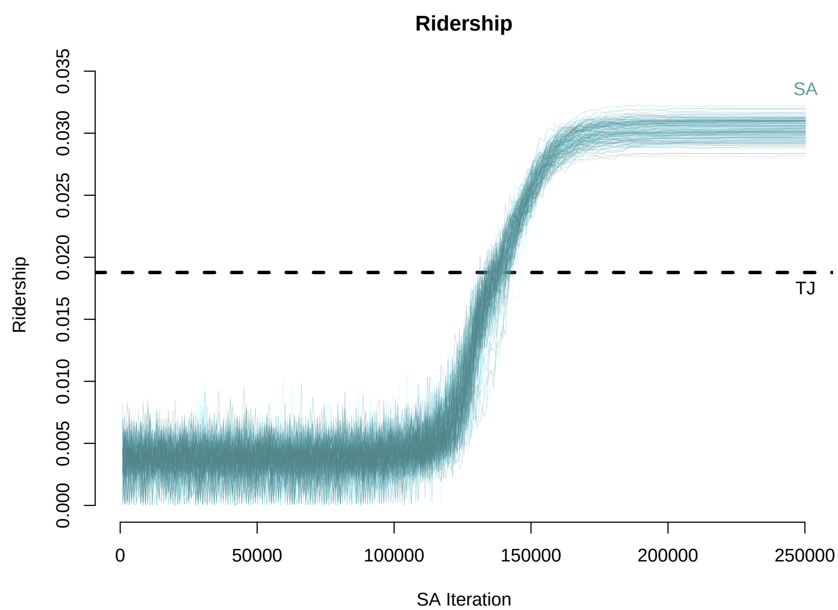
Notes: This map prints the geographic environment with 2km×2km grid cells used to compute optimal networks. The central DKI area is in light blue. The size of each circle is proportional to the number of trips that depart from that grid cell.

Figure A.14: Welfare and Ridership Over the Course of the Simulated Annealing Algorithm

(a) Welfare



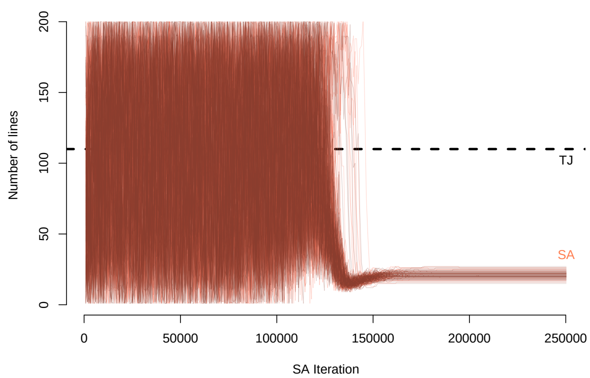
(b) Ridership



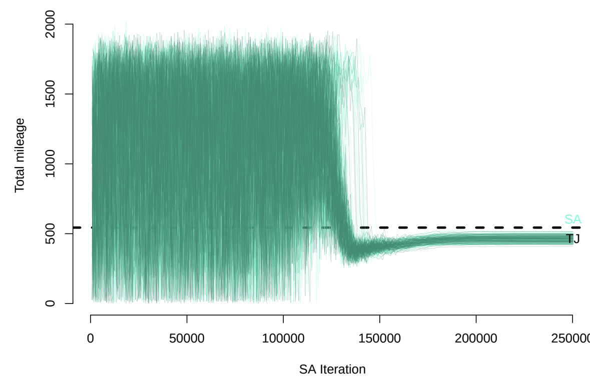
Notes: Each line shows the progression of welfare (panel a, denominated in minutes of travel time equivalent variation relative to the current TJ network) and ridership (panel b) throughout the 200 runs of the simulated annealing algorithm.

Figure A.15: Evolution of Network Characteristics

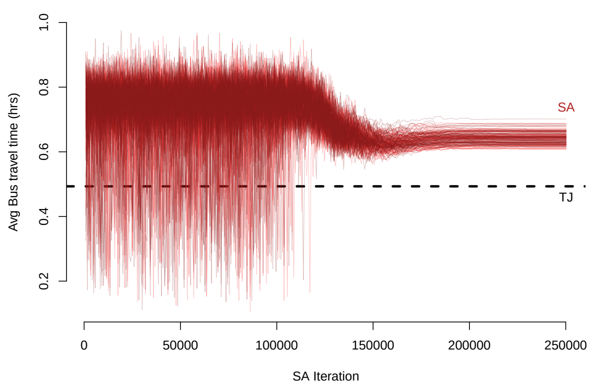
(a) Number of lines



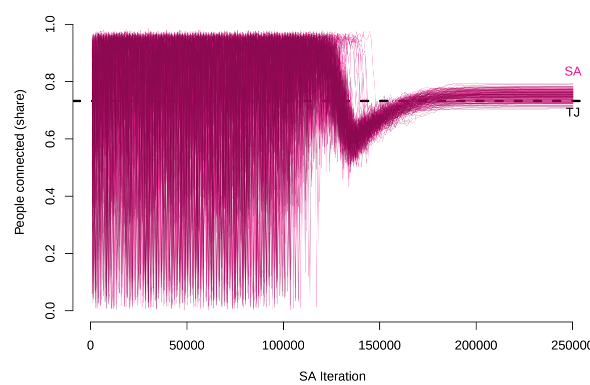
(b) Total mileage



(c) Average bus travel time (in hrs)

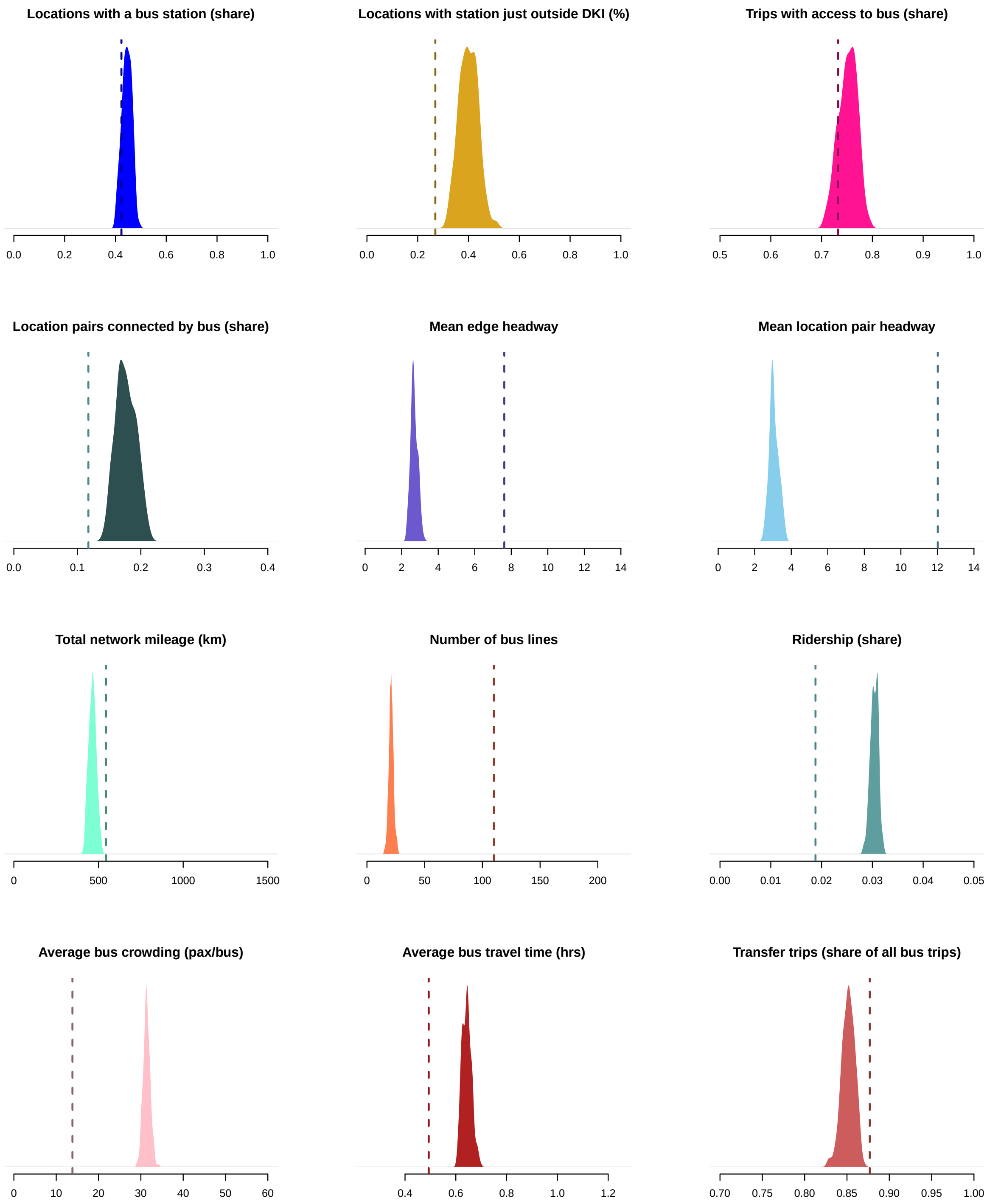


(d) Trips with feasible connection (share)



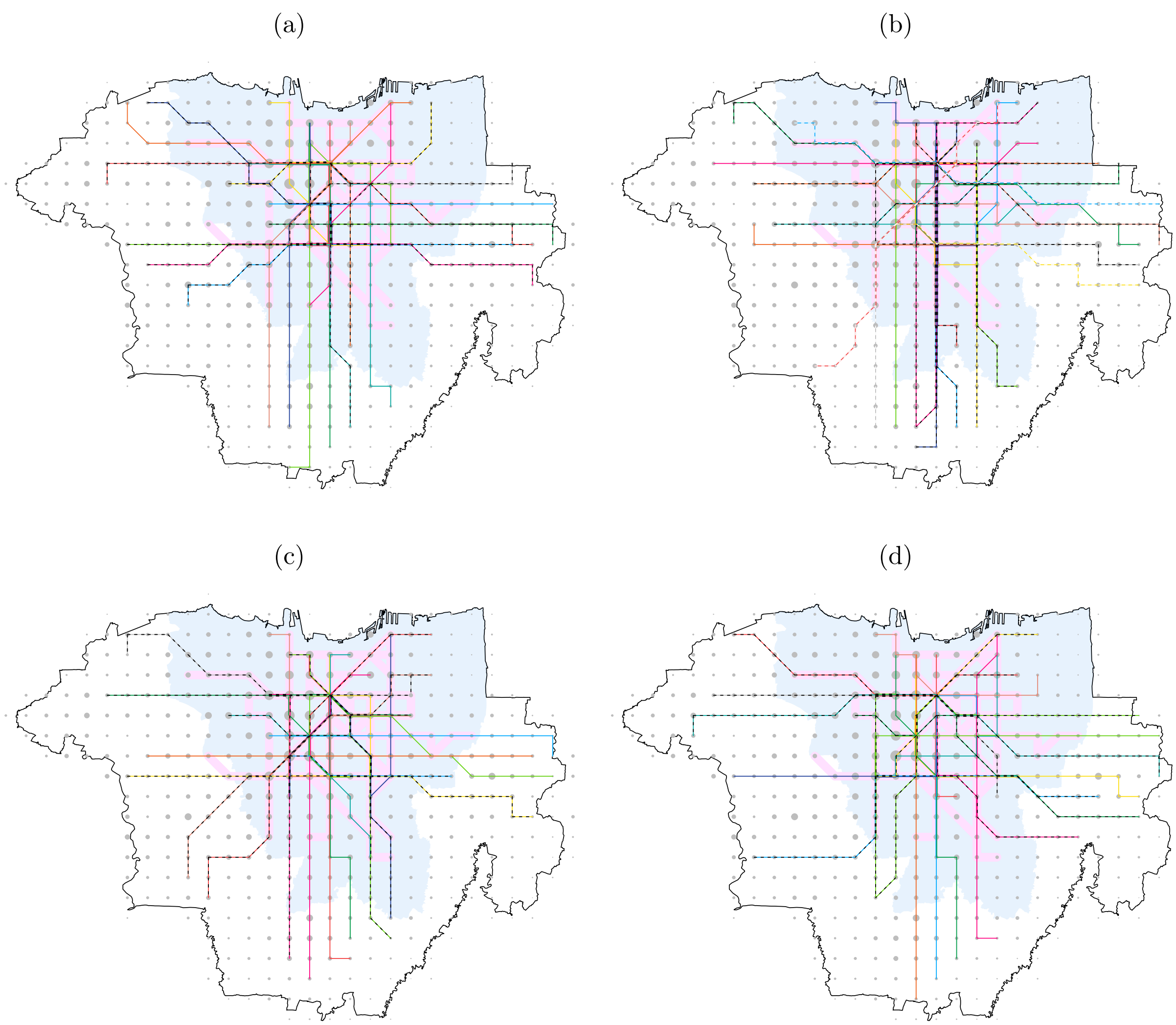
Notes: This figure plots the evolution of selected network characteristics across the 251,089 simulated annealing iterations, for all 200 independent runs. It shows that the algorithm spends roughly the first half of its iterations to explore the full space of available networks, before converging to a fairly stable set of optimal network characteristics.

Figure A.16: Distributions of Optimal Network Properties



Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final network from the simulated annealing algorithm, over the 200 parallel simulated annealing runs. The vertical line plots the measure for the current TransJakarta network for comparison.

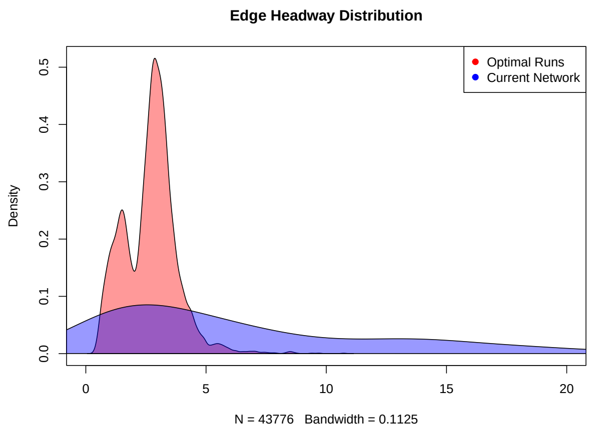
Figure A.17: Optimal Network Examples



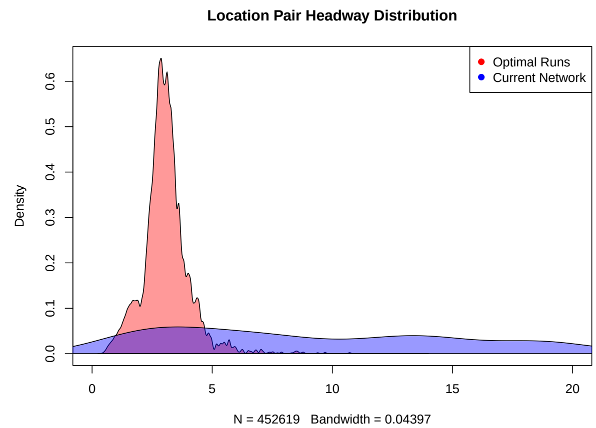
Notes: This figure depicts four additional randomly selected networks from the 200 networks obtained via SA. These represent random draws from the planner's distribution of optimal networks. Each network is obtained from an independent simulated annealing run.

Figure A.18: Distribution of Headways (in minutes)

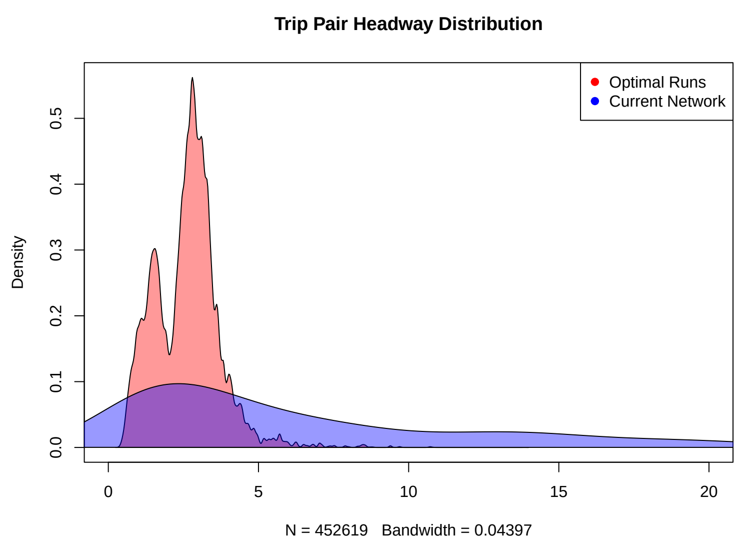
(a) Edge level



(b) Location pair level



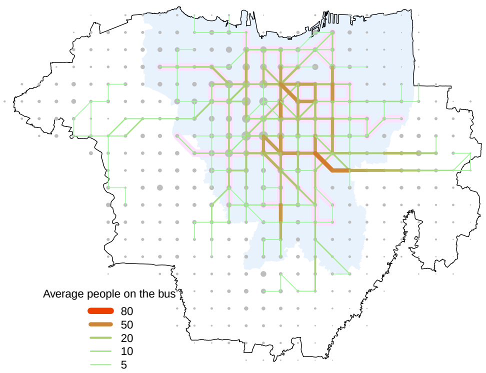
(c) Location pair level (weighted by number of trips)



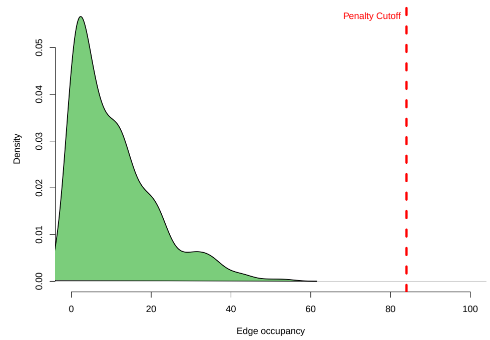
Notes: Distribution of headway times (in minutes) for the 200 optimal runs (in red) and the current TJ network (in blue).

Figure A.19: Average Bus Occupancy in Current and Optimal Networks

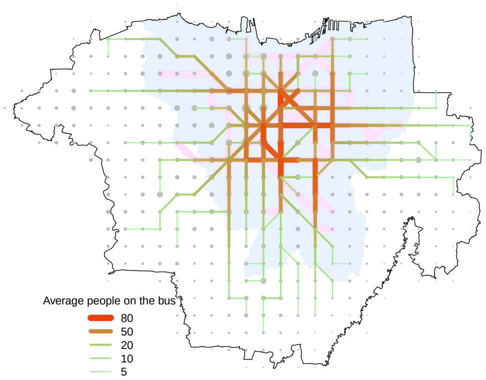
(a) Bus Occupancy (Current Network)



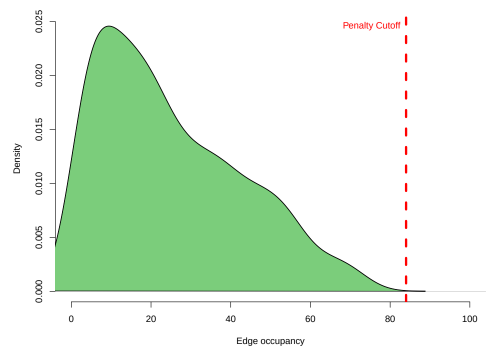
(b) Distribution (Current Network)



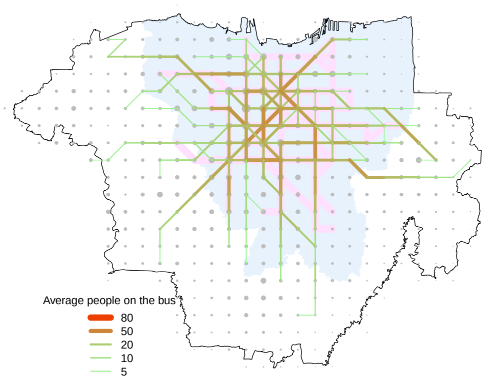
(c) Bus Occupancy (Single SA Run)



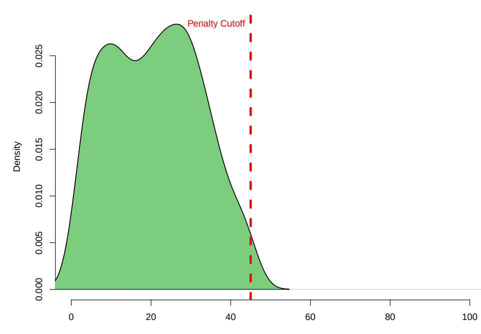
(d) Distribution (Single SA Run)



(e) Bus Occupancy (More Stringent Penalty)

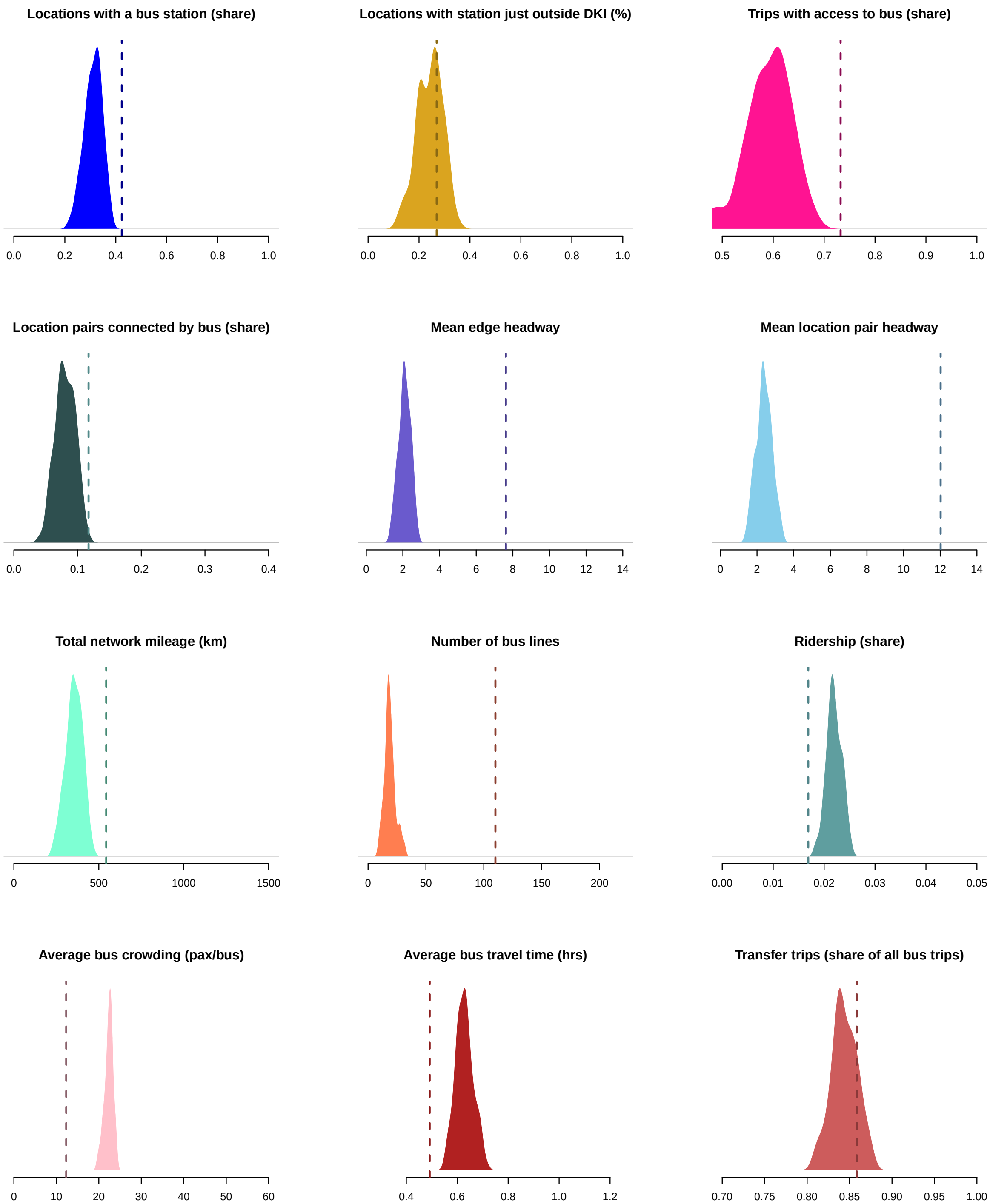


(f) Distribution (More Stringent Penalty)



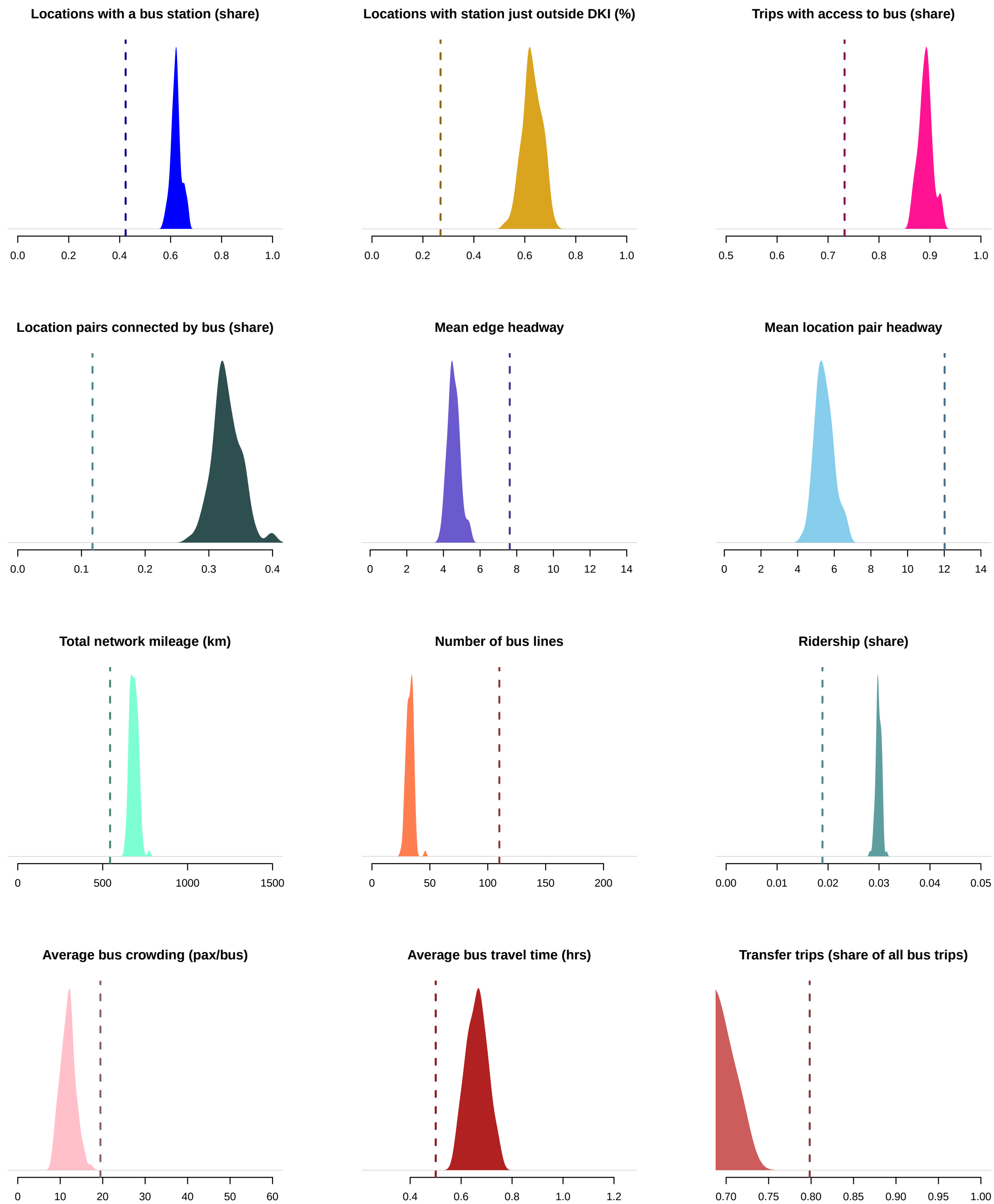
Notes: This figure depicts edge-level bus occupancy, defined as the average number of riders on a typical bus traveling across an edge, for the current TransJakarta network (a), for a randomly selected sampled network (b), and for a random sampled network with stricter (lower) bus capacity. To construct this graph, we use model-implied ridership percentages to compute how many people take each bus line and where they get on and off the line. We assume buses run 5AM-10PM each day. Panels (b), (d) and (f) plot the distribution of edge-level occupancy and the assumed bus capacity. See Appendix A.7.2 for details. The counterfactual in panels (e) and (f) uses a bus capacity that is 1.85× lower to account for within-day variation in demand. We obtain this number as follows. We use the aggregate ridership R_h by hour in the current network. Peak ridership (at 7 AM) is 85% larger than the average. We then lower all bus capacities by a factor of 1.85.

Figure A.20: Distributions of Optimal Network Properties: More Stringent Occupancy Penalty



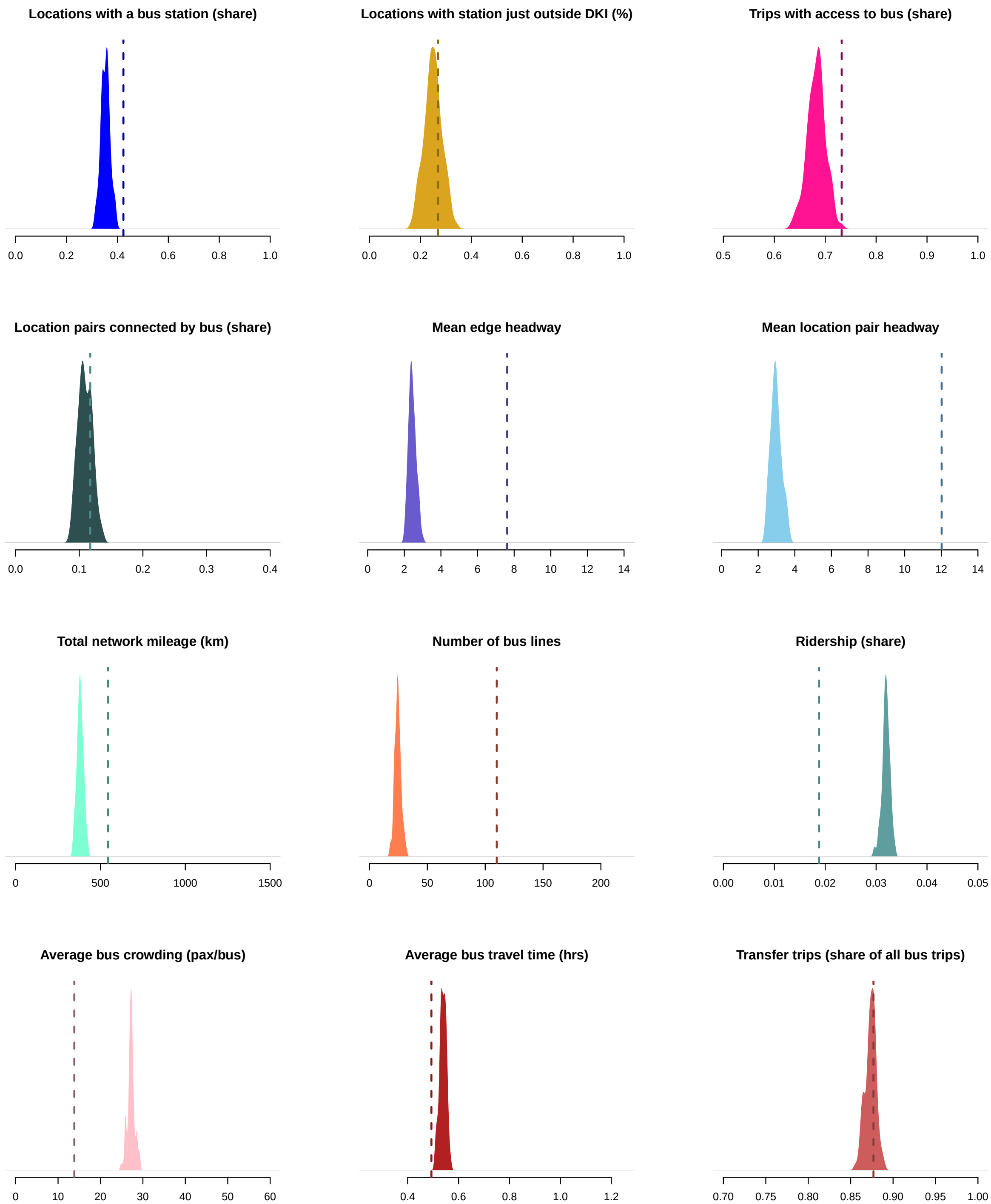
Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from 100 runs of the simulated annealing algorithm where the bus capacity is 1.85× lower than in the baseline runs to account for within-day variation in demand. We obtain this number as follows. We use the aggregate ridership R_h by hour in the current network. Peak ridership (at 7 AM) is 85% larger than the average. We then lower all bus capacities by a factor of 1.85.

Figure A.21: Distributions of Optimal Network Properties: Heterogeneous Preferences by Poverty



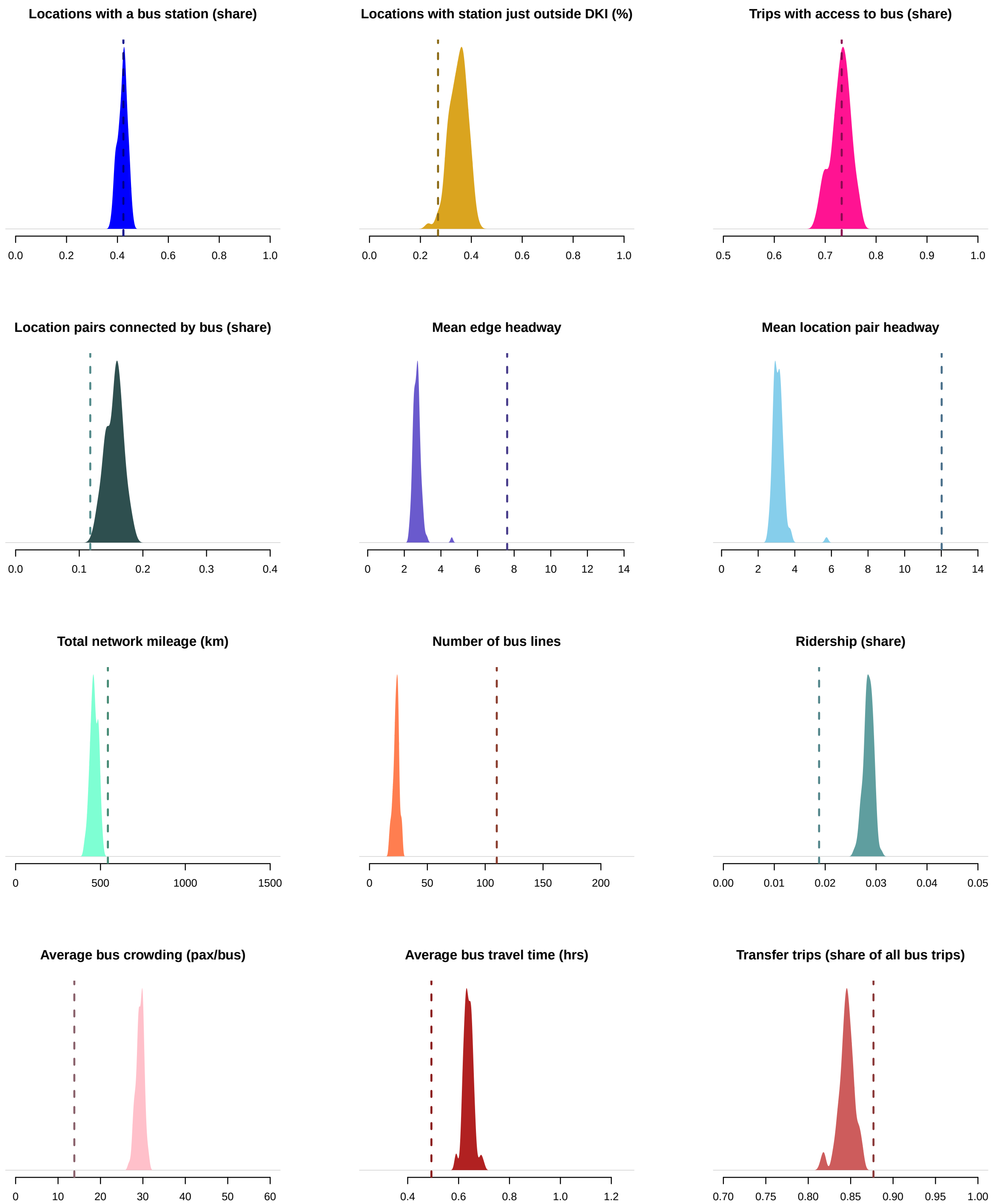
Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from 100 runs of the simulated annealing algorithm where preference parameters are allowed to vary for locations above and below median poverty. The planner puts equal weight on equivalent variation welfare expressed in travel time on the bus for the two groups. Bus crowding is computed based on joint preferences to ease computational burden.

Figure A.22: Distributions of Optimal Network Properties: Ridership as Objective



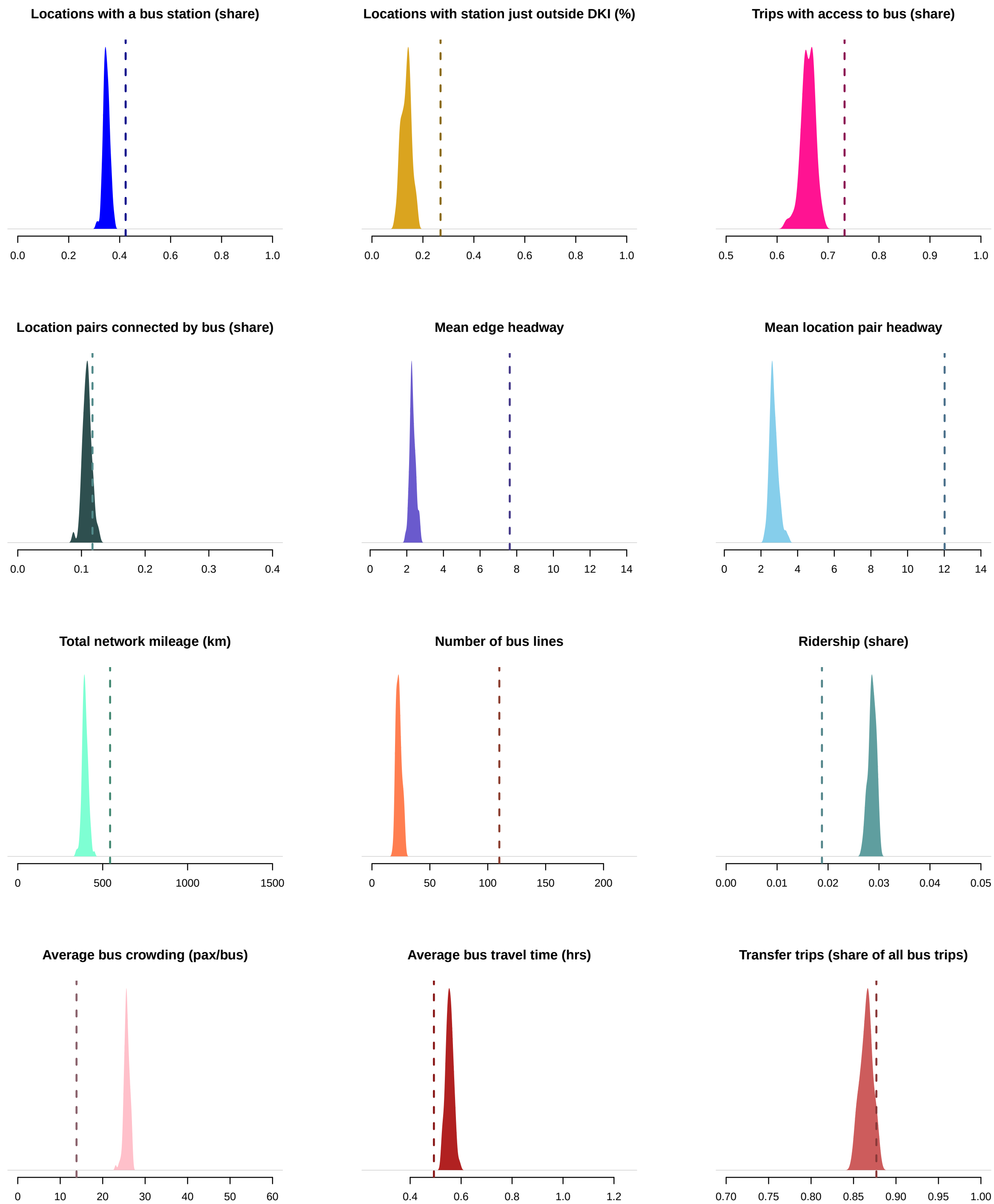
Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from 100 runs of the simulated annealing algorithm where the objective is total bus ridership instead of commuter welfare. The vertical line plots the measure for the current TransJakarta network for comparison.

Figure A.23: Distributions of Optimal Network Properties: Welfare Bias Towards Low Poverty Regions



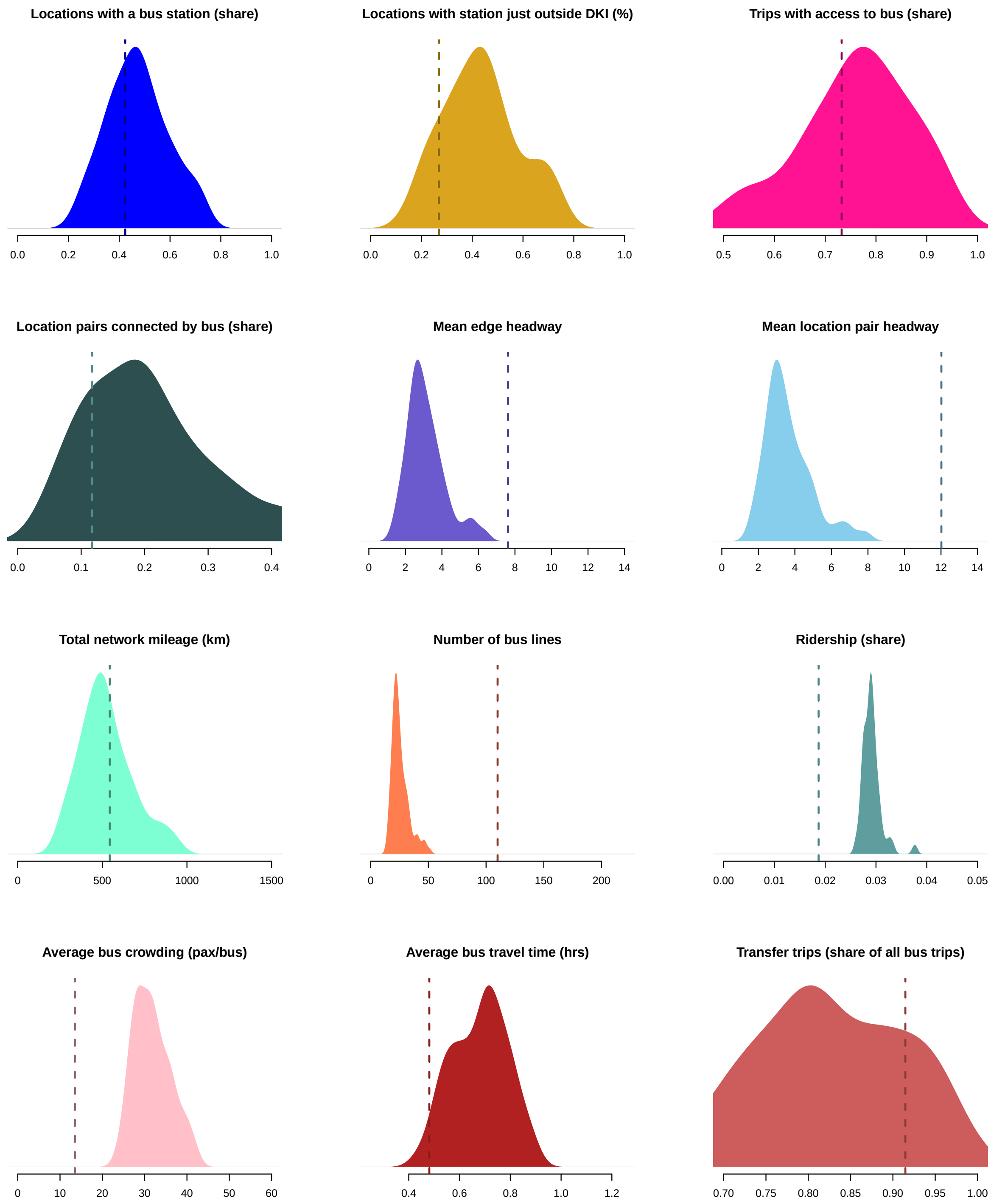
Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from 100 runs of the simulated annealing algorithm, in which the social planner is assumed to care more about low poverty regions. Grid cells with below-median poverty receive a welfare weight of 1.5×, while all grid cells above get a weight of 0.5×. The vertical line plots the measure for the current TransJakarta network for comparison.

Figure A.24: Distributions of Optimal Network Properties: Zero Welfare Weights Outside Central Jakarta (DKI Jakarta)



Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from 100 runs of the simulated annealing algorithm where we use a modified welfare function that puts zero welfare weights on commuters who live outside DKI Jakarta (the light blue region in maps). The vertical line plots the measure for the current TransJakarta network for comparison.

Figure A.25: Distributions of Optimal Network Properties with Bootstrapped Preference Parameters



Notes: Each graph plots the kernel density graph of a specific network characteristic, for the final networks from the simulated annealing algorithm with preference parameters drawn from the joint bootstrapped distribution. The vertical line plots the measure for the current TransJakarta network for comparison.

A.2 Appendix Tables

Table A.1: Number of Transjakarta Routes by Type

	(1) No. of routes
BRT	43
Non-BRT (inner city)	84
Non-BRT (outer city)	12
Royaltrans	13
Mikrotrans	45
Others	17
Total	214

Notes: A snapshot of the TransJakarta network in August 2019.

Table A.2: Robustness with 500m Square Grids: Impact of Network Expansion on Travel Time, Wait Times, and **Bus Ridership**

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.028***			-0.039			0.162***		
	(0.008)			(0.023)			(0.024)		
E2: New Direct Line (quicker)		-0.273***			0.069*			0.245***	
		(0.025)			(0.034)			(0.032)	
E3: Additional Busses			-0.018*			0.278***			0.115***
			(0.008)			(0.023)			(0.020)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	4,962,030	4,964,566	1,202,193	4,962,030	4,964,566	1,202,193	4,962,030	4,964,566	1,202,193
R^2	0.982	0.980	0.998	0.927	0.925	0.998			
Mean outcome	31.9	43.3	11.9	34.8	28.1	21.1	69.4	48.3	122.2
Median outcome	29.4	41.2	9.9	27.6	22.7	18.3	32.4	21.2	48.1
Unique origin x destination pairs	29,518	29,657	8,027	29,518	29,657	8,027	29,518	29,657	8,027
Unique origins	185	185	190	185	185	190	185	185	190

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.024*			0.296***			0.785***		
	(0.012)			(0.040)			(0.187)		
E2: New Direct Line (quicker)		-0.711***			0.225***			1.356***	
		(0.072)			(0.033)			(0.105)	
E3: Additional Busses			-0.086***			0.425***			0.359***
			(0.015)			(0.019)			(0.103)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	755,706	765,671	435,225	755,706	765,671	435,225	755,706	765,671	435,225
R^2	0.993	0.990	0.998	0.967	0.968	0.992			
Mean outcome	27.8	50.7	12.2	10.1	9.8	7.8	12.0	2.9	6.4
Median outcome	21.4	49.4	10.5	8.7	8.4	7.0	0.0	0.0	0.0
Unique origin x destination pairs	6,656	6,749	3,890	6,656	6,749	3,890	6,656	6,749	3,890
Unique origins	110	110	139	110	110	139	110	110	139

Notes: Version of Table 1 using 500-meters square grids. The two panels report results for BRT and non-BRT events, respectively. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.3: Robustness With 2000m Square Grids: Impact of Network Expansion on Travel Time, Wait Times, and **Bus Ridership**

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.017			-0.042			0.126***		
	(0.011)			(0.034)			(0.022)		
E2: New Direct Line (quicker)		-0.298***			0.064			0.151***	
		(0.036)			(0.049)			(0.029)	
E3: Additional Busses			-0.092**			0.328***			0.122***
			(0.028)			(0.068)			(0.036)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	676,039	672,111	205,889	676,039	672,111	205,889	676,039	672,111	205,889
R^2	0.984	0.979	0.996	0.940	0.938	0.996			
Mean outcome	37.5	46.3	16.2	48.5	38.7	21.4	465.3	384.7	841.3
Median outcome	36.7	45.1	12.6	39.9	29.2	18.3	315.3	246.5	459.1
Unique origin x destination pairs	3,868	3,869	1,405	3,868	3,869	1,405	3,868	3,869	1,405
Unique origins	68	68	70	68	68	70	68	68	70

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.038*			0.281***			0.547*		
	(0.017)			(0.057)			(0.222)		
E2: New Direct Line (quicker)		-0.655***			0.163*			1.660***	
		(0.119)			(0.060)			(0.437)	
E3: Additional Busses			-0.120**			0.389***			0.264
			(0.039)			(0.024)			(0.243)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	53,315	54,412	19,078	53,315	54,412	19,078	53,315	54,412	19,078
R^2	0.989	0.983	0.998	0.938	0.940	0.991			
Mean outcome	45.6	66.2	20.1	10.2	9.5	7.1	117.7	27.6	55.7
Median outcome	32.8	66.2	19.3	8.7	8.4	7.0	25.8	0.0	14.1
Unique origin x destination pairs	422	435	156	422	435	156	422	435	156
Unique origins	16	16	16	16	16	16	16	16	16

Notes: Version of Table 1 using 2000-meters square grids. The two panels report results for BRT and non-BRT events, respectively. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.4: Impact of Network Expansion on Travel Time, Wait Times, and **Bus Ridership**. Spatially Clustered Standard Errors.

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.036***			-0.051			0.158***		
	(0.008)			(0.031)			(0.025)		
E2: New Direct Line (quicker)		-0.294***			0.071			0.268***	
		(0.030)			(0.045)			(0.036)	
E3: Additional Busses			-0.027*			0.309***			0.092***
			(0.012)			(0.039)			(0.024)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	3,154,672	3,143,019	793,965	3,154,672	3,143,019	793,965	3,154,672	3,143,019	793,965
R^2	0.981	0.980	0.998	0.922	0.921	0.997			
Mean outcome	33.2	44.2	12.9	36.6	28.0	21.1	111.3	76.9	210.4
Median outcome	31.0	42.4	10.6	29.4	22.0	18.3	55.9	39.1	94.6
Unique origin x destination pairs	18,433	18,449	5,319	18,433	18,449	5,319	18,433	18,449	5,319
Unique origins	148	148	148	148	148	148	148	148	148

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.030***			0.294***			1.016***		
	(0.008)			(0.051)			(0.162)		
E2: New Direct Line (quicker)		-0.716***			0.208***			1.320***	
		(0.076)			(0.033)			(0.268)	
E3: Additional Busses			-0.089**			0.417***			0.450**
			(0.027)			(0.022)			(0.138)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	306,722	313,994	144,763	306,722	313,994	144,763	306,722	313,994	144,763
R^2	0.992	0.987	0.998	0.964	0.964	0.992			
Mean outcome	34.8	49.3	13.0	10.9	10.8	7.7	19.2	7.5	26.0
Median outcome	24.8	47.2	11.0	7.0	8.9	7.0	0.0	0.0	0.0
Unique origin x destination pairs	2,588	2,652	1,263	2,588	2,652	1,263	2,588	2,652	1,263
Unique origins	57	57	64	57	57	64	57	57	64

Notes: This table replicates Table 1, with standard errors clustered two-way at the level of 2km hexagonal grids at the origin and destination location. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.5: Robustness: Impact of Network Expansion on Travel Time, Wait Times, and Bus Ridership, **Excluding Observations after MRT Opening**

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.033***			0.022			0.189***		
	(0.009)			(0.032)			(0.038)		
E2: New Direct Line (quicker)		-0.352***			0.185***			0.362***	
		(0.034)			(0.051)			(0.064)	
E3: Additional Busses			-0.035**			0.315***			0.086***
			(0.012)			(0.034)			(0.020)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	2,212,872	2,200,389	530,810	2,212,872	2,200,389	530,810	2,212,872	2,200,389	530,810
R^2	0.988	0.987	0.998	0.941	0.940	0.997			
Mean outcome	33.8	45.1	12.9	36.4	25.8	21.5	113.1	75.7	211.9
Median outcome	31.4	42.9	10.6	29.3	19.1	18.3	55.5	35.8	92.0
Unique origin x destination pairs	16,820	16,834	4,360	16,820	16,834	4,360	16,820	16,834	4,360
Unique origins	148	148	148	148	148	148	148	148	148

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E1: New Direct Line	0.065**			0.299***			1.042***		
	(0.020)			(0.046)			(0.144)		
E2: New Direct Line (quicker)		-0.617***			0.204***			1.362***	
		(0.067)			(0.051)			(0.324)	
E3: Additional Busses			-0.088**			0.416***			0.093
			(0.026)			(0.024)			(0.124)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson
Obs.	175,779	179,716	82,418	175,779	179,716	82,418	175,779	179,716	82,418
R^2	0.994	0.993	0.999	0.970	0.970	0.993			
Mean outcome	41.1	52.9	13.3	12.8	11.3	7.7	23.4	6.5	9.3
Median outcome	24.9	51.8	11.5	7.0	8.9	7.0	0.0	0.0	0.0
Unique origin x destination pairs	2,283	2,344	923	2,283	2,344	923	2,283	2,344	923
Unique origins	57	57	63	57	57	63	57	57	63

Notes: Version of Table 1 excluding observations after March 12, 2019 – the day the sole Jakarta metro line (MRT) opened operation. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.6: Heterogeneity by Poverty Level: Impact of Network Expansion on Bus Ridership

	Bus Ridership					
	(1)	(2)	(3)	(4)	(5)	(6)
New Direct Line	0.215***	0.326***		0.972*	1.400**	
	(0.030)	(0.057)		(0.383)	(0.467)	
New Direct Line x High Poverty Origin	-0.096**	-0.053		-0.079	0.489	
	(0.032)	(0.061)		(0.366)	(0.417)	
E3: Additional Busses			0.074*			0.281
			(0.036)			(0.242)
E3: Add'l Busses x High Poverty Origin			-0.048			0.591**
			(0.034)			(0.229)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	3,154,672	3,143,019	793,965	306,722	313,994	144,763
Event type	BRT 1	BRT 2	BRT 3	non-BRT 1	non-BRT 2	non-BRT 3
Median outcome x Low Poverty	49.5	39.4	102.7	0.0	0.0	0.0
Median outcome x High Poverty	69.2	38.6	86.6	0.0	0.0	0.0

Notes: This table reports heterogeneity by whether the origin grid is high- or low-poverty for the bus ridership impacts in Table 1. We use poverty and population data at the kelurahan level from (SMERU, 2014) and PODES 2010, which we assign at the grid cell level based on surface area intersection. The interaction variable is an indicator for within-sample above median poverty rate. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.7: Robustness With 500m Square Grids: Impact of Network Expansion on Aggregate Trip Volume

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All Trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.047***			-0.134***			0.106***			-0.057		
	(0.006)			(0.027)			(0.018)			(0.119)		
E2: New Direct Line (quicker)		-0.212***			-0.122***			0.199***			0.046	
		(0.027)			(0.028)			(0.027)			(0.084)	
E3: Additional Busses			0.012			0.197**			0.094*			-0.007
			(0.032)			(0.060)			(0.040)			(0.314)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	1,850,929	1,854,359	310,496	1,850,929	1,854,359	310,496	1,850,929	1,854,359	310,496	1,850,929	1,854,359	310,496
R^2	0.988	0.986	0.999	0.949	0.949	1.000						
Mean outcome	31.6	39.7	11.8	35.3	34.1	15.9	83.4	53.8	98.3	237.6	176.5	572.1
Median outcome	28.7	38.6	10.1	30.0	29.4	17.4	38.6	27.8	63.0	0.0	0.0	0.0
Unique origin x destination pairs	21,813	21,852	3,960	21,813	21,852	3,960	21,813	21,852	3,960	21,813	21,852	3,960
Unique origins	156	156	123	156	156	123	156	156	123	156	156	123

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All Trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.023			0.278***			0.736***			-0.239		
	(0.015)			(0.039)			(0.216)			(0.177)		
E2: New Direct Line (quicker)		-0.738***			0.233***			1.276***			0.071	
		(0.084)			(0.035)			(0.147)			(0.129)	
E3: Additional Busses			-0.087***			0.353***			0.394**			-0.223
			(0.023)			(0.018)			(0.128)			(0.138)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	371,173	374,650	173,907	371,173	374,650	173,907	371,173	374,650	173,907	371,173	374,650	173,907
R^2	0.996	0.993	0.998	0.979	0.979	0.996						
Mean outcome	28.5	46.5	11.5	10.1	10.2	8.2	10.3	3.3	9.4	446.4	704.3	1238.0
Median outcome	22.1	46.7	10.1	8.7	8.4	8.4	0.0	0.0	0.0	0.0	0.0	0.0
Unique origin x destination pairs	4,728	4,770	2,481	4,728	4,770	2,481	4,728	4,770	2,481	4,728	4,770	2,481
Unique origins	90	90	110	90	90	110	90	90	110	90	90	110

Notes: This table replicates Table 2 using 500-meter square grids instead of 1000 hexagonal grids. The two panels report results for BRT and non-BRT events, respectively. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.8: Impact of BRT and Non-BRT Network Expansions on **Aggregate Trip Volume**. Spatially Clustered Standard Errors.

(a) BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All Trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.057***			-0.157***			0.108***			-0.008		
	(0.011)			(0.040)			(0.016)			(0.052)		
E2: New Direct Line (quicker)		-0.214***			-0.107**			0.208***			-0.073	
		(0.027)			(0.035)			(0.034)			(0.052)	
E3: Additional Busses			-0.006			0.174***			0.046			0.166
			(0.033)			(0.047)			(0.040)			(0.112)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039	1,184,781	1,172,746	197,039
R^2	0.986	0.985	0.999	0.948	0.948	0.999						
Mean outcome	32.6	42.0	13.4	38.5	33.9	14.4	122.9	93.7	195.5	1895.5	1729.5	10117.3
Median outcome	30.6	42.2	10.9	33.5	29.3	12.5	61.9	54.8	153.7	0.0	0.0	6412.4
Unique origin x destination pairs	13,802	13,675	2,497	13,802	13,675	2,497	13,802	13,675	2,497	13,802	13,675	2,497
Unique origins	124	124	89	124	124	89	124	124	89	124	124	89

(b) Non-BRT

	log Min Travel Time			log Bus/hr (origin)			Bus Ridership			All trips		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
E1: New Direct Line	0.028**			0.260***			0.999***			-0.042		
	(0.010)			(0.055)			(0.198)			(0.056)		
E2: New Direct Line (quicker)		-0.737***			0.205***			1.337***			0.031	
		(0.101)			(0.038)			(0.321)			(0.056)	
E3: Additional Busses			-0.097**			0.375***			0.425*			-0.177*
			(0.035)			(0.028)			(0.177)			(0.076)
Orig × Dest FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orig × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dest × Week FE Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Obs.	156,656	160,409	61,377	156,656	160,409	61,377	156,656	160,409	61,377	156,656	160,409	61,377
R^2	0.995	0.989	0.999	0.973	0.973	0.996						
Mean outcome	35.9	44.3	12.4	11.2	11.6	7.9	13.0	7.6	40.7	3948.6	6060.7	6880.5
Median outcome	24.5	43.6	10.8	7.0	8.9	7.1	0.0	0.0	0.0	635.5	956.9	1017.5
Unique origin x destination pairs	1,906	1,950	860	1,906	1,950	860	1,906	1,950	860	1,906	1,950	860
Unique origins	49	49	53	49	49	53	49	49	53	49	49	53

Notes: This table replicates Table 2, with standard errors clustered two-way at the level of 2km hexagonal grids at the origin and destination location. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.9: Demand Estimation Robustness and Extensions

	(1) Benchmark	(2) Poverty: Low	(3) Poverty: High	(4) Constant σ_{od}	(5) γ: Slope = 15	(6) γ: Slope = 45
BRT: Wait time $\alpha_{\text{wait}}/\alpha_{\text{time}}$	2.1	1.7	3.6	2.4	2.2	2.1
	[1.5, 3.4]	[1.3, 2.5]	[1.9, 6.5]	[2.1, 4.0]	[1.6, 3.7]	[1.5, 3.4]
BRT: Travel time α_{time}	1.0	1.0	1.0	1.0	1.0	1.0
	[1.0, 1.0]	[1.0, 1.0]	[1.0, 1.0]	[1.0, 1.0]	[1.0, 1.0]	[1.0, 1.0]
BRT: Transfer shifter μ/α_{wait} (minutes)	4.8	5.6	-0.2	6.2	4.3	4.9
	[0.8, 7.4]	[1.8, 7.5]	[-6.6, 3.3]	[1.3, 9.6]	[0.4, 7.2]	[0.9, 7.2]
BRT: Attention cutoff η	1.36	1.26	1.49	1.43	1.32	1.39
	[1.25, 1.46]	[1.18, 1.38]	[1.34, 1.86]	[1.34, 1.65]	[1.23, 1.43]	[1.27, 1.47]
Non-BRT: Wait time $\alpha_{\text{wait}}/\alpha_{\text{time}}$	3.41	2.51	5.08	4.52	3.53	3.40
	[2.72, 5.01]	[1.56, 4.67]	[3.15, 7.96]	[3.29, 8.58]	[2.74, 5.40]	[2.75, 5.02]
Non-BRT: Transfer shifter μ/α_{wait} (minutes)	2.22	3.84	-1.91	-0.02	1.69	2.33
	[-1.35, 4.57]	[-2.09, 10.34]	[-11.91, 1.12]	[-5.89, 4.12]	[-1.87, 4.08]	[-1.18, 4.24]
Non-BRT: Attention cutoff η	1.43	1.44	1.27	1.42	1.40	1.44
	[1.42, 1.44]	[1.44, 1.44]	[1.20, 2.56]	[1.41, 1.44]	[1.39, 1.42]	[1.43, 1.45]
Logit parameter σ	0.13	0.19	0.03	0.20	0.11	0.13
	[0.04, 0.32]	[0.07, 0.38]	[0.01, 0.07]	[0.05, 0.78]	[0.04, 0.30]	[0.04, 0.28]

Note: This table replicates the estimation from Table 4, column 2, under several different assumptions. The first column repeats the benchmark demand estimation. The second and third columns report estimates from the model restricted to above- and below-median poverty grid cells, respectively. The fourth column uses a constant $\sigma_{od} = \sigma$ instead of the normalization $\sigma_{od} = \sigma \frac{\overline{D}}{D_{od}}$. The fifth and sixth columns use the inattention slope in equation (4) set to 15 and 45, respectively, rather than 30 as in the main estimation. 95% confidence intervals in column 2-6 are based on re-estimating the model 200 times to match perturbed moments, see the notes for Table 4.

Table A.10: Moment Dependence on Parameters (Jacobian)

	Event 1 (α^{1B})	Event 2 (α^{2B})	Event 3 (α^{3B})
α_{Wait}	1.1	-2.0	0.2
α_{Time}	0.0	9.2	0.5
μ_{Transfer}	-0.1	-0.4	-0.0

Notes: The (i, j) entry in this matrix reports how the j-th moment varies locally in the model when the i-th parameter changes, namely $\sigma \partial m_j(\theta)/\partial \theta_i$, for moments $m_j \in \{\alpha^{1B}, \alpha^{2B}, \alpha^{3B}\}$ and parameters $\theta_i \in \{\alpha^{\text{BRT}}_{\text{wait}}, \alpha_{\text{time}}, \mu^{\text{BRT}}_{\text{transfer}}\}$. The partial derivative is scaled by the logit parameter σ.

Table A.11: Sensitivity Measure: Estimated Parameter Dependence on Moment Values

	Event 1 (α^{1B})	Event 2 (α^{2B})	Event 3 (α^{3B})
α_{Wait}	4.4	-0.5	25.0
α_{Time}	0.7	1.4	16.7
μ_{Transfer}	-3.5	-0.3	23.5

Notes: The (i, j) entry in this matrix reports the (Andrews et al., 2017) sensitivity measure $(\widehat{SE}_i)^{-1}\partial\widehat{\theta}_i/\partial m_j^{\text{data}}$ where the scaling factor $\widehat{SE}_i$ is the estimated standard error of $\widehat{\theta}_i$.

Table A.12: Optimal Network Characteristics: Robustness to Simulated Annealing Run Time

Statistic	Current Network	Baseline Optimal	One Day Run Optimal
Locations with a bus station (share)	0.42	0.45	0.45
Locations with station just outside DKI (%)	0.27	0.4	0.4
Trips with access to bus (share)	0.73	0.75	0.75
Location pairs connected by bus (share)	0.12	0.18	0.18
Mean edge headway	7.62	2.69	2.84
Mean location pair headway	12.02	3.056	3.26
Total network mileage (km)	543.6	464.01	485.13
Number of bus lines	110	21	23.44
Welfare (minutes travel time)	0	10.15	8.56
Ridership (%)	1.88	3.039	2.85
Average bus crowding (pax/bus)	13.84	31.37	30.081
Average bus travel time (hrs)	0.49	0.64	0.65
Transfer trips (share of all bus trips)	0.88	0.85	0.85

Notes: Comparison of characteristics of optimal networks to different run times of the simulated annealing algorithm. In our baseline results, we run the SA algorithm for 251,089 iterations (taking roughly 7 days of computing time). In some robustness runs, we run the SA algorithm for approximately 46,041 iterations (taking about 1 day). This table shows that networks achieved from the two approaches are generally similar, yet the shorter runs achieve slightly lower welfare and ridership.

Table A.13: Optimal Network Characteristics: Robustness to Planner's Parameter β

Statistic	Current Network	$\beta = 4 \times 10^2$	$\beta = 4 \times 10^3$	$\beta = 4 \times 10^4$	$\beta = 4 \times 10^5$
Locations with a bus station (share)	0.423	0.430	0.443	0.445	0.445
Locations with station just outside DKI (%)	0.269	0.380	0.398	0.400	0.400
Trips with access to bus (share)	0.732	0.739	0.750	0.751	0.752
Location pairs connected by bus (share)	0.117	0.163	0.175	0.176	0.176
Mean edge headway	7.618	2.840	2.920	2.926	2.926
Mean location pair headway	12.020	3.222	3.326	3.336	3.337
Total network mileage (km)	543.604	450.953	462.873	463.834	463.773
Number of bus lines	110	20.965	20.965	20.975	20.975
Ridership (share)	0.019	0.030	0.030	0.030	0.030
Average bus crowding (pax/bus)	13.836	30.817	31.296	31.357	31.376
Average bus travel time (hrs)	0.493	0.638	0.643	0.643	0.643
Transfer trips (share of all bus trips)	0.877	0.852	0.852	0.851	0.852

Notes: This table reports optimal networks characteristics for different values of β, the inverse standard deviation of planner shocks. Higher levels of β correspond to idiosyncratic shocks with lower variance.

Table A.14: Properties of Current and Optimal Networks with Bootstrapped Preference Parameters

Statistic	Network in 2016	Current Network	Optimal Networks	Difference Optimal – Current
Locations with a bus station (share)	0.17	0.42	0.47	0.049
				[0.12]
Locations with station just outside DKI (%)	0.0077	0.27	0.43	0.16
				[0.15]
Trips with access to bus (share)	0.43	0.73	0.76	0.027
				[0.11]
Location pairs connected by bus (share)	0.022	0.12	0.2	0.087
				[0.099]
Mean edge headway (minutes)	3.03	7.62	3.14	-4.48
				[1.047]
Mean location pair headway (minutes)	3.86	12.02	3.65	-8.37
				[1.31]
Total network mileage (km)	180.093	543.6	523.75	-19.85
				[157.42]
Number of bus lines	19	110	25.51	-84.49
				[7.36]
Ridership (%)	0.83	1.87	2.91	1.044
				[0.19]
Average bus crowding (pax/bus)	14.74	13.51	31.74	18.23
				[4.33]
Average bus travel time (hrs)	0.48	0.48	0.68	0.2
				[0.11]
Transfer trips (share of all bus trips)	0.87	0.91	0.83	-0.088
				[0.086]

Notes: This table replicates Table 5 using 100 bootstrapped runs of the optimisation algorithm, each with an independent draw from the joint distribution of estimated preference parameters from column 2, Table 4. In column 4 we report the mean and standard deviation of the difference between optimal and current networks, computed over all optimal network draws. See Figure A.25 for the distributions of properties of optimal networks.

Table A.15: Optimal Networks Local Comparative Statics

Statistic	Current Network	Baseline Optimal	Local derivative $dF^*/d\alpha_W$	Local derivative $dF^*/d\alpha_T$	Local derivative $dF^*/d\mu_T$
Panel A: Coverage measures					
Locations with a bus station (share)	0.42	0.45	0.0042	-0.001	-0.0017
		(0.0013)	[-0.022; 0.032]	[-0.039; 0.039]	[-0.017; 0.013]
Location pairs connected by bus (share)	0.12	0.18	-0.0048	-0.016	-0.0004
		(0.0011)	[-0.025; 0.018]	[-0.047; 0.017]	[-0.012; 0.011]
Total network mileage (km)	543.6	464.064	5.096	10.9	-3.56
		(1.66)	[-25.36; 38.58]	[-35.0041; 60.29]	[-22.79; 14.18]
Panel B: Speed measures					
Bus time relative to quickest possible bus route	1.33	1.24	-0.17	-0.36	-0.029
		(0.0085)	[-0.33; 0.0088]	[-0.63; -0.11]	[-0.12; 0.05]
Panel C: Directness measures					
Connected directly (share of all connected pairs)	0.21	0.15	-0.00019	-0.0014	-0.0034
		(0.00064)	[-0.01; 0.0098]	[-0.019; 0.015]	[-0.01; 0.0025]

Notes: This table reports how network shape measures (rows) change locally as a function of parameter changes (columns). It replicates Table 6 for local parameter changes, using the sample analogue of equation (19). Column 3 reports local changes to both $\alpha^{\text{BRT}}_{\text{wait}}$ and $\alpha^{\text{non-BRT}}_{\text{wait}}$, column 5 reports local changes to both $\mu^{\text{BRT}}_{\text{transfer}}$ and $\mu^{\text{non-BRT}}_{\text{transfer}}$. Derivatives are scaled down by a factor of 10^8 to aid legibility. Note that a positive change in the transfer shifter corresponds to a *lower* transfer penalty. 90% bootstrapped confidence intervals (columns 3-5) in parentheses.

A.3 Data Processing

A.3.1 Bus GPS Data Processing

This section explains how we process bus GPS data to find bus station arrival times, which we use when describing the wait time distribution and when assigning bus transactions to bus stations. We use GPS data every 5 or 10 seconds available for most TransJakarta buses between January 2017 and March 2020. We also use bus trip logs entered manually by bus dispatchers. For each trip, this data contains the bus code, the bus route code (with direction), and the trip start time. **2,798** TransJakarta buses appear in the GPS data.

Combined data on GPS and trip logs is significantly better starting in 2018. In 2017, **22.2%** of bus days contain both GPS data and trip logs, **76.2%** contain only GPS data but no logs, and the rest contain only trip logs without GPS data. However, in 2018 - March 2020, **70.7%** of bus days contain both GPS data and trip logs, **22.8%** contain only GPS data but no logs, and the rest contain only trip logs without GPS data.

We developed three algorithms to identify when a bus arrives at a given bus station. When both GPS data and trip logs are available, we map match the bus GPS locations to the path of the bus route (from separate data), starting from the trip start time recorded in the trip log, and find arrival times for all bus stations along that route. (The algorithm automatically identifies bus trips where the "return" trip log is missing, which happens **15.4%** of the time.)

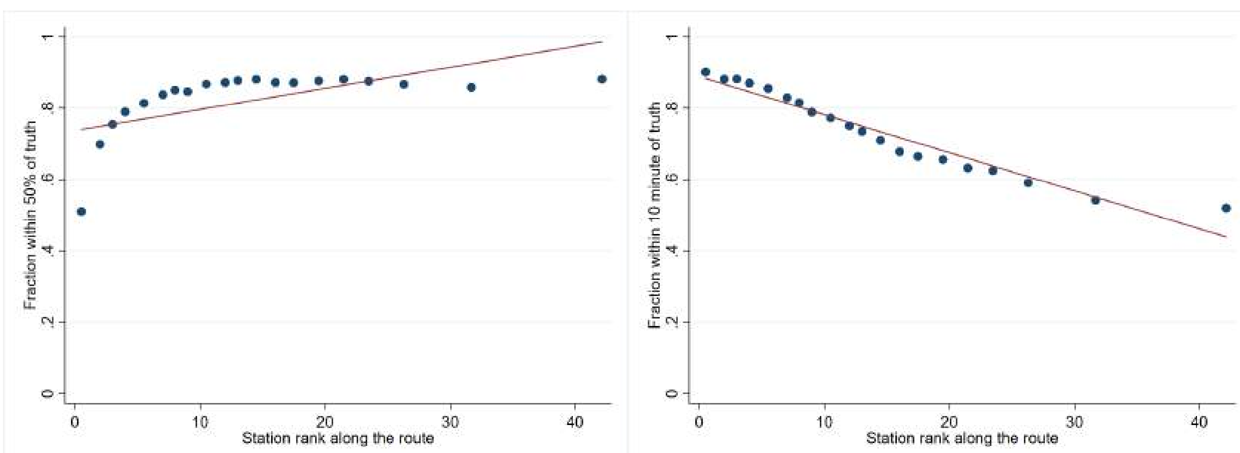
When only GPS data is available, the algorithm proceeds in two steps. First, given a bus and a date, it ranks bus routes in decreasing order of overlap with the GPS traces for that bus day and generates a set of candidate routes where the traces overlap at least **30%** with the route. Second, the algorithm map matches the GPS traces to the best-fit bus route, trying multiple candidates, starting from the first GPS point that is near to the first station of a candidate bus route.

When only trip log data is available, the algorithm proceeds in two steps. First, we predict *station* arrival time as a function of trip start time, bus route, and time of the day. We estimate this model using the output of the first algorithm for the same route, on bus-days when both GPS and trip log data is available. Figure A.26 shows that we achieve high accuracy using this model. Second, we apply these predicted times using the trip start time from the log.

Figure A.26: Accuracy of Predicting Station Arrival Time Based Only on Trip Start Time

(a) Relative Error

(b) Absolute Error



We end up with a total of **7,315,854** trips out of which **5,568,890 (76.1%)** are identified when both GPS data and trip logs are available, **859,776 (11.8%)** are identified when only GPS data is available and the remaining **12.1%** are identified when only trip log data is available.

A.3.2 Bus Travel Times

In this section, we describe how we compute travel times between an origin station o, a destination station d, along a route r. These travel times are used in the reduced form analysis to compute bus route travel times for defining Events 1 and 2, and in the model to characterize the bus network choice set of any given traveler.

For every triplet (r, o, d) we consider all trips along r and the time they take to go from o and d. We then take the median travel time within this set, over all trips between 7 AM and 7 PM in our study period.

Figure A.27 shows that there is only a very small amount of variation in "delay" (median travel time per kilometer, or inverse speed) for trips starting at different times of the day between 7 AM - 7 PM. The variation is even smaller for BRT routes. Moreover, we can also observe that delay is mostly stable over the years in our sample. This supports our choice of using medians of travel times of trips starting between 7 AM - 7 PM, over the entire data period (January 2017 - March 2020). Travel time and delay in different years are strongly correlated after including route, origin and destination fixed effects (Table A.16).

Figure A.27: Bus Travel Delay by Departure Time and Year

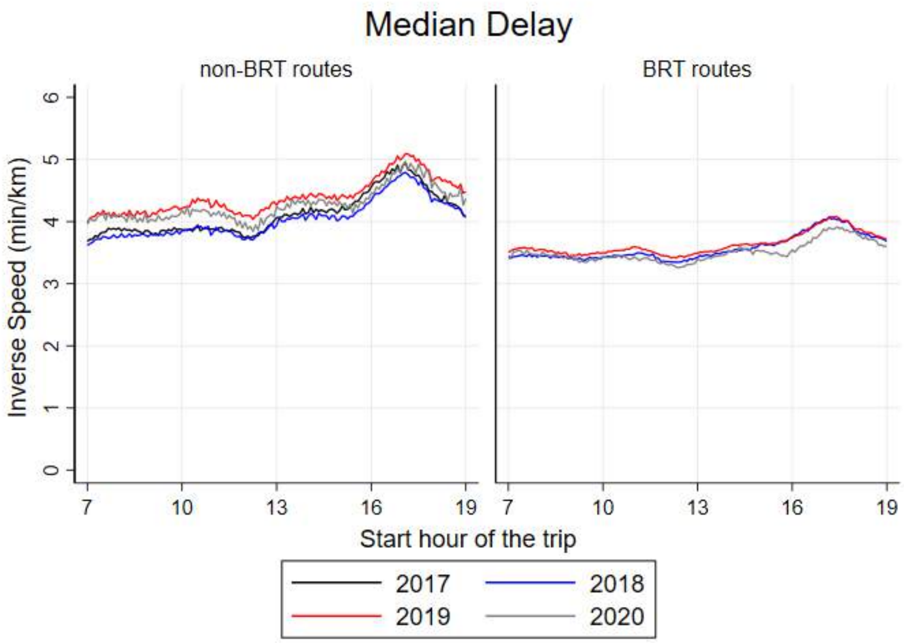


Figure Notes: This figure reports median delay (inverse speed, in minutes per km) by departure time, separately for BRT routes and for non-BRT routes using data from January 2017 - March 2020 for routes that were active throughout this period. To construct this figure, we assign each trip to 5-minute bins by their departure time at first station on the route. We then find the time taken by the bus to complete each trip and the distance traveled for the entire route (start to end). Using this, we calculate the inverse speed for each trip (time/distance). Within each 5-minute bin, we finally plot the median of inverse speed of trips throughout the day, grouping the trips by year.

Table A.16: **Bus Travel Time Correlation Across Time**

	Log(Median Delay) 2017	Log(Median Delay) 2018	Log(Median Travel Time) 2017	Log(Median Travel Time) 2018
	(1)	(2)	(3)	(4)
Log(Median Delay) 2019	0.716***	0.794***		
	(0.048)	(0.038)		
Log(Median Travel Time) 2019			1.031***	1.009***
			(0.007)	(0.002)
Constant	0.368***	0.222***	-0.193***	-0.083***
	(0.062)	(0.047)	(0.043)	(0.018)
R^2	0.476	0.685	0.961	0.983
N	26990	38624	26990	38624

Table Notes: We organize the data at the route-origin-destination level. The outcomes variables are the log of median delay (inverse speed) and log of median travel time. Standard errors are clustered three-way by route, origin and destination, and reported in parentheses: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A.3.3 Bus Wait Time Distribution

In the model, we assume that wait times are exponentially distributed. In this section we analyze how wait times for different TransJakarta routes are distributed in reality, using the GPS data from 2019.

We first calculate bus headways (difference in minutes between two consecutive buses) for every station, route, and direction. Then, we calculate the frequency of headway occurrence by route and minute. Assuming that TransJakarta passengers are "non-planning" and thus arrive at bus stations at a (locally) constant rate, we calculate the implied histogram of wait times by taking the reverse cumulative total of headway frequency. We compute the wait time distribution for 45 routes (18 BRT and 27 non-BRT) that have only two main route versions (i.e. where the top two trip variants account for > 95% of all trips on that route).

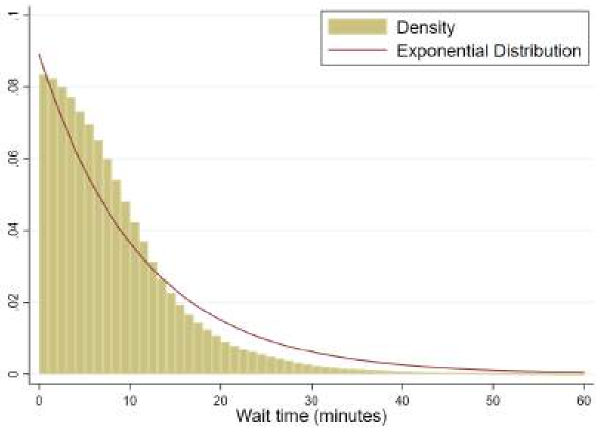
Panels A and B in Figure A.29 plot the wait time distribution for non-BRT route 1E. The wait time distribution is approximately exponential, except for wait times over 30 minutes, where the empirical distribution puts lower weight than the exponential. Panel B plots results for all the routes in our sample, showing that there is a tight linear relationship between a route's mean wait times and the coefficient of variance of wait time. BRT and non-BRT routes share the same relationship. In other words, BRT routes are not less (or more) uncertain than non-BRT routes, conditional on average wait time.

Figure A.28: Sample Arrival Time Monitor at a TransJakarta Station

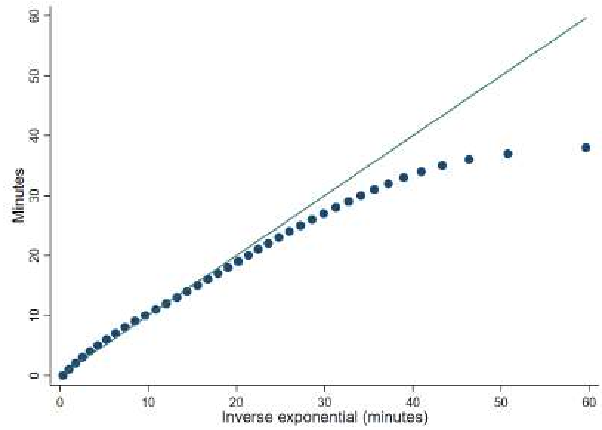
Notes: During the study period, all BRT bus stops were equipped with screens that displayed real-time estimated bus wait times. (The estimates were based on real-time bus GPS location data.)

Figure A.29: Wait Time Distribution Within and Across Routes

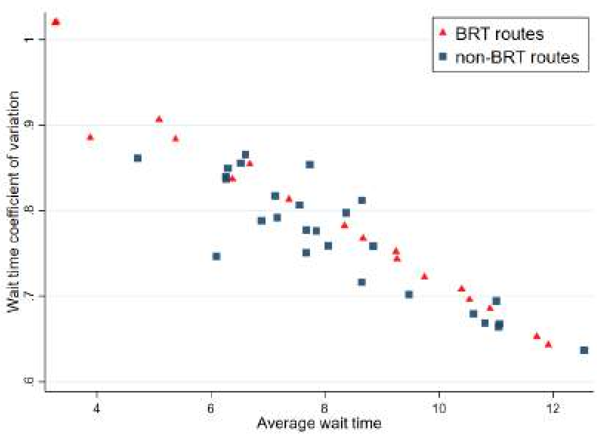
(a) Wait Time Distribution (route 1E)



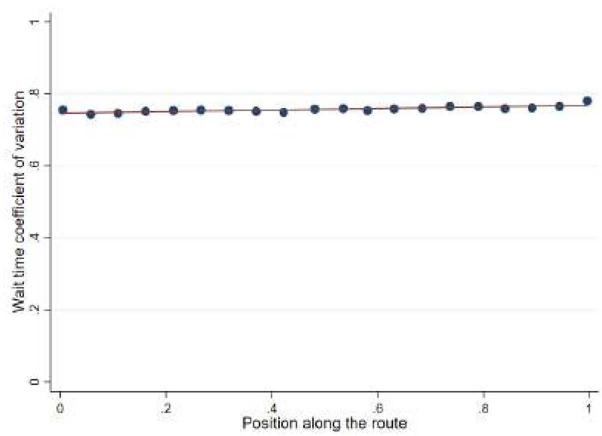
(b) Wait Time Q-Q Plot (route 1E)



(c) Wait Time Reliability on BRT and Non-BRT Routes



(d) Within-Route Reliability



Notes: Panel (A) and (B) report the wait time distribution for non-BRT route 1E. To create these graphs, we calculate bus headways (defined as the duration in minutes between two consecutive buses of the same route and direction at a particular station) using the bus GPS data. To obtain the wait time distribution from the headway distribution, we assume that passengers arrive at the station at a constant rate. In panel (A), we fit an exponential distribution to the empirical wait time distribution. Panel (C) reports the mean of wait time (in minutes) and the coefficient of variation of wait time, separately for BRT routes and for non-BRT routes. Panel (D) reports the coefficient of variation of wait time over position along the route (0 for first station, 1 for last station). For all panels, we use data from January to October 2019 and restrict the sample to only include arrival times during morning peak hours (8 AM-12 PM). Each Transjakarta route may have several route variants that differ slightly in length and stations reached. Therefore, for panels (C) and (D), we restrict the sample to routes where two trip variants account for above 95% of all trips on that, consisting of 45 routes (18 BRT and 27 non-BRT routes).

A.3.4 TransJakarta Ridership Data Processing

We use two main TransJakarta ridership data sources. First, we use transactions ("taps") in BRT bus shelters, where passengers pass through turnstiles to enter the bus shelter. In theory, passengers also need to "tap out" when they exit a bus shelter. This is only enforced in **35.9**% of bus shelters (accounting for **34.0**% of shelter ridership).

Second, we use transaction data from non-BRT buses. When a passenger gets on the bus from a (non-BRT) bus station on the side of the road, they pay inside the bus using their card or cash. When a passenger uses cash, the bus attendant uses their own card on the card reading machine. In 2019, 57% of transactions were cash. To assign these transactions to bus station, we use the bus station arrival time from the GPS or trip log data (section A.3.1).

Origin-Destination Ridership Flows. To construct origin-destination ridership flows at each point in time, we proceed in several steps. We focus on a sample of cards for which we observe ridership behavior over time. At each step, we construct weights assuming that the sample of cards and transactions that we use is representative.

First, we drop "administrative" cards that are likely used by bus attendants or other TransJakarta employees. We label a card as administrative on a given day if it is used repeatedly throughout the day on the same bus (for non-BRT transactions), or at a BRT shelter. Administrative cards account for **2.1%** of all BRT shelter transactions, and **63.2%** of all non-BRT bus transactions. We assume that travel behavior for non-admin cards is representative of all trips in the system.

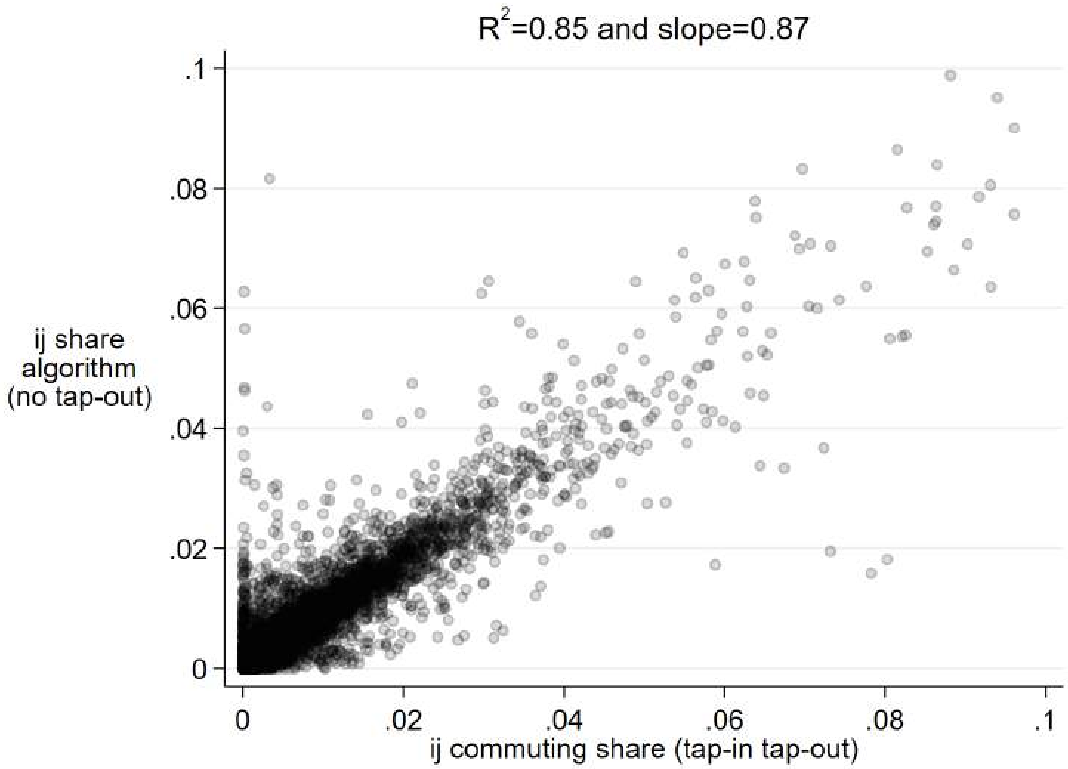
Second, we process "serial" taps. We combine consecutive taps using the same card into a single transaction, likely capturing groups traveling together and using a single card. **8.1%** of transactions have two or more consecutive taps.

We exclude 15 non-BRT routes from ridership coding for reasons related to the way payments were made. Each non-BRT bus accepts payments with smart cards provided by a specific bank. We lack detailed micro-transaction data corresponding to one of the banks. Overall, tap data from dropped routes accounts for 14.96% of total tap data in 2017-2020. Coverage is 44.11% of total routes in 2017 and 72.14% in 2020.

Algorithm to Infer Trip Destination and Validation. Ideally, we want to determine a commuter's origin-destination based on the tap-in and tap-out locations of their trip. However, tap-out information is not always available across all routes and stations in TransJakarta. Commuters were not required to tap-out on non-BRT routes, and tap-out was enforced only at some BRT bus stops. Given these limitations, we developed an algorithm to infer a commuter's destination for each ridership tap-in. The algorithm consists of three main rules for assigning a trip's destination. The first rule is to use the actual tap-out location as the destination, when the tap-out occurs within 4 hours after the tap-in. For the second rule, we proxy for the destination using the next trip's tap-in, if that next trip takes place on the same day or the next day. For the third rule, for each smart card, each month, we computed the top 2 stations with the most taps. If the trip origin is one of the top 2 stations, we proxy for the trip destination using the other top 2 stations. This method is only applied for frequent commuters, defined as those with more than 10 taps in a month and more than 75% of total taps located within its top-2 stations.

We validate the algorithm by comparing ridership flow destination shares originating from station i towards station j (destination) using data with only tap-in entry (algorithm) and data with both tap-in and tap-out entry (actual). Figure shows that the two measures are highly correlated.

Figure A.30: Comparison of Actual and Inferred (o, d) Ridership Shares



A.3.5 Veraset Smartphone Location Data Trip Processing

We use the smartphone trip processing algorithm from Kreindler (2024) to convert raw GPS data into individual trips and common location for each device in the data.

For each individual i, the algorithm first classifies individual trips and "stays," periods of time when the device is observed to be stationary in a given area. Then, the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is used to cluster all locations for any given devices. The most common cluster is labeled as "home."

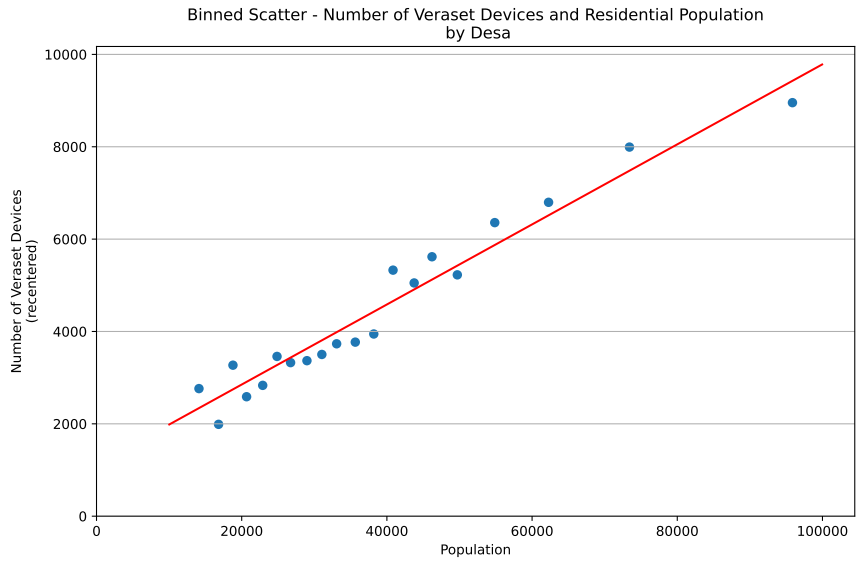
The sample of trips used in the analysis is all weekday trips starting and ending at a known locations (i.e. locations classified by DBSCAN as belonging to a cluster), excluding trips starting before 5AM or after 11PM. We also drop trips that are unusually long or short (in both duration and distance), "swiggly" trips (where ratio of the largest distance between any two points on the trip to path length is less than 0.3), and "short loop" trips (trips that are less than 2km where the ratio of origin-destination distance to path length is less than 0.3).

Given the selection concerns when using smartphone location data (Blanchard et al., 2021), we examine how representative the users in our data are of the general Jakarta population. First, we compare the distribution of users' home locations obtained from the data to that of residential population from the PODES survey (Figure A.31). The number of devices in our data with home locations in each desa is correlated with populations in the desa, suggesting that distribution of Veraset devices is consistent with population distribution. However, the coverage of our data (number of devices per total population) is slightly lower in areas with higher population density and proportion of population under poverty.

We construct two main data sets using the Veraset trips data. First, we construct a panel of trip flows at the origin grid by destination grid by week level. Each week, for each device in the data, we re-weight all its weekday trips that week to represent a single typical weekday. (This is essentially using inverse weights given by the number of weekdays when we observe the decide, except that we also account for partial days.) We also use a single overall adjustment factor such that the sum of all device weights equals approximately 14 million, the number of individuals over 15 years old in the study area. To construct trips, we consider any individual-day with sufficient GPS data frequency and include all the trips that can be identified in the data on that day, and use the trip detection algorithm from Kreindler (2024).

Second, we construct a cross-section of typical (o, d) trip flows throughout the period. We apply the same procedure as for the panel, except that we first pool all data for the entire two year period.

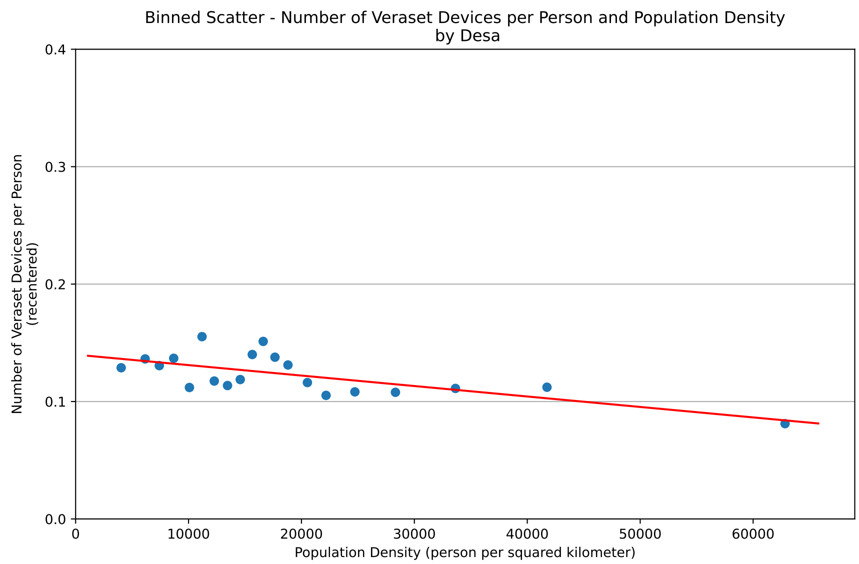
Figure A.31: Binned Correlation Between Veraset Devices and Residential Population



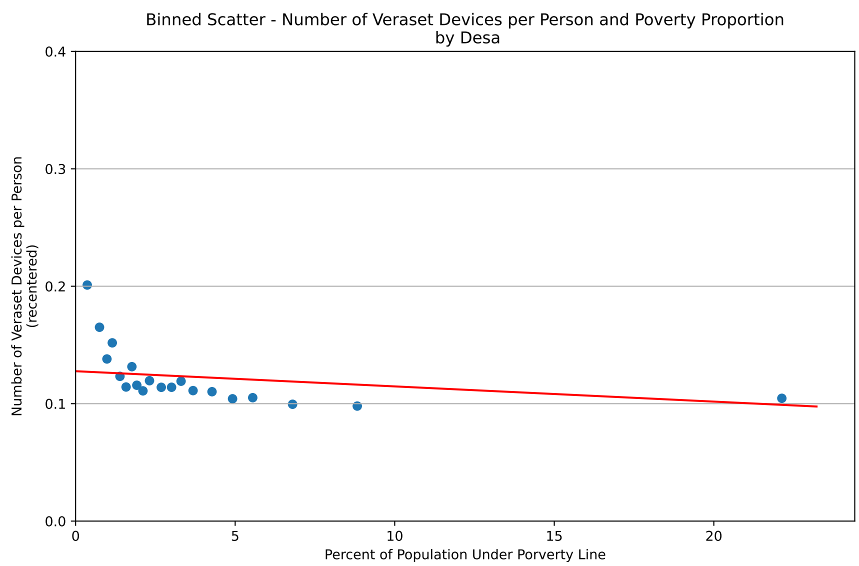
Notes: Each observation is an urban neighborhood (kelurahan or desa), $N = 538$. Population data is from the 2011 PODES survey, the most recent source for population at this level of geographical detail. For each desa, we count all devices which have their "home" location assigned inside the desa.

Figure A.32: Representativeness of the Veraset Trip Data

(a) Population Density



(b) Poverty



Notes: Binscatter plots analyzing the representativeness of the Veraset trip data at the kelurahan level. The sample excludes kelurahans below the 10th percentile of the distribution of PODES population. These locations tend to represent commercial or leisure areas. In these locations, the ratio of Veraset to PODES population is high. Panel A compares the ratio of Veraset home locations in a kelurahan to the kelurahan's population density. Panel B uses a poverty indicator from SMERU (2014).

A.4 Reduced Form Estimation

This section describes the differences-in-difference specifications.

For BRT events, we define $Post^{iB}_{odt}$, $i = 1, 2$, to be a time-varying indicator for the new direct connection between BRT grid cells o and d having been switched on in the past 10 months.[69] The coefficient α^{iB} in equation 1 captures the overall effect of a new direct connection being added between o and d in the first 10 months after its launch. The sample of observations is all odt that have a direct or transfer connection between BRT grid cells o and d, and o and d each have at least one treated observation.[70]

For non-BRT events, $Post^{iN}_{odt}$, $i = 1, 2$, is a time-varying indicator for the new direct connection between grid cells o and d having been switched on in the past 10 months. We restrict to origin grids o that are never BRT, o and d are connected either directly or by transfer, and o and d each have at least one treated observation. (In the latter case, the first leg is necessarily non-BRT, while the second leg can be BRT or non-BRT.)

In Event 3, grid-cell pairs o and d that are already directly connected get more buses from a new route because it overlaps with the existing route for the portion between o and d. Specifically, $Post^{3}_{odt}$ is an indicator for the first event of an additional direct route launched between o and d taking place, in the ten months before week t.

A positive coefficient α^{3N} captures the degree to which ridership or all trips between o and d increases after more buses are added to the route between o and d due to an additional direct route.[71] The sample is all odt such that BRT grid cells o and d are directly connected at time t, o and d each have at least one treated observation, and o and d are only connected by direct connections before the event.[72]

For non-BRT events, $Post^{3N}_{odt}$ is defined analogously. The sample is all odt that are connected directly at time t, o is never a BRT grid cell, o and d each have at least one treated observation, and o and d are only connected by direct routes before the event.

[69] We impose that the origin and destination grids have BRT stations, while the new route may be a non-BRT route traveling between these locations. It is often the case that non-BRT route travel along BRT corridors and stop at BRT stations for a portion of the route.

[70] This sample includes o, d pairs that are treated at some point, as well as o, d pairs that are never treated, yet there exist o', d' such that o, d' and o', d are treated.

[71] As for event types 1 and 2, the route itself can be non-BRT as long as it passes through BRT grid cells o and d.

[72] We make this last restriction to focus on the case where the change in the choice set is very simple. Results where we include o, d pairs that have both direct and transfer connections before the event are similar.

A.5 Attention Probabilities

We incorporate partial inattention by assuming that the agent notices each arrival from option k with independent probability p_k. In other words, with probability $1-p_k$ the agent fails to notice the first arrival for option k. This leads to an "effective" arrival rate for option k of $\widetilde{\lambda}_k = \phi(p_k)\lambda_k$ with $0 \leq \phi(p) \leq 1$. $\phi(p)$ is given by $-\frac{p\log(p)}{1-p}$, which is increasing and concave in p, $\phi(0)=0$, and $\phi(1)=1$. To see this, note that the effective arrival rate is a weighted sum of the first arrival rates, the second arrival, etc. Dropping the k subscript, we can write:

$$
\begin{aligned}
\widetilde{\lambda} &= p\lambda + (1-p)p\frac{\lambda}{2} + (1-p)^2 p\frac{\lambda}{3} + ... \\
&= p\lambda(\sum_{k=0}^{\infty} k^{-1}(1-p)^k) \\
&= -\frac{p\log(p)}{1-p}\lambda.
\end{aligned}
$$

The expression on the second line is the Mercator series for $p-1$.

A.6 Demand Model Derivations

Proof of Proposition 2. Here we derive expressions for the probability π_k to choose option k, and for expected utility $\mathbb{E}\max_k u_k$. We also derive expressions for expected travel time and expected wait time.

In general, assume that we have independent random variables $X_1, X_2, \ldots, X_N$, then

$$
\Pr(k \in \arg\max_j X_j) = \int_{-\infty}^{\infty} f_k(x) \prod_{i\neq k} F_i(x)dx, \tag{12}
$$

where f denotes a pdf function, and F denotes a cdf function. Expected utility is

$$
\mathbb{E}\max_k X_k = \int_{-\infty}^{\infty} x \sum_k f_k(x) \prod_{i\neq k} F_i(x)dx. \tag{13}
$$

The probability density that x is the maximum is the sum over k that the k-th options is $X_k = x$ and that all other options are strictly lower than x.

In our model, have $u_k = v_k - \alpha_{\text{wait}} w_k$ where w_k is exponentially distributed with parameter λ_k. Assume that $\alpha_{\text{wait}} = 1$, if necessary replacing all λ_k by $\lambda_k/\alpha_{\text{wait}}$. The pdf and cdf for exponential variables are given by

- $F_k(u) = \exp(\lambda_k(u - v_k))$ for $u \leq v_k$ and 1 above, and

- $f_k(u) = \lambda_k \exp(\lambda_k(u - v_k))$ for $u \leq v_k$ and 0 above.

Choice Probabilities. Assume that options are ranked such that $v_1 < v_2 < \cdots < v_N$. Replacing the pdf and cdf in (12) and separating the integral by intervals delimited by the v_k's, the probability that option k is optimal is:

$$
\begin{aligned}
\pi_k = \Pr(k \in \arg\max) &= \int_{-\infty}^{\infty} f_k(u) \prod_{i \neq k} F_i(u) du \\
&= \lambda_k \int_{-\infty}^{v_1} e^{\lambda_k(u-v_k)} \times \prod_{i \geq 1, i \neq k} e^{\lambda_i(u-v_i)} du \\
&+ \lambda_k \int_{v_1}^{v_2} e^{\lambda_k(u-v_k)} \times \prod_{i \geq 2, i \neq k} e^{\lambda_i(u-v_i)} du \\
&\dots \\
&+ \lambda_k \int_{v_{k-1}}^{v_k} e^{\lambda_k(u-v_k)} \times \prod_{i > k} e^{\lambda_i(u-v_i)} du.
\end{aligned}
$$

(For any $u > v_k$, the probability that k is optimal is zero.)

We use the notation: $\Lambda_i = \sum_{\ell \geq i} \lambda_\ell$ and $M_i = \sum_{\ell \geq i} \lambda_\ell v_\ell$. We have

$$
\begin{aligned}
\lambda_k^{-1} \pi_k &= e^{-M_1} \int_{-\infty}^{v_1} e^{u\Lambda_1} du + \cdots + e^{-M_k} \int_{v_{k-1}}^{v_k} e^{u\Lambda_k} du \\
&= \sum_{i=1}^{k} e^{-M_i} \int_{v_{i-1}}^{v_i} e^{u\Lambda_i} du \\
&= \sum_{i=1}^{k} e^{-M_i} \frac{e^{v_i \Lambda_i} - e^{v_{i-1}\Lambda_i}}{\Lambda_i},
\end{aligned}
$$

were we use the convention $v_0 = -\infty$.

Expected Utility. Plugging the exponential pdf and cdf formulae in (13) we get

$$
\begin{aligned}
\mathbb{E} \max_k u_k &= \sum_{i=1}^{N} \int_{v_{i-1}}^{v_i} u \sum_{k \geq i} f_k(u) \prod_{j \geq i, j \neq k} F_j(u) du \\
&= \sum_{i=1}^{N} \int_{v_{i-1}}^{v_i} u \sum_{k \geq i} \lambda_k \exp\left(\sum_{j \geq i} \lambda_j (u - v_j) \right) du.
\end{aligned}
$$

The $\exp(\cdot)$ term can be factored out of the sum as it does not depend on k, so we get

$$\begin{aligned}
\mathbb{E}\max_k u_k &= \sum_{i=1}^{N} \int_{v_{i-1}}^{v_i} u\Lambda_i e^{u\Lambda_i - M_i} du \\
&= \sum_{i=1}^{N} \Lambda_i^{-1} e^{-M_i} \int_{v_{i-1}}^{v_i} u\Lambda_i e^{u\Lambda_i} d(u\Lambda_i) \\
&= \sum_{i=1}^{N} \Lambda_i^{-1} e^{-M_i} \left[e^{\Lambda_i v_i}(\Lambda_i v_i - 1) - e^{\Lambda_i v_{i-1}}(\Lambda_i v_{i-1} - 1) \right] \\
&= \underbrace{\sum_{i=1}^{N} e^{-M_i} \left[e^{\Lambda_i v_i} v_i - e^{\Lambda_i v_{i-1}} v_{i-1} \right]}_{v_N} - \underbrace{\sum_{i=1}^{N} \Lambda_i^{-1} e^{-M_i} \left[e^{\Lambda_i v_i} - e^{\Lambda_i v_{i-1}} \right]}_{\lambda_N^{-1}\pi_N}
\end{aligned}$$

The first sum is telescopic and evaluates to $e^{-M_N + \Lambda_N v_N} v_N = v_N$. This concludes the proof of part 2 in Proposition 2.

Expected Travel Time and Expected Wait Time. Travel time is non-random, so expected travel time is given by:

$$\mathbb{E}T_k^{\text{time}} \equiv \mathbb{E}\left(T_k^{\text{time}} \mid k \in \arg\max\right) = \sum_k \pi_k T_k^{\text{time}}. \tag{14}$$

By a similar argument, $\mathbb{E}v_k = \sum_k \pi_k v_k$. We use this result to derive expected wait time:

$$\begin{aligned}
\mathbb{E}u_k &= \mathbb{E}v_k - \alpha_{\text{wait}} \mathbb{E}T_k^{\text{wait}} \\
v_N - \pi_N \alpha_{\text{wait}}/\lambda_N &= \sum_k \pi_k v_k - \alpha_{\text{wait}} \mathbb{E}T_k^{\text{wait}} \\
\Rightarrow \mathbb{E}T_k^{\text{wait}} &= \pi_N \lambda_N^{-1} - \alpha_{\text{wait}}^{-1} \left(\sum_k \pi_k (v_N - v_k) \right).
\end{aligned}$$

A.7 Optimal Network Design

A.7.1 Optimization Environment: Predicted Bus Travel Times

In this section, we describe how we estimate bus travel times for every edge in the grid cell environment, in a manner that makes these edge costs consistent with our entire GPS data on bus travel times. The key challenge is that certain origin-destination pairs are not connected by bus in the current network.

We proceed in two steps. First, we use data that we collected on driving times between every pair of grid cells, and we project bus log travel time – for those origin-destination-route combinations available – onto log driving time for that origin-destination pair. Second, we estimate edge-specific bus travel times that "micro-found" these predicted bus travel times. We estimate edge times in the routing model from Allen and Arkolakis (2022), which allows us to express the predicted bus travel time between any origin and destination as the noisy shortest route in our grid cell environment network. We repeat this exercise separately for BRT and non-BRT.

Step 1. Predicted bus travel time for all origin-destination pairs. We obtain driving travel times between tens of thousands of pairs of locations in Jakarta from a commercial provider of route data derived from smartphones and other GPS-enabled devices. We obtained data for the entire Jakarta region in the year 2020.[73] We then spatially interpolate this data to construct driving time T_{od}^{drive} between every origin and every destination.

We then predict bus travel times between all (o, d) pairs. We start with our data on median bus travel times T_{odr}^{bus} between o and d on route r (only for o, d pairs where this data is available). We then estimate the following linear regression

$$\log(T_{odr}^{\text{bus}}) = \alpha_0 + \alpha_1 \log(T_{od}^{\text{drive}}) + \epsilon_{odr}.$$

We then construct the prediction $\widehat{T}_{od}^{\text{bus}}$ for all origin-destination pairs, separately for BRT and non-BRT bus travel times.

Step 2. Estimate edge-level bus travel times. In our counterfactual simulations, we need to predict bus travel time for routes following any path. The model primites are *edge-specific* bus travel times, which we sum up over any possible route. We now describe how we estimate edge-specific bus travel times using our data on predicted bus travel times for all origin-destination pairs.

Given that we do not have specific bus routes for $\widehat{T}_{od}^{\text{bus}}$, we model these travel times as noisy shortest routes over the underlying network (Allen and Arkolakis, 2022).

For every edge ij in the grid cell network – this includes horizontal, vertical and diagonal neighboring grids – denote by T_{ij} the bus travel time parameter in the model. We denote by p a path between o and d, and by T_p the sum of edge-wise travel times along the path. Assume that a bus commuter traveling from o to d can choose any finite path between o and d and faces realized travel time $\widetilde{T}_p = T_p + \epsilon_p$, where ϵ_p is an iid Gumbel-distributed idiosyncratic shock with parameter ν. These shocks capture in a stylized way the fact that bus routes

[73] Through additional partial city coverage from earlier years, we confirmed that this data captures pre-COVID traffic congestion patterns and that aggregate congestion appears flat since 2016.

between o and d do not always follow the shortest path, with small ν corresponding to larger deviations. The commuter selects the path with the shortest realized travel time. By properties of multinomial logit, the expected bus travel time between o and d is

$$\widetilde{T}_{od} \equiv \mathbb{E} \max_p \widetilde{T}_p = -\nu^{-1} \log \Big(\sum_p \exp(-\nu T_p) \Big)$$

Allen and Arkolakis (2022) prove that the matrix $B = (\exp(-\nu \widetilde{T}_{od}))_{od}$ is given by $B = (I - A)^{-1}$ where $A = (\exp(-\nu T_{ij}))_{ij}$ is the travel time adjacency matrix, which has entries equal to zero whenever ij is not an edge.

We next estimate T_{ij} for all ij edges, as well as ν, using data on $\widehat{T}^{\text{bus}}_{od}$. We minimize the following objective function

$$\min_{\nu,(T_{ij})_{ij}} \sum_{o,d} \psi_{od} \Big(\log(\widehat{T}^{\text{bus}}_{od}) - \log(\widetilde{T}_{od}) \Big)^2$$

using weights ψ_{od} equal to the inverse squared distance between grid cells o and d. We estimate $\widehat{\nu} = 118$ for BRT and $\widehat{\nu} = 116$ for non-BRT, suggesting that bus routes are best explained as nearly shortest-route paths. (After this point, we no longer use ν.) We use the resulting $\widehat{T}_{ij}$ for BRT and non-BRT as bus travel time in all our counterfactual network exercises.

Figure A.33 prints the empirical fit of this exercise.

A.7.2 Ridership Equilibrium With Bus Capacity Penalties

Here we set up the formal model where commuters incur time penalties when bus occupancy exceeds bus capacity. Consider a route r and two consecutive stations s and s' on r. The route-edge occupancy level $B_{ss'r}$ is given by the expected number of passengers on a bus on route r between s and s'

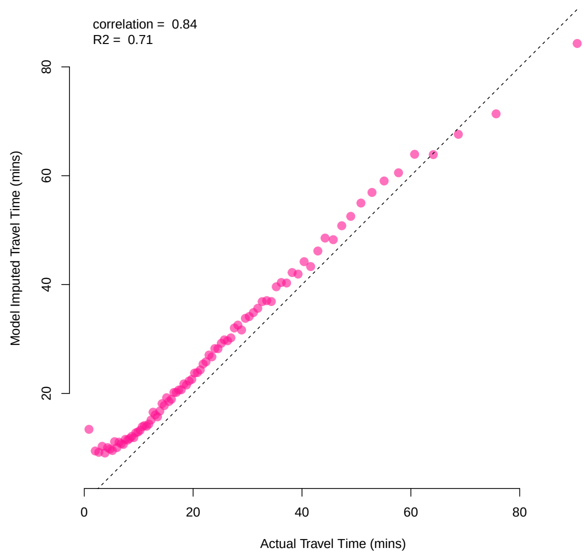
$$B_{ss'r} = \frac{1}{D_r} \sum_{o \le s < s' \le d} R_{odr}, \tag{15}$$

where R_{odr} is the daily number of commuters traveling on route r between any pair of stations o and d, and D_r is the daily number of bus departures on route r. The sum is over all stations o on route r that precedes s (in the direction from s to s'), and all d that come after s'.

We assume that the perceived travel time on route r and edge ss' is given by

$$T^p_{ss'r} = T_{ss'r} \cdot \kappa(B_{ss'r}), \tag{16}$$

Figure A.33: Bus Travel Time Model Fit



Notes: This binscatter plot assesses GMM model fit. It correlates our data on bus travel times T^{bus}_{odr} at the origin-destination-route level, restricting to origin-destination pairs for which this data is available, on the x-axis, with model-predicted expected bus travel time $\widetilde{T}_{od}$ on the y-axis.

where $T_{ss'r}$ is the baseline (objective) travel time, and $\kappa(B)$ is a function equal to 1 when bus occupancy is less than bus capacity, $B \leq \overline{B}$, and convex and increasing for $B > \overline{B}$. We parameterize κ as

$$\kappa(B) = \max\left(1, \quad \exp\left(0.2 \cdot (B - 84)\right)\right). \tag{17}$$

We assume there is no penalty up to $\overline{B} = 84$ persons per bus (the average bus capacity in TransJakarta's fleet at the end of our study period), and the penalty increases rapidly afterward. Perceived travel time with 90 people on the bus is already more than three times as high as objective travel time.

We define the perceived travel time between any two stations o, d on route r as the sum of edge-wise perceived travel times along that route, i.e. $T^p_{odr} = \sum_{k=1}^{n} T^p_{s_k s_{k+1} r}$ where $s_1 = o$ and $s_n = d$. As commuters get on and off, crowding on a given route fluctuates at different points on the route. Our model assumes that commuter disutility is additive over route segments, based on the crowding level at each point.

A ridership equilibrium is defined by a set of crowding levels (for each route r and each pair of consecutive stations s, s') such that optimal commuter choices lead to ridership patterns and a set of crowding levels that is exactly the assumed one. That is, it involves a fixed point, whereby crowding $B_{ss'r}$ is determined by equation (15) based on route-specific ridership flows R_{odr}, and ridership patterns R_{odr} are given by choice probabilities based on equations (2), (3) and (5), where we replace travel time T_{odr} with perceived travel time T^p_{odr} given by (16), which depends on $B_{ss'r}$.

We compute a ridership equilibrium by iterating until we reach a fixed point. For computational reasons, in our simulated annealing algorithm, we approximate the equilibrium by a single one-step updating rule. We compute route choices and ridership assuming no penalties, then compute occupancy levels and re-compute ridership once more. This allocation is very close to the equilibrium. For the 200 bus networks we obtain from the simulated annealing algorithm, we compare the one-step allocation with the fixed-point equilibrium. 170 of the 200 networks are already at equilibrium because all edge-occupancy levels are below $\overline{B}$. Even when they differ, the maximum deviation is negligible, equivalent to less than a second of wait time for every original TransJakarta passenger. The maximum ridership difference is 0.03% of daily bus ridership.

A.7.3 Model Welfare Bound

Here we prove a slightly stronger version of Proposition 3. Without loss of generality, we assume $W^{\max} = 0$, so all other welfare levels are non-positive. Denote $\mathcal{N}^{\neg\max}$ the set of networks excluding the welfare maximizing network (picking one arbitrarily if there are ties). We rewrite expected welfare as

$$\mathbb{E}_\pi W = \sum_{N \in \mathcal{N}^{\neg\max}} x_N \pi(N) = \sum_{N \in \mathcal{N}^{\neg\max}} x_N \frac{\exp(\beta x_N)}{1 + \sum_{M \in \mathcal{N}^{\neg\max}} \exp(\beta x_M)}. \tag{18}$$

We focus on minimizing $\mathbb{E}_\pi W$ with respect to welfare levels $x_N \leq 0$ for all $N \in \mathcal{N}^{\neg\max}$.

Lemma 1. *To minimize $\mathbb{E}_\pi W$, all welfare levels are interior, meaning $-\infty < x_N < 0$ for all $N \in \mathcal{N}^{\neg\max}$.*

Proof. Write $\mathbb{E}_\pi W$ as a function the welfare of one of the networks $N \in \mathcal{N}^{\neg\max}$, fixing the other:

$$\mathbb{E}_\pi W = f(x) = \frac{x \exp(\beta x) + \Sigma_2}{1 + \exp(\beta x) + \Sigma_1},$$

where

$$\Sigma_1 \equiv \sum_{M \in \mathcal{N}^{\neg\max} \setminus \{N\}} \exp(\beta x_M) \geq 0$$

$$\Sigma_2 \equiv \sum_{M \in \mathcal{N}^{\neg\max} \setminus \{N\}} x_M \exp(\beta x_M) \leq 0$$

The function f is continuously differentiable with endpoints $f(-\infty) = \frac{\Sigma_2}{1+\Sigma_1} < f(0) = \frac{\Sigma_2}{2+\Sigma_1} \leq 0$. This means $x = 0$ cannot minimize f. We next rule out that $x = -\infty$ can minimize f. It is easy to check that $f'(x) \propto 1 + \Sigma_1 - \beta\Sigma_2 + \exp(\beta x) + \beta x(1 + \Sigma_1)$, which is negative for sufficiently negative x. This means that f is minimized for $-\infty < x < 0$. $\square$

Lemma 2. *To minimize $\mathbb{E}_\pi W$, all welfare levels are equal to the solution of*

$$(n-1)\exp(\beta x) = -(1 + \beta x),$$

where $n = |\mathcal{N}|$ is the total number of networks.

Proof. Using standard results for logit probabilities, the partial derivative of (18) is

$$\begin{aligned}
\frac{d}{dx_N}\mathbb{E}_\pi W &= \frac{d}{dx_N} x_N \pi(N) + \frac{d}{dx_N}\Big(\sum_{M \neq N} x_M \pi(M)\Big) \\
&= \Big(\pi(N) + x_N \pi(N)(1 - \pi(N))\beta\Big) + \Big(\sum_{M \neq N} -x_M \pi(N)\pi(M)\beta\Big) \\
&= \pi(N)\Big(1 + \beta x_N - \sum_M \beta x_M \pi(M)\Big)
\end{aligned}$$

where the sums in the first two rows are over $M \in \mathcal{N}^{\neg\max} \setminus \{N\}$.

Setting this to zero, and noting that $\pi(N) > 0$ for all N and that the last summation term in the last row does not depend on N, we find that all optimal x_N are equal. Setting all $x_N = x$ and using the logit expression for π_M we get

$$1 + \beta x = (n-1)\beta x \frac{\exp(\beta x)}{1 + (n-1)\exp(\beta x)}.$$

This is equivalent to the expression in the lemma. $\square$

To finish up the proof of Proposition 3, note that the function $g(x) = (n-1)\exp(\beta x) + 1 + \beta x$ is increasing in $x \leq 0$, that $g(0) > 0$, and it is easy to check that plugging in $x < 0$ that satisfies $1 + \beta x = -\log(n)$ gives $g(x) = \frac{n-1}{n}e^{-1} - \log(n-1) < 0$.

This implies that at the common welfare level $X_N = x^*$ for all $N \in \mathcal{N}^{\neg\max}$ we have $1 + \beta x^* > -\log(n)$. Minimized expected welfare then satisfies

$$\begin{aligned}
\min \mathbb{E}_\pi W &= \frac{x^*(n-1)\exp(\beta x^*)}{1 + (n-1)\exp(\beta x^*)} \\
&= \frac{-x^*(1 + \beta x^*)}{1 - 1 - \beta x^*} = \frac{1 + \beta x^*}{\beta} = \\
&> -\frac{\log(n)}{\beta} \qquad \square
\end{aligned}$$

A.7.4 Analytic Results for Optimal Allocations Model

Formula for Local Comparative Statics. Here we derive an expression for the derivative of the expected optimal property $f^*(\theta)$ with respect to a scalar θ:

$$D_\theta f^*(\theta) = \sum_N \pi(N;\theta)\Big(\beta D_\theta W(N;\theta)\big(f(N,\theta) - f^*(\theta)\big) + D_\theta f(N;\theta)\Big), \tag{19}$$

where $D_\theta W$ and $D_\theta f$ are the gradients of W and f with respect to θ, respectively. The first term captures the change in f^* due to changes in the probabilities $\pi(N;\theta)$, while the second term captures the direct effect on f of changing θ. When f does not depend directly on θ, as for the network statistics we analyze later, the gradient scales linearly with β and with the scale of welfare W. Hence, we focus on the sign of $D_\theta f^*(\theta)$ and compare the local effects of different parameters in θ. Once again, we can estimate $D_\theta f^*(\theta)$ using a sample of networks from the π distribution.

To prove this, we first use the logit formula to show that

$$D_\theta \pi(N,\theta) = \beta\pi(N,\theta)\Big(D_\theta W(N,\theta) - \mathbb{E}_{N'} D_\theta W(N',\theta)\Big) \tag{20}$$

The term in parentheses says that when θ changes, the probability of a given network N increases if and only if the derivative of welfare W with respect to θ at N is higher than its expectation over all possible networks. Using this expression and rearranging sums several times we get

$$\begin{aligned}
D_\theta f^*(\theta) &= \sum_N D_\theta \pi(N,\theta) f(N,\theta) + \pi(N,\theta) D_\theta f(N,\theta) \\
&= \sum_N \beta\pi(N,\theta)\Big(D_\theta W(N,\theta) - \mathbb{E}_{N'} D_\theta W(N',\theta)\Big) f(N,\theta) + \pi(N,\theta) D_\theta f(N,\theta) \\
&= \Big[\sum_N \beta\pi(N,\theta) D_\theta W(N,\theta) f(N,\theta)\Big] - \beta\Big[\mathbb{E}_{N'} D_\theta W(N',\theta)\Big]\underbrace{\Big[\sum_N \beta\pi(N,\theta) f(N,\theta)\Big]}_{f^*(\theta)} + \\
&\qquad + \Big[\sum_N \pi(N,\theta) D_\theta f(N,\theta)\Big] \\
&= \sum_N \beta\pi(N,\theta) D_\theta W(N,\theta) f(N,\theta) - \beta\Big[\sum_{N'} \pi(N',\theta) D_\theta W(N',\theta)\Big] f^*(\theta) + \sum_N \pi(N,\theta) D_\theta f(N,\theta) \\
&= \sum_N \pi(N,\theta)\Big[\beta D_\theta W(N,\theta)\Big[f(N,\theta) - f^*(\theta)\Big] + D_\theta f(N,\theta)\Big].
\end{aligned}$$

A.7.5 The Simulated Annealing (SA) Algorithm Asymptotic Result

In this section, we define the Metropolis-Hastings (MH) and (modified) Simulated Annealing (SA) algorithms formally, and include a brief discussion of mixing times. This material draws on (Levin and Peres, 2017).

The MH algorithm asymptotically samples from the planner's distribution π in the following sense:

Proposition 4. *Assume that the proposal distribution Ψ is irreducible, aperiodic, and satisfies $\Psi(N \mid N') > 0 \iff \Psi(N' \mid N) > 0$. The Metropolis-Hastings algorithm with transition probabilities given by (11) is an ergodic Markov chain, meaning it has a unique stationary distribution, and that unique stationary distribution is the planner's probability distribution π. As $k \to \infty$, the state N_k converges in distribution to π.*

This is a classic result, see for example (Levin and Peres, 2017), section 3.2.

Mixing Times for MH. The mixing time of a Markov chain formalizes how many steps it takes to get "close" to the stationary distribution. For the k-th step of MH, we define the worst-case distance to the stationary distribution π as

$$d(k) = \max_{N_1 \in \mathcal{N}} \left\| \nu_k(N_1) - \pi \right\|_{TV},$$

where $\nu_k(N_1)$ is the distribution of N_k when MH starts at N_1 and $\left\|\alpha - \beta\right\|_{TV} = \max_{A \subseteq \mathcal{N}} |\alpha(A) - \beta(A)|$ is the total variation distance between two probability distributions α and β over $\mathcal{N}$.

The convergence theorem for ergodic Markov chains states that $d(k)$ falls exponentially in k. Specifically, there exist constants $C > 0$ and $\gamma \in (0, 1)$ such that $d(k) \leq C\gamma^k$.

The mixing time is typically defined as $t_{\text{mix}} = \min\{k : d(k) \leq \frac{1}{4}\}$.[74]

The convergence theorem implies that the constant γ is the crucial determinant of mixing times. The closer γ is to 1, the slower the chain is guaranteed to mix.

The concepts of slow and fast mixing are defined with respect to a family of problems underlying the MH algorithm (or any Markov chain) that depend on a growing parameter n that captures the problem size. In our setting, n could be, for example, the number of grids in the geographic environment. The object of interest is then how the expectation of the mixing time depends on n. When this dependence is itself exponential, we say that MH mixes slowly. Mixing times that are polynomial in n are considered fast (see footnote 58 for an example).

[74] Neither the distance measure nor the value of the $\frac{1}{4}$ constant (which is used by convention), matter for the substantive mixing time properties of a chain.

The problem with the MH algorithm is that it can exhibit slow mixing when, loosely speaking, the welfare function W has multiple local maxima (defined with respect to the proposal function Ψ).

We next define the SA algorithm, which is one of several algorithms that build on MH with the aim to improve its convergence properties.

The Modified Simulated Annealing Algorithm SA runs for a fixed number of steps K. SA begins at an arbitrary network N_1. Given network N_k at step k, it draws a candidate network N' from a probabilistic proposal distribution $\Psi(N' \mid N_k)$. SA accepts the network N' with probability:

$$\Pr(N_{k+1} = N' \mid N_k) = \min\left(1, \frac{\exp\left(\beta_k W(N')\right)\Psi(N_k \mid N')}{\exp\left(\beta_k W(N_k)\right)\Psi(N' \mid N_k)}\right), \tag{21}$$

and keeps $N_{k+1} = N_k$ otherwise. That is, the candidate network is always accepted if it has higher welfare (with an adjustment for proposal probabilities), and otherwise it is accepted with a probability that is increasing in $W(N')$ and decreasing in β_k. The term β_k is called the "inverse temperature" and it follows a deterministic schedule. It grows exponentially from some initial value β_1 to $\beta_K = \beta$. (Hence, at the end of the SA algorithm the inverse temperature coincides with the planner's parameter β). That is, $\beta_k = \beta_1 \rho^k$ for an appropriate $\rho > 1$.

This algorithm also asymptotically samples from the planner's distribution π.

Proposition 5. *Assume that the final inverse temperature parameter is set to $\beta_K = \beta$, the logit parameter from the social planner's problem and that the proposal distribution Ψ is irreducible, aperiodic, and satisfies $\Psi(N \mid N') > 0 \iff \Psi(N' \mid N) > 0$. As the number of steps $K \to \infty$, the final state N_K of the simulated annealing algorithm converges in distribution to π, the planner's distribution over allocations.*

The proof uses standard Metropolist-Hastings algorithm arguments. It mirrors a classic proof that as the number of steps increases, the outcome of SA converges to the global optimum.

As the number of steps K grows, the SA algorithm can be approximated by a sequence of MH algorithms with increasing inverse temperatures. In particular, the number of steps with an inverse temperature nearly equal to β grows to infinity, and hence the endpoint of the SA algorithm is asymptotically distributed according to π. These arguments can be made precise, see Nikolaev and Jacobson (2010).

A.7.6 The Candidate Network Proposal Function Ψ

How to produce new candidate networks is critical for the success of the SA algorithm. Starting from a network N_k, we obtain a proposed network N' by applying one of the "modifier" operations described below. We first select one of the four categories with equal probability, then a type of modifier within the category with equal probability, and finally, we generate a network N' according to the selected modifier, uniformly randomly.

Local Bus Modifiers

1. Exchange one bus between two randomly drawn bus routes.
2. Exchange a randomly drawn 10% of buses between two randomly drawn bus routes.

Global Bus Modifiers

1. Give or take away buses from a randomly selected bus route. Redistribute buses among randomly chosen other lines to stay at the constraint of 1,500 total buses.

Local Route Modifiers

1. Draw a bus route at random at add one random new adjacent stop to one end of the route.
2. Draw a bus route at random and take away one random stop at one end of the route.
3. Draw a bus route at random, pick two locations A and B on the route and "straighten" the route by replacing the intermediate stops by the shortest path between A and B.
4. Draw a bus route at random and add a detour to it. Pick two stops on the bus route, pick one new location on the map at random and let the route go between the two stops through the new location.

Global Route Modifiers

1. Create a random new bus route. Pick two locations at random and create a bus route on the shortest path between these locations. Assign a random number of buses to the new route, redistributed from randomly chosen other bus routes.
2. Delete a randomly drawn route and replace it with a random new bus route: pick two locations at random and create a bus route on the shortest path between these locations. Assign a random number of buses to the new route, redistributed from randomly chosen other bus routes.
3. Delete a randomly drawn existing bus route, where routes with less ridership are more likely to be chosen. Redistribute the buses among other, randomly chosen, bus routes.
4. Delete a randomly chosen set of 10 bus routes at once. Redistribute the buses among other, randomly chosen, bus routes.
5. Add 10 random bus lines at once. Assign a random number of buses to the new routes, redistributed from randomly chosen other bus routes.

Approximating Proposal Probabilities Ratio In our implementation of the simulated annealing algorithm, we use the following approximation when using equation (21)

$$\frac{\Psi(N_k \mid N')}{\Psi(N' \mid N_k)} = 1.$$

We make this approximation because computing the proposal function ratio term precisely is computationally intensive. $\Psi(N' \mid N_k)$ is the probability that network N' is proposed when starting from network N_k. To compute this, we need to enumerate all possible modifications starting from N_k. (Given that we stratify, we only need to do this within each category and type of modification.)

For most modifiers described above, this ratio is likely to be close to 1. For example, this ratio is approximately equal to 1 for the proposal function that exchanges a bus between two randomly drawn bus routes. There are two types of exceptions. First, certain modifiers that we use are not reversible. Second, for the modifiers that add or delete routes, the ratio is systematically far from 1. We argue that approximating the ratio with 1 will lead our algorithm to converge to a stationary distribution that puts more weight on expansive networks. This would only strengthen our main results that optimal networks are more expansive than the current TransJakarta network.

To see this more clearly, denote by R the (very large) number of possible routes in our square grid cell environment, and by $r(N)$ the number of routes in network N. For ease of exposition, assume that only modifiers that add or remove one route are allowed, and assume that N' is obtained from N_k by adding a randomly chosen route. Then the proposal function ratio is given by

$$\frac{\Psi(N_k \mid N')}{\Psi(N' \mid N_k)} = \frac{\frac{1}{r(N_k)+1}}{\frac{1}{R-r(N_k)}} = \frac{R - r(N_k)}{r(N_k)+1}.$$

This is significantly larger than 1, meaning that transitions towards networks with many routes is favored in equation (21). It easy to prove that if we use the ratio equal to 1, the Markov chain will converge to a modified stationary distribution given by

$$\widetilde{\pi}(N) \propto \pi(N) \cdot \binom{R}{r(N)+1}^{-1}$$

In other words, the modified stationary distribution puts less weight on expansive networks. Note that this adjustment effectively compensates for the very large number of expansive networks, which is of the same magnitude as the adjustment factor.

A.7.7 Details on Statistical Test of Mixing

To conduct the test of statistical mixing outlined in the main test, we need an upper bound on the number of hypothetically feasible networks $\bar{n}$. This number is extremely large, but can nevertheless be characterised and employed for our purposes.

We assume that each hypothetical network consists of at most N lines, each of length ℓ, arranged over 418 grid cells, each with 8 neighbors. B total buses are distributed across these lines. The total number of potential networks $\bar{n}$ is then at most equal to

$$\underbrace{(418 * 8^{\ell-1})^N}_{\text{line designs}} \times \underbrace{\frac{(B+N-1)!}{B!(N-1)!}}_{\text{bus allocations}} \tag{22}$$

where the second term is just the number of combinatorial ways to allocate B indistinguishable balls into N distinguishable urns.

We are interested in

$$\begin{aligned}\log \bar{n} &= N \log 418 + N(\ell-1)\log 8 + \log((B+N-1)!) - \log(B!) - \log((N-1)!) \\ &\approx N \log 418 + N(\ell-1)\log 8 + (B+N-1)\log(B+N-1) \\ &\quad - B\log(B) - (N-1)\log(N-1)\end{aligned}$$

where we have used Ramanujan's formula that $\log(x!) \approx x \log x$.

In our simulations, we assume that there can be at most $N = 200$ networks of maximum length of $\ell = 60$ stops and a total of $B = 1500$ buses. Plugging these in, we arrive at an upper bound of $\log \bar{n} \approx 26,258$.

Overall, the upper bound from Proposition 3 is:

$$\frac{\log \bar{n}}{\beta} \approx \frac{26,258}{4.0 \times 10^5} \approx 0.0656.$$

A.7.8 A Simple Model of Optimal Wait Times

In this section, we describe a simple model of optimal wait times when the planner allocates buses between two partly overlapping routes to different areas of the city.

The city center (location 0) is connected by two bus routes to two locations $i = 1$ (near center) and $i = 2$ (periphery), with bus travel times $T_1 < T_2$ and number of commuters $V_1 > V_2$. (We assume no commuters travel between 1 and 2.)

The bus route to location 2 passes through location 1, so both bus routes connect the city center to location 1.

The planner allocates B_i buses to route i to maximize aggregate ridership, subject to a budget constraint $B_1 + B_2 = B$. Bus arrival rates between the city center and the two locations are

$$\lambda_1 = \frac{B_1}{T_1} + \frac{B_2}{T_2},$$
$$\lambda_2 = \frac{B_2}{T_2}.$$

The expected wait time to location i is $W_i = 1/\lambda_i$, which is convex in B_i.

We assume that the share of commuter who take the bus is linear in wait time, $\pi(W_i) = \pi_0 - \pi_1 W_i$ with $\pi_0 < 1$ and $\pi_1 > 0$. The key assumption is that the share is bounded even when wait times are zero, which is also a feature of our full demand model. This means that in the limit, very small wait times are not useful. Their benefit is bounded, and they are "expensive" to achieve because wait time is inversely proportional to B_i.

The planner's problem is hence

$$\max_{B_1, B_2} V_1 \pi(W_1) + V_2 \pi(W_2) \qquad \text{s.t. } B_1 + B_2 = B$$

With a linear demand function π, the first order conditions imply that at an interior optimum, the ratio of wait times between the city center and the two destinations is

$$\frac{W_2^*}{W_1^*} = \sqrt{\frac{V_1 \cdot \left(\frac{1}{T_1} - \frac{1}{T_2}\right)}{V_2 \cdot \frac{1}{T_2}}} = \sqrt{\frac{V_1}{V_2}\left(\frac{T_2}{T_1} - 1\right)}.$$

Wait times to reach the periphery are relatively higher when the ratio of commuters V_1/V_2 is higher. The ratio of travel times may attenuate this effect when $T_2 < 2T_1$. Intuitively, buses on both routes decrease wait times to get to location 1, although buses allocated to route $i = 2$ are less efficient because they spend more time traveling beyond location 1. Hence, the relative marginal benefit of adding a bus to route 1 is proportional to $\frac{1}{T_1} - \frac{1}{T_2}$, which can be smaller than $\frac{1}{T_2}$ when the time to the periphery is not much higher than to location 1. The square root moderates both of these forces, reflecting the fact that it is increasingly "expensive" to reduce wait times that are already small.

This simple model shows how several core features of demand and geography interact to determine the optimal degree of dispersion in wait times in the bus network.

A.8 Impact on Air Pollution and Congestion

In this section we estimate the impact of TransJakarta network improvements on air pollution and road traffic congestion using back-of-the-envelope calculations.

A.8.1 Impact on Road Traffic Congestion

Consider the impact of 1% of trips diverted from other travel modes to TransJakarta buses. We will estimate the reduction in road traffic volume, and then use a congestion elasticity to convert the change in volume to a change in travel time. For context, recall that our main simulations, optimal networks increase ridership by 1.16 percentage points (Table 5).

Reduction in Road Traffic Volume. The change in traffic volume depends on what modes the new TransJakarta passengers are substituting away from, and the relative volumes of those modes.[75]

In our benchmark specification, we assume that a 1% shift toward TransJakarta from other modes lower road traffic volumes by 1%. We obtain this number by measuring initial modal shares and assuming that substitution is proportional to initial shares. The 2018 JUTPI household survey measured modal shares in the larger JaBoDeTaBek area, home to around 30 million people (JUTPI2, 2019). The modal shares were 14.5% car, 78.9% motorcycle or motorcycle taxi, 1.35% TransJakarta, 2.9% other buses, and 1.7% train. We assume a volume of 1 for motorcycles, 2.5 for cars, and zero for other buses and trains, which means that we are assuming that there is no reduction in road traffic volume from train and other bus switchers. This leads to a reduction of road traffic volume of

$$1.15\% = 14.5 \times 2.5\% + 78.9 \times 1\% + (2.9 + 1.7) \times 0\%$$

We obtain a 1% reduction in road traffic volume by rounding down to obtain a more conservative number.

Congestion Elasticity. (Gaduh et al., 2022) estimate the elasticity of road travel time to road lanes, and obtain a value of 0.29.[76] We set the elasticity of road travel time to traffic volume to be the same. Akbar and Duranton (2017) and Kreindler (2024) estimate this elasticity for Bogota, Columbia, and Bangalore, India, and find maximum elasticities of

[75] For this calculation, we ignore the equilibrium aspect of road traffic congestion. Due to lighter congestion, some commuters would switch away from TransJakarta.

[76] This is the parameter δ_1 in their notation, given by $\delta_1 = \lambda\theta = 0.055 \times 5.36$.

around 0.20.

Using the number for Jakarta, this means that a 1% bus modal share increase increases travel times by 0.3%. We assume that an average trip takes 30 minutes at baseline and that all non-train modes benefit from the lower congestion. The total daily travel time savings due to lower congestion are

$$\begin{aligned} 58,820 \text{ hours} = &(39,891,951 \text{ trips}) \times (0.983 \text{ share of trips using roads}) \times \\ &(0.5 \text{ hours trips duration}) \times (0.003 \text{ travel time reduction}) \end{aligned}$$

One way to benchmark these results is using the equivalent variation benefits from optimal networks cited in section 5 of 11.2 minutes of bus travel time per initial bus trip. This adds up to

$$139,994 \text{ hours} = (39,891,951 \text{ trips}) \times (0.019 \text{ initial TransJakarta share}) \times (11.2/60 \text{ EV gain})$$

The benefit due to lower traffic congestion (58,820 hours of time saved) is approximately 42% of the (direct) equivalent variation benefit from re-organizing the bus network (equivalent to 139,994 hours of time on the bus saved). This calculation assumes that time on the bus and time in traffic are valued similarly.

At the end of this section, we return to these estimates and monetize them using a back-of-the-envelope calibration of the value of time at 50% of the wage in Jakarta.

Limitations. Our simple estimates aim to give a rough sense of possible impacts on congestion but have several limitations. First, we do not have the data to estimate the substitution between buses and other modes. We have also assumed, based on the null effect of TransJakarta expansion on total trips, that all new bus trips replace other modes. If, in fact, some of the bus trips are new, this would diminish the congestion effects. We used an overall elasticity of travel times with respect to congestion, thus ignoring the spatial heterogeneity in bus ridership and traffic congestion. It is possible that bus users select from routes with high marginal congestion as in Anderson (2014), or vice-versa. Some commuters might value time on congested conditions higher than the value of time on the bus, leading to higher values than we find here.

A.8.2 Impact on Air Pollution

Our main approach is to use estimates from Gendron-Carrier et al. (2022), who quantify the impact of the opening of subway systems on air pollution. We begin with a short description of air pollution and emissions patterns in Jakarta.

Vehicle emissions in Jakarta. (Mahalana et al., 2022) measured real-world emissions for over $93,000$ vehicles in Jakarta at 18 sites using devices installed at the side of the road, and matching readings with license plate information and vehicle registration information.

Table A.17 below lists median emissions by vehicle type for three different pollutants, expressed in grams of pollutant per kilogram of fuel. TransJakarta buses, which are certified to the Euro II standard, are less polluting than other buses for all pollutant types. Private vehicles emit fewer nitrogen oxides (NOx) than TransJakarta buses per kilogram of fuel, but more than twice the carbon monoxide (CO) and three times the hydrocarbons (HC). While motorcycles account for more than the majority of vehicles in Jakarta, this study has a small sample of motorcycle because the measurement devices were located near toll booths, where motorcycles are prohibited. Based on 135 motorcycle readings in the data, the report finds suggestive evidence that motorcycles pollute significantly more than private vehicles and than buses per kg of fuel. These figures do not adjust for the smaller fuel consumption per kilometer for private vehicles and motorcycles, nor for the higher occupancy of buses.

Overall, these results suggest that shifting commuters in Jakarta towards using TransJakarta buses is likely to reduce emissions and pollution.

Table A.17: Median emissions by vehicle category (in g/kg fuel).

	Nitrogen Oxides (NOx)	Carbon Monoxide (CO)	Hydrocarbon (HC)	Sample Size
TransJakarta buses	24	16	3	789
Other buses	56	24	6	1,504
Private vehicles	5	34	9	77,989
Motorcycles	45	156	67	135

Note: The source is Mahalana et al. (2022). The numbers in this table are estimated based on the bar charts from Figures 7, 14, and 15.

The impact of public transport on air pollution. Gendron-Carrier et al. (2022) measure air pollution using satellite measurements of Aerosol Optical Depth (AOD). They show that a new subway system lowers AOD by -0.0182 in a sample of 42 cities where they have ridership data (Table 8). The same event increases ridership per $1,000$ people in the city by 36.8857. The implied effect of public transport ridership is that each subway passenger per $1,000$ population lowers AOD by 4.934×10^{-4}.

In Jakarta, the population of our study area is $18,384,186$, and the weekday total number of trips is $39,891,951$. Consider a 1% increase in the mode share of TransJakarta. The increase in ridership per $1,000$ inhabitants is

$$21.70 = \frac{1\% \text{ bus mode share} \times 39,891,951 \text{ trips}}{18,384,186 \text{ inhabitants}} \times 1,000$$

The implied effect on AOD is

$$21.70 \times -4.934 \times 10^{-4} = -0.0107$$

Gendron-Carrier et al. (2022) estimate that one unit of AOD is equivalent to $114.6\mu g/m^3$ of PM10 and to $60.6\mu g/m^3$ of PM2.5 (columns 2 and 8, Appendix Table A.2). This means that the -0.0107 AOD effect above is equivalent to a reduction in PM10 of $1.227\mu g/m^3$ and a reduction of $0.648\mu g/m^3$ in PM2.5. The mean annual concentration of PM2.5 in the Jakarta area in 2021 was around $30\mu g/m^3$, so this represents a 2.16% decrease in air pollution.

Limitations. The ridership impact in Gendron-Carrier et al. (2022) is measured in an 18 month period after the opening of the system, while the level of ridership after 5 years is roughly double. If we interpret the ridership effects implied by our model as the long-run effects, we should use this longer-term ridership numbers and hence divide all our estimates by a factor of approximately 2. By using the estimates from this paper, we are also assuming that subway and bus systems have similar effects on private mode usage and hence air pollution. Finally, we are using the estimates from a global sample of cities to the case of Jakarta.[77]

A.8.3 Monetizing Congestion and Air Pollution Effects

Direct effects. As derived above, the direct effects of implementing an optimal TransJakarta network is equivalent to saving around $139,994$ hours per day (in bus travel time) in the entire study area, or $36,398,440$ hours per year assuming 260 workdays per year.

We use an average wage of 3.5 USD per hour in Jakarta and assume that the value of time on the bus is 50% of the wage, leading to 63.7 million USD per year in direct gains.

Congestion effects. The reduction in congestion calculated above ($58,820$ hours of time in traffic per day), valued the same way as time on the bus, leads to estimated gains of 26.7

[77] Gendron-Carrier et al. (2022) show that the air pollution effects are larger in cities with high pollution levels at baseline, such as Jakarta.

million USD per year.

Air pollution effects. The monetary value of health savings from air pollution reductions are more difficult to estimate. We follow Gendron-Carrier et al. (2022), who use two methodologies. The first focuses only on benefits from lower infant mortality, while the second uses the WHO Global Burden of Disease methodology to capture reductions in mortality and morbidity throughout the life-cycle.

We replicate here the first methodology, and note that the gains based on the second methodology are around 20 times larger. We use the estimate in Gendron-Carrier et al. (2022) that $1\mu g/m^3$ PM10 leads to 10 infant deaths averted per $100,000$ births, based on their literature review of the causal impact of air pollution on infant mortality. The yearly birth rate for Indonesia is 17 per $1,000$ people, which implies $312,531$ births per year in our study area. This means that the number of infant deaths averted by an optimal TransJakarta network is

$$38.34 = (1.227\mu\text{g/m}^3 \text{ of } PM10) \times \frac{10}{100,000} \times (312,531 \text{ births per year})$$

We use a Value of Statistical Life (VSL) of 6 million USD in the United States with a 0.6 elasticity of VSL with respect to country GDP. This leads to a value of 2.57 million USD per life saved in Jakarta.[78] In total, this means gains of $98.41 = 38.34 \times 2.57$ million USD per year.

[78]This is based on a GDP per capita of 18k USD for the Jakarta area in 2022, relative to 76k USD for the United States.